

A Framework of Building Reliable Large Language Models to Mitigate Hallucination

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Abstract

Although large language models (LLMs) have demonstrated remarkable capabilities, they remain susceptible to generating factually incorrect, fabricated, deceptive, and inconsistent content, referred to as “**Hallucination**”, potentially undermining the reliability of LLMs and misleading users, which has become a major bottleneck for LLM deployment. LLM hallucination primarily stems from the training phase, where LLMs are prone to learning frequent textual patterns rather than truthfully understanding the meanings of texts, exacerbating biases of unreliable outputs in three aspects: 1) LLMs themselves cannot always respond honestly, while 2) their outputs often contain factual errors in single responses and 3) contextual inconsistency in multi-turn interactions, collectively forming the fundamental research challenges for constructing **Reliable LLMs**.

Therefore, this thesis proposes a systematic framework to construct reliable LLMs for hallucination mitigation. The work introduces a formal definition of LLM reliability: a reliable LLM should factually leverage knowledge to generate responses, honestly express uncertainty for unsure queries, and maintain contextual consistency throughout multi-turn interactions. Accordingly, we conduct research along the three aforementioned aspects: **Factuality**, **Honesty**, and **Consistency**.

First, this work analyzes non-factual responses in LLMs and attributes them to ambiguous knowledge boundaries, and hence proposes a novel factuality alignment framework characterized by “from perception to expression”, which enables LLMs to accurately perceive their precise knowledge boundary representations and

then explicitly leverage the perceptions to align with factual expressions via PPO. Experimental results demonstrate that the proposed factual alignment framework substantially improves the factuality of LLMs, highlighting a novel paradigm of “from perception to expression” on LLM alignment for future work.

Subsequently, this work investigates LLM honesty in two key aspects: self-awareness and confidence estimation. Since the evaluation of honesty varies across task scenarios, this thesis introduces two underexplored and challenging benchmarks of multilingual confidence estimation and awareness of mathematical reasoning solvability for LLMs. We conduct extensive empirical analyses on two benchmarks, derive several insightful findings, and propose corresponding optimization strategies. The two benchmarks can effectively evaluate LLM honesty in multilingual and mathematical tasks, carrying great significance for the global deployment and complex reasoning capability of LLMs.

Finally, this work focuses on the scenario of LLM-based search agents regarding the contextual inconsistency issue introduced through multi-turn interactions with external environments during retrieval. We identify the major factors contributing to contextual inconsistency and construct a context refiner. Then we incorporate the context refiner into the RL training pipeline of search agents, achieving further improvements in both accuracy and efficiency, inspiring a novel paradigm of “refining context and then generating” for LLM agents.

Overall, the framework in this thesis constructs reliable LLMs from the three perspectives of factuality, honesty, and consistency, significantly alleviating LLM hallucinations. Furthermore, the well-grounded framework establishes a strong foundation for navigating future development of trustworthy AI systems.

摘要

儘管大語言模型（LLM）已展現出卓越能力，卻仍易生成事實錯誤的、編造的、欺騙性的，以及前後不一致的內容，此類現象稱為“**幻覺**”。該問題會損害大型語言模型的可靠度、誤導使用者，已成為其落地部署的主要瓶頸。大模型的幻覺問題主要源於訓練階段：模型更傾向學習高頻固定文本模式而非真實理解文本意思，這就導致了不可靠輸出，體現於三個層面：1) 大模型本身無法總是誠實作答；2) 大模型單輪輸出結果常存在事實謬誤；3) 在多輪輸出中會出現上下文不一致。以上三者共同構成研發可靠大型語言模型的核心難題。

因此，本論文提出一套建構可靠大模型的系統化框架，用以緩解模型幻覺問題。本文對大模型的可靠度給出正式定義：可靠的大模型應能精準運用知識生成事實性的回覆、對未知問題誠實表達不確定性，並在多輪互動過程中維持上下文語義的一致性。據此，本研究從**事實性**、**誠實性**與**一致性**三大維度展開探討。

首先，本文分析大模型的事實錯誤輸出，並將其歸因於模型知識邊界模糊。對此，本研究提出一套全新「從感知到表達」的事實性對齊框架，讓大模型能夠精準感知自身知識邊界的表徵，再通過近端策略優化，將感知結果對齊為事實性表達。實驗結果顯示，本論文所提之事實性對齊策略可大幅提升大模型的事實性，為後續研究提供一套具備發展潛力的「從感知到表達」的模型對齊範式。

其次，本研究從自我意識與置信度評估兩大核心探究大模型的誠實性。考量到誠實性評估會隨任務場景而異，本文引入兩個尚未被研究且具挑戰性的基準：多語言置信度評估、數學推理可解性評估。我們針對兩大基準展開大量分析，歸納出多項具啟發性的結論，並提出相對應的優化策略。此兩套基準可有效評估大模型在多語言與數學任務中的誠實性，對模型全球化部署與複雜推理能力發展具重要意義。

最後，本文聚焦大模型的檢索智能體場景，探討其與外部環境多輪互動時衍生的上下文不一致問題。研究歸納出造成上下文不一致的關鍵因素，並建構上下文提純器。隨後將該提純器整合至檢索智能體的強化學習訓練流程，進一步提升模型準確度與效率，為大模型智能體開創「先優化上下文、再生成回覆」的全新範式。

整體而言，本論文所建構的研究框架，從事實性、誠實性、一致性三大層面實現可靠大模型的構建，有效緩解大模型的幻覺問題。本研究具備扎實理論基礎的整體框架，亦為後續可信人工智慧系統的發展奠定穩固根基。

Preface

The research works in this thesis have been published in the conference proceedings.

The work in Chapter 3 has been publicized in Boyang Xue, Fei Mi, Qi Zhu, Hongru Wang, Rui Wang, Sheng Wang, Erxin Yu, Xuming Hu, Kam-Fai Wong. UAlign: Leveraging Uncertainty Estimations for Factuality Alignment on Large Language Models. Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). 2025: 6002-6024 [218].

The first benchmark of multilingual confidence estimation in Chapter 4 has been publicized in: Boyang Xue, Hongru Wang, Rui Wang, Sheng Wang, Zezhong Wang, Yiming Du, Bin Liang, Wenxuan Zhang, Kam-Fai Wong. MlingConf: A Comprehensive Study of Multilingual Confidence Estimation on Large Language Models. Findings of the Association for Computational Linguistics: ACL 2025. 2025: 2535-2556 [219]. The second benchmark of mathematical reasoning is under review by ACL ARR March 2026.

The work in Chapter 5 has been publicized in: Boyang XUE, Bin Wu, Shuofei Qiao, Sheng Wang, Rui Wang, Yiming Du, Hongru WANG, Jeff Z. Pan, Emine Yilmaz, Kam-Fai Wong, Aldo Lipani. Mitigating Context Interference for Reliable and Efficient Search Agents. Findings of the Association for Computational Linguistics: ACL 2026. 2026 [219].

In addition, one preliminary work of Chapter 5 has been publicized in: Boyang Xue, Weichao Wang, Hongru Wang, Fei Mi, Rui Wang, Yasheng Wang, Lifeng

Shang, Xin Jiang, Qun Liu, Kam-Fai Wong. Improving Factual Consistency for Knowledge-Grounded Dialogue Systems via Knowledge Enhancement and Alignment. Findings of the Association for Computational Linguistics: EMNLP 2023. 2023: 7829-7844 [220].

Note that some details of the aforesaid chapters might be slightly different from the corresponding previous publications due to different experimental settings.

Biographical Sketch

Boyang XUE was born in Baoji, Shaanxi, China. He earned his B.S. degree in the School of Artificial Intelligence and Automation at Huazhong University of Science and Technology in 2020. Afterwards, he worked as a research assistant in the Department of Systems Engineering and Engineering Management at The Chinese University of Hong Kong, supervised by Prof. Xunying Liu from September 2020 to August 2021. Then, he enrolled as a Ph.D. student in the same group of the Department of Systems Engineering and Engineering Management, The Chinese University of Hong Kong, from August 2021 to September 2022. Then he has interned at Huawei Noah’s Ark Lab (Shenzhen) from November 2022 to July 2023. From August 2023, he pursued his Ph.D. study under the supervision of Prof. Kam-Fai Wong. He also cherished the great opportunity to visit the AI Center of University College London and the AI Center of ETH Zurich, working closely with Prof. Emine Yilmaz, Prof. Aldo Lipani, and Prof. Ryan Cotterell.

Boyang’s research focus consists of large language models, AI agents, trustworthy AI, and speech recognition. He has published papers at top conferences in the NLP research field, including ACL, EMNLP, NAACL, etc. He also actively contributes to the development of the AI community. His long-term goal is to continuously enhance the LLMs’ ability to understand and interact with the real world, while ensuring the safety and trustworthiness of deployment.

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My doctoral journey at the Chinese University of Hong Kong is drawing to a close. I am deeply grateful to everyone who has accompanied and supported me throughout my academic pursuit. This experience shall forever remain one of the most enjoyable and unforgettable chapters in my life.

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In addition, I am deeply thankful to the University and the SEEM Department for providing me with an outstanding academic platform. I feel incredibly fortunate to have studied at the most beautiful campus in Hong Kong. The five-year doctoral training has fostered my growth in academia, personality, and life. The department is a great home to me. I am also grateful to Prof. Andrew Xunying

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Besides, I am truly fortunate to have two precious research visiting experiences during my PhD study in London and Zurich with different but wonderful academic and cultural lifestyles. I got the chance to work with Prof. Emine Yilmaz and Prof. Aldo Lipani at the AI Center of University College London. Both of them made a significant impact in both academic and industrial fields. They are kind-hearted to care about my overseas life and offer me invaluable guidance for my career planning. I feel truly blessed to be accompanied by Prof. Emine and other group members in celebrating my 28th birthday. During my work with Prof. Ryan Cotterell at AI Center of ETH Zurich, I was always impressive for his profound academic accomplishments, rigorous scholarly style, and humility. I am equally honored to have met Prof. Jeff Pan at the University of Edinburgh, who also provided generous guidance for my research. Meeting so many distinguished and kind mentors has filled my heart with immense warmth and gratitude.

Next, I feel extremely fortunate to have met so many outstanding seniors and peers in my research group, including both alumni and current members, for their constant encouragement and support. The academic exchanges with them have continuously driven my progress, and their excellence has also inspired me to pursue more achievements. Beyond research, we have supported one another in daily life and forged many precious friendships. I would like to express my sincere gratitude to all my labmates: Xingshan Zeng, Lingzhi Wang, Bin Liang, Wai-Chung

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Looking back on my years of academic pursuit, I would like to express my sincere gratitude to my parents. Since I left home at a young age to further my studies, we have spent far more time apart than together. For years, I devoted myself to pursuing my academic dreams, yet habitually neglected your feelings. In our daily phone chats, I mostly talked about my own life and struggles, rarely taking the initiative to ask about your daily lives and health. All too often, I even dismissed your advice casually. Even so, you have always tolerated my willfulness and given me the very best within your power. You are also the ones who take the greatest pride in my pursuit of a doctoral degree. I have now come to realize that the best reward I can offer you lies not in remarkable achievements, but in sincere and regular care and conversations, more company, a stable career, and a healthy

life. I will truly grow from an ignorant student into a mature and responsible adult. May you both enjoy good health, peace, and lifelong happiness.

The new chapter of my life is to begin. I will pursue continuous excellence in my career, love, family, and life, and eventually become the ideal person I want. I will always keep passion and anticipation for life, and make steady progress in my career each day. The only way to do great work is to love what I do.

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Chapter 1

Introduction

1.1 Motivation

Large Language Models (LLMs) have demonstrated remarkable capabilities across a broad spectrum of tasks. However, despite their widespread adoption and impressive performance, LLMs still face a critical challenge that severely limits their further deployment: the generation of factually incorrect, fabricated, deceptive, and inconsistent content, commonly referred to as “**Hallucination**” issue [65, 72, 153, 45]. Although in recent years, LLMs have experienced unprecedented advancements, evolving from basic text generation applications (e.g., ChatGPT [135]) to advanced intelligent agents (e.g., Claude Code [7], GPT-5.5 [141]) capable of interacting with the external world and handling various complex tasks [206, 30], the hallucination issue remains a persistent bottleneck that constrains the capability of LLMs. This issue not only undermines the reliability of LLMs but also poses potential risks of misleading users, especially causing concerns in some high-stakes scenarios like health care [199] and finance [205], as exemplified in Figure 1.1, making hallucination mitigation one of the most critical challenges in LLM development.

The essence of LLM hallucination lies in the inherent limitations of the training mechanism [216, 65, 234, 45]. During the training phase, LLMs tend to learn

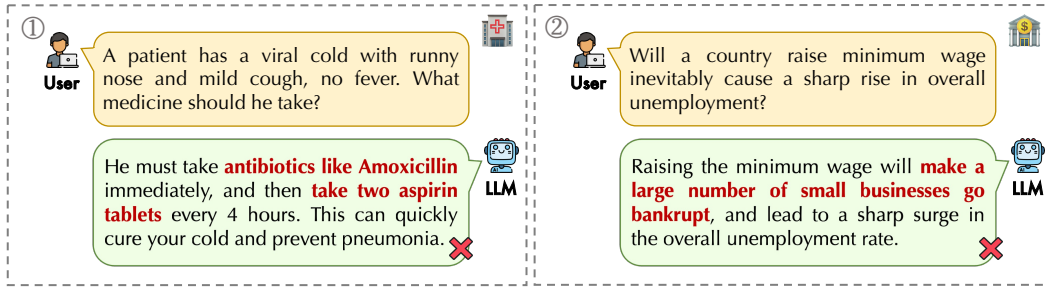


Figure 1.1: Two typical examples of hallucinations in health care and economics.



Figure 1.2: Main causes of LLM hallucination regarding pattern matching or behavior cloning.

frequent textual patterns on large-scale corpora [45, 108, 177, 234, 218] as illustrated in Figure 1.2, rather than truthfully understanding the meaning of texts, grasping objective facts, or establishing a clear boundary between known and unknown knowledge [6, 227, 233]. Specifically, during pre-training, LLMs learn massive amounts of deterministic texts of knowledge by maximum likelihood estimation (MLE) training for next token prediction. During post-training, LLMs merely imitate the response patterns contained in the instruction dataset, even equipped with reinforcement learning, which essentially amounts to **behavior cloning** or **pattern matching**. This intrinsic characteristic exacerbates the unreliability of LLM outputs in three key aspects, as illustrated in Figure 1.3: First, LLMs themselves lack the ability to always respond honestly or express uncertainty when confronted with queries beyond their knowledge scope [222, 101, 227, 6, 233]. Second, in single responses, their outputs are prone to contain factually incorrect knowledge [95, 187, 155, 218, 234]. Third, in multi-turn interactions, LLMs often fail to maintain contextual consistency, generating responses that contradict previous statements or deviate from the core context [209, 70, 242, 220, 186, 92]. These three aspects collectively form the fundamental research challenges of this work to construct **Reliable**

Dishonesty	Non-Factual	Inconsistency
 Do you know who invented the telephone and when was it invented?  Yes I know. The telephone was invented by Alexander Graham Bell in 1876.  <i>Unreliability of LLM itself</i>	 Who invented the telephone and when was it invented?  The telephone was invented by Alexander Graham Bell in 1876.  <i>Unreliability in Single Output</i>	 Who invented the telephone and when was it invented?  The telephone was invented by Alexander Graham Bell in 1876. ...  The first successful telephone was actually created in 1880 <i>Unreliability across Multi-turn Outputs</i>

Figure 1.3: *Unreliability of LLM outputs in three key aspects.*

LLMs, which implies that if no deliberate effort is made to enhance reliability during LLM training [114], LLMs will always suffer from hallucinations.

Existing research on LLM hallucination mitigation has explored various approaches, including data enhancement [110, 108, 177], fine-tuning strategies [233, 45, 108, 177], and retrieval-augmented generation (RAG) [42, 97, 56, 155]. However, most of these studies focus on a single dimension of hallucination and lack a framework that comprehensively considers the multiple aspects of LLM unreliability. Therefore, this dissertation proposes a systematic framework to construct reliable LLMs for hallucination mitigation. To provide a clear theoretical basis for this research, we first introduce a rigorous formal definition of LLM reliability: a reliable LLM should not only generate factually accurate responses by proficiently leveraging existing knowledge (**Factuality**) but also honestly express uncertainty when faced with unsure queries (**Honesty**), and maintain contextual consistency throughout multi-turn interactions (**Consistency**). Guided by this definition, this work conducts research along the three aforementioned aspects of LLM reliability in a progressive and hierarchical manner, contributing to each dimension of reliability and proposing targeted solutions as demonstrated in Figure 1.4.

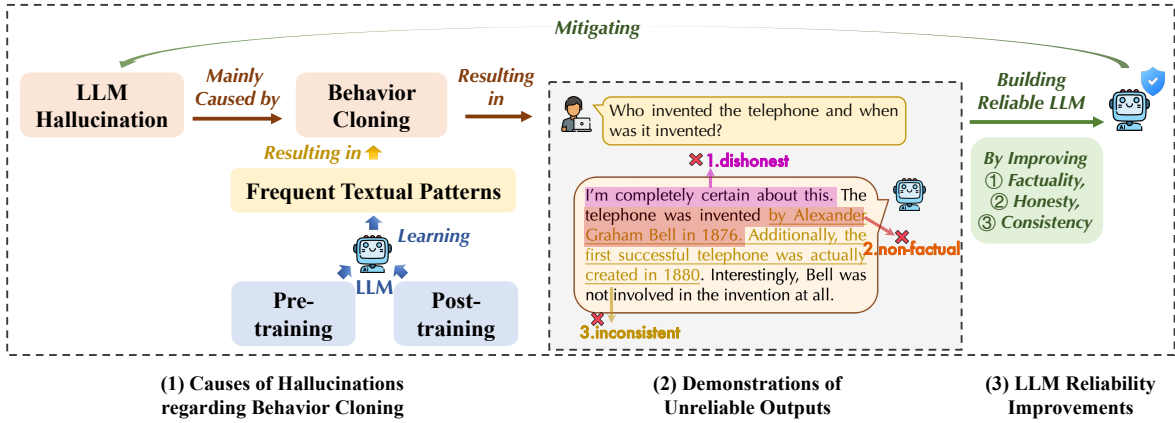


Figure 1.4: Framework of building reliable LLMs to mitigate hallucinations.

1.2 Contributions

The contributions illustrated in Figure 1.5 of this thesis are summarized below.

1.2.1 LLM Factuality

Factuality is the most basic requirement for reliability. First, **we conduct an in-depth analysis of non-factual responses generated by LLMs and attribute the cause to ambiguous knowledge boundaries**—LLMs cannot clearly perceive which knowledge they have mastered and which they have not, leading to hallucinated generations. The analysis highlights the great significance of focusing on the knowledge boundary information of LLMs.

To address this issue, **we propose a novel factuality alignment framework characterized by “from perception to expression”**, which enables LLMs to accurately perceive knowledge boundary representations and then explicitly leverage these perceptions to align their outputs with factuality through Proximal Policy Optimization (PPO). Specifically, we jointly employ confidence and semantic entropy to represent precise knowledge boundaries, and incorporate the metrics into prompts to elicit LLMs’ factual expressions.

Finally, **we conduct experiments that demonstrate that our factuality align-**

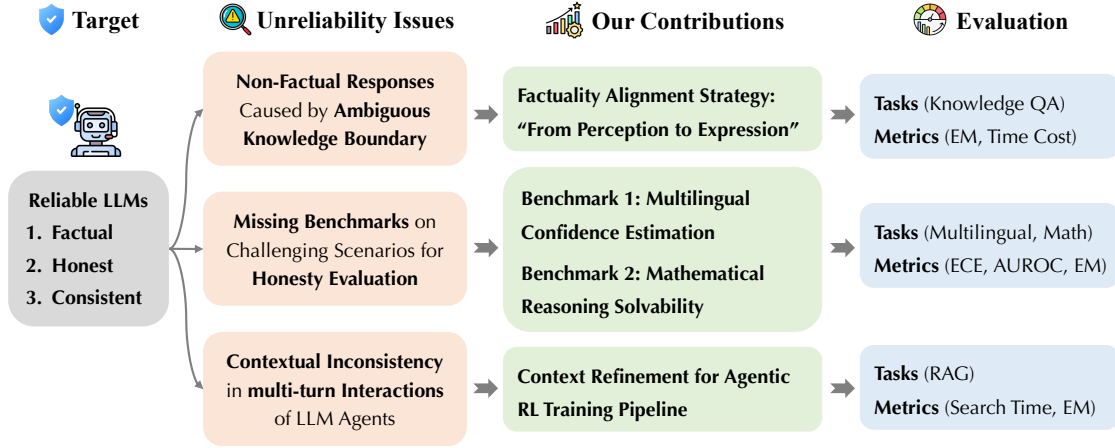


Figure 1.5: Overview of unreliability issues, as well as our contributions and evaluations to build reliable LLMs regarding factuality, honesty, and consistency.

ment method can substantially improve the factuality and generalization of LLMs over various baselines on both in-domain and out-of-domain knowledge QA tasks, validating the effectiveness of our proposed method.

To the best of our knowledge, **this is the first to explicitly leverage knowledge boundary for LLM alignment, heralding a paradigm of “from perception to expression” for future research of LLM alignment.**

1.2.2 LLM Honesty

Honesty is a key manifestation of reliability in practical interactions. First, **we investigate LLM honesty from two key aspects: self-awareness and confidence estimation**, conducting LLM honesty evaluation from qualitative and quantitative perspectives, respectively. Recognizing that the evaluation of LLM honesty varies significantly across different task scenarios, **we point out the issue of insufficient benchmarks to cover complex and practical scenarios.** To fill this gap, we introduce two underexplored but challenging benchmarks: a multilingual confidence estimation benchmark **MlingConf** and a mathematical reasoning solvability **ReliableMath** benchmark with expert check.

The **MlingConf** benchmark contains both language-agnostic (LA) and language-specific (LS) scenarios to explore the language dominance effects. Experiments present valuable findings, especially **language dominance effects of English on LA tasks, and query-related languages on LS tasks**, respectively. Hence, **we further propose a native-tone prompting method for multilingual confidence calibration**, which significantly enhances the honesty and accuracy compared to the use of any single language prompts for LS tasks. The benchmark has great significance for promoting the reliable global LLM deployment.

The **ReliableMath** benchmark is the first to evaluate the awareness of step-by-step reasoning on both solvable and unsolvable math problems. We conduct substantial experiments on various LLMs, showcasing several valuable findings. We propose an honesty alignment strategy to **effectively enable LLMs to critically think and respond on both in-domain math and out-of-domain knowledge tasks**, providing insights for enhancing the LLM’s reliability on reasoning tasks.

1.2.3 LLM Consistency

In this thesis, LLM consistency is investigated in more complex application scenarios of LLM-based search agents. Search agents often interact with external environments through multi-turn conversations to retrieve information and generate responses, which may introduce contextual inconsistency. **We first reveal that contextual inconsistency primarily derives from the latest documents in multi-turn search agents, and therefore introduce a distillation-based context refiner** to dynamically eliminate contextual inconsistency for search agents.

We then incorporate the context refiner into the reinforcement learning (RL) training pipeline of search agents, which can further enhance both reliability and efficiency of search agents, highlighting the necessity of context refinement to LLM agents, inspiring a paradigm of “refine context and then generate” for future work.

1.2.4 Summary

The systematic framework proposed in this dissertation constructs reliable LLMs from the three core perspectives of factuality, honesty, and consistency, effectively alleviating the hallucination problem of LLMs. The research work not only proposes targeted solutions for each dimension of LLM reliability but also introduces novel benchmarks and paradigms that enrich the research landscape of LLM hallucination mitigation. Furthermore, the well-grounded framework and the associated benchmarks in this work lay a strong foundation for navigating the future development of trustworthy AI systems in practical applications.

1.3 Thesis Organization

The remainder of this thesis is organized as follows. Chapter 2 provides a brief description of the background knowledge and related work of this thesis. Chapter 3 demonstrates a novel alignment strategy regarding knowledge boundaries for factuality improvements. Chapter 4 introduces the two proposed challenging benchmarks for LLM honesty evaluation on multilingual confidence estimation and awareness of mathematical reasoning solvability. Chapter 5 proposes a context refinement method for LLM agents to address contextual inconsistency. Finally, Chapter 6 concludes this thesis and discusses possible directions for future work.

Chapter 2

Background Study

Chapter 2 is organized as follows. Section 2.1 investigates the related work on LLM hallucination regarding the causes of data, model architecture, training, and inference in Section 2.1.1, as well as the previous study on LLM reliability in Section 2.1.2. Section 2.2 presents the preliminary of the LLM generation process, which is the basis for analyzing and addressing the unreliable outputs in this thesis.

2.1 Literature Review

2.1.1 LLM Hallucination

Different from existing surveys [65, 235, 153, 72], this work investigates the causes and relevant mitigation methods of LLM hallucinations from four dimensions in a novel perspective: data, model architecture, training algorithms, and inference time, as illustrated in Figure 2.1, which collectively covers the full spectrum of potential LLM hallucination triggers.

Training Data

The quality and composition of training data are widely recognized as a fundamental source of LLM hallucinations. Models trained on noisy, outdated, or

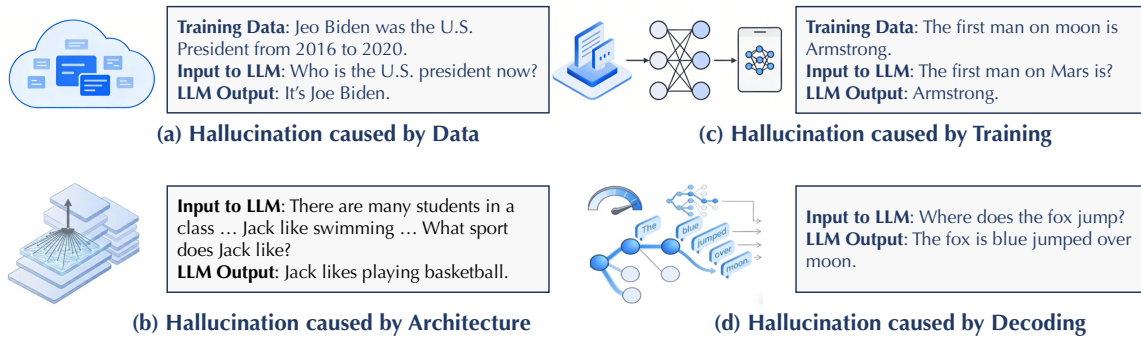


Figure 2.1: Examples of causes of LLM hallucinations in data, model architecture, training algorithm, and decoding.

factually inconsistent corpora are significantly more prone to generating fabricated content [72, 216, 65, 153, 235]. Research suggests that noise and the “long-tail” nature of pre-training data are primary culprits, where LLMs perform well on high-frequency facts but are highly susceptible to hallucinations regarding long-tail entities that appear rarely in pre-training corpora [169]. Additionally, the vast amount of unverified web data used for training—containing rumors, outdated facts, and biases—is inadvertently internalized by models during unsupervised learning [72]. Several studies suggest that if instruction tuning data is misaligned with the facts learned during pre-training, the model may learn to prioritize “style” over “truth”, leading to confident but false responses [45, 213, 240].

One major line of work addresses this by augmenting model inputs with verified external knowledge at inference time. Retrieval-Augmented Generation (RAG) was proposed to enable LLMs to ground their outputs in verifiable external evidence rather than relying solely on potentially stale or erroneous parametric knowledge [42]. In response to these data quality gaps, researchers have proposed targeted mitigation strategies. Pile of Law, a 256GB legal dataset, was introduced to demonstrate how rigorous source filtering [60], including removing outdated statutes and cross-referencing with official legal databases, can reduce hallucinations in domain-specific LLMs. The importance of proactive data cleaning was

further emphasized in [201], finding that even basic deduplication and noise reduction can cut hallucination rates.

Model Architecture

Transformer-based autoregressive model architectures introduce inherent biases that limit how models store and retrieve objective truths, manifesting most acutely in long-form generation and complex reasoning tasks, thus contributing to hallucinations. The attention mechanism can lead to “source confusion,” where the model over-focuses on local statistical patterns rather than maintaining global logical consistency [120]. Since models rely on fixed parameters for knowledge, the RAG architectural shift mitigates hallucinations by allowing the model to query external, non-parametric knowledge bases rather than relying solely on “internal memory” [97]. Research into State Space Models (SSMs) and long-context Transformers has highlighted the “lost in the middle” phenomenon [92, 113], where traditional architectures fail to maintain factual consistency over long sequences, triggering hallucinations due to information decay.

Training Algorithm

Hallucinations are often ingrained during training, with critical vulnerabilities emerging across pre-training and post-training. The optimization objectives used during training often inadvertently encourage models to hallucinate. First, pre-training is the foundational stage where LLMs acquire general world knowledge. However, the maximum likelihood estimation (MLE) training forces the model to predict the next token regardless of whether it has the underlying factual support, essentially “forcing” the model to guess [110]. InstructGPT [135, 142] and related studies [243, 149, 20, 243] found that Reinforcement Learning from Human Feedback (RLHF) can induce “sycophancy,” where the LLM prioritizes being helpful or pleasing to the human evaluator over being factually accurate. RLHF encourages

models to provide definitive answers even when they lack sufficient knowledge, which are more likely to generate hallucinations in response to questions [178], while fine-tuning LLMs on new knowledge may encourage hallucinations [45].

Alleviating hallucinations in post-training is a standard procedure to improve LLMs’ helpfulness and factuality [9]. The main goal of post-training is to guide human preference through SFT, RLHF [142, 9] or RLAIIF [10]. Therefore, many works explore applying RL to improve LLMs’ factuality through Proximal Policy Optimization (PPO) [157] with the trained reward model [107, 212] or Direct Preference Optimization (DPO) [150] with the constructed preference dataset [234, 108] to align with factuality preferences annotated by human beings. LLMs are encouraged to reject unknown questions using the constructed preference data by leveraging knowledge feedback [212]. Some works also explore alignment methods for LLM safety [228, 230, 229]. From a different angle, [132] proposes mitigating hallucination through knowledge distillation, wherein a teacher model provides smoothed soft labels to a student model during supervised fine-tuning to amplify hallucination. Together, these approaches highlight the importance of designing training objectives that are explicitly sensitive to hallucinations.

Inference Time

The inference stage is currently the most active area for hallucination mitigation through post-processing and decoding interventions, which is a lightweight and promising way, categorized into decoding and prompting strategies.

For decoding strategies, high temperature is prone to produce hallucinations for open-ended generations when decoding [96]. Self-consistency by sampling multiple reasoning paths and using a majority vote can filter out inconsistent, “random” hallucinations [198]. ITI [98] identified specific “truthful” activation directions within the model’s hidden states and applied shifts during inference to enhance factual accuracy. DoLa [21] discovered that factual knowledge is often localized in

specific layers. By contrasting logits from different layers, they successfully amplified factual signals and suppressed hallucinations.

There are also various prompting-based methods. Chain-of-Thought (CoT) [200] showed that breaking down complex tasks into steps reduces logical hallucinations. SelfCheckGPT [123] developed a zero-resource method that detects hallucinations by checking the consistency of multiple stochastic samples from the model. [73] proposed self-reflection techniques where LLMs verify their own reasoning chains to identify and correct hallucinations.

2.1.2 LLM Reliability

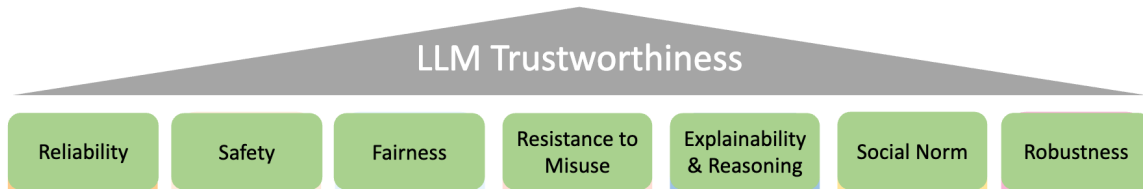


Figure 2.2: *Taxonomy of LLM trustworthiness from [114].*

Reliability is a foundational requirement of trustworthy LLMs, equally important as safety and fairness, as categorized in Figure 2.2 from the previous survey [114]. The meaning of reliability is many-sided. Building upon previous studies, we classify reliability into three key dimensions of factuality, honesty, and consistency, which collectively encompass the full range of issues in this field.

LLM Factuality

If an LLM provides incorrect or misleading information, it can lead to misunderstandings, spread false views, or even cause harm, especially for those domains that demand high factual accuracy [238], such as health [172, 175], law [66], and finance [205]. A surge of research has been directed towards evaluating LLMs’ factuality [15, 110, 123, 129]. Beyond evaluation, efforts to improve the factual

knowledge of LLMs have been notable. Strategies have ranged from retrieving information from external knowledge bases to continual pretraining and supervised finetuning [95, 163, 187, 177, 234, 177]. However, current factuality alignment methods lack the utilization of LLM’s perception of its knowledge boundaries, preventing the model from genuinely learning factual expressions.

LLM Honesty

Being honest is described as being “free of deception, morally upright or virtuous, among other traits” [28]. This thesis investigates LLM honesty in two aspects of self-awareness and confidence estimation. In traditional machine learning fields, confidence calibration is widely studied [51, 27]. For LLMs, honesty is described and evaluated in various studies [8, 82, 33, 109, 6], including truthfulness, calibration, self-awareness, and non-deceptiveness. Recent studies suggest that an honest model should accurately express its knowledge and humbly acknowledge its limitations without deception or being inconsistent [19, 222]. However, there is still a lack of benchmarks for evaluating LLM honesty in many complex scenarios.

LLM Consistency

LLMs are prone to giving inconsistent outputs [70]. These inconsistent answers can create confusion among users and reduce user trust. The issue of inconsistency has attracted increasing attention in many fields, such as abstract summarization [87, 126, 209, 242, 131], data-to-text generation [31, 176], and dialogue systems [32, 55, 152, 164, 143]. However, Current research remains insufficient in exploring LLM agent scenarios involving complex multi-turn interactions.

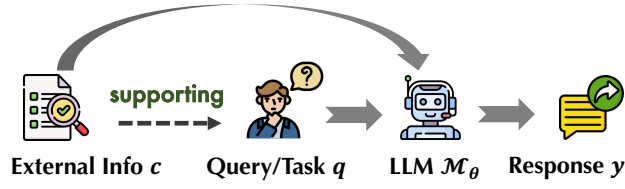


Figure 2.3: Illustration of LLM generation process.

2.2 Preliminary

2.2.1 LLM Generation Process

In this section, we present a simplified formulation of the LLM generation process to support the analysis and mitigation of unreliable outputs in subsequent chapters. Previous works identify the concept of *Internal/External Knowledge* for LLMs [189, 74], where internal knowledge $\mathcal{K}_I$ is learned from the pretrained corpus and encoded within the model parameters, and external knowledge $\mathcal{K}_E$ is accessed through external tools. For an LLM $\mathcal{M}$ with parameters θ , the output y is jointly determined by its internal parametric knowledge $\mathcal{K}_I \in \theta$, the external supporting information $c \in \mathcal{K}_E$ (generally can be tools, history information or retrieved documents), and the input task q as $y = \mathcal{M}_\theta(q, c)$, where $x = [q, c]$ generally refers to input context of LLMs.

The reliability of LLM output in response is jointly determined by the parameter θ and context c . As $\mathcal{K}_I$ embedded in parameters θ cannot be directly accessed, the manifestation of $\mathcal{K}_I$ is conditioned on the context of x and d . If $\mathcal{K}_I$ lacks the knowledge necessary to answer x yet $\mathcal{M}$ is still required to respond, it can generate factually incorrect or dishonest outputs. The environment generally returns plenty of information in $d \in \mathcal{K}_E$, which inevitably introduces redundancy, noise, and conflicts. Such extraneous content can in turn distort the consistency of LLMs. The analysis provides a direction to improve factuality, honesty, and consistency regarding either processing input context or adjusting LLM internal parameters,

respectively. In the following chapters, we will present how to process the LLM parameters or input contexts to build reliable LLMs for hallucination mitigation.

Chapter 3

Building Reliable LLMs regarding Factuality

Unity of knowledge and practice.

知行合一

— Yangming Wang 王陽明

Chapter 3 is organized as follows. Section 3.1 presents an overview of LLM factuality and our proposed factuality alignment method. Section 3.2 further analyzes the problem of ambiguous knowledge boundaries on LLMs, which cause non-factual generations. Section 3.3 introduces the preliminary of this chapter, including the knowledge boundary, LLM alignment, and the specific notations. Section 3.4 presents the proposed factuality alignment method in detail. In Section 3.5, we present the experimental results and analysis of the proposed method. Finally, Section 3.6 summarizes this chapter.

3.1 Overview

This chapter focuses on exploring approaches to enhance the factuality of LLMs and mitigate hallucinations [187, 72, 65], with the overarching goal of con-

structuring more reliable LLMs. LLM factuality is generally defined as whether the LLM produced contents conform to objective real-world knowledge or to formally established facts [187], with a particular focus on knowledge-intensive tasks.

Driven by the scaling law of LLMs [84], the continual expansion of pretraining data has become a mainstream trend among major AI companies. Pre-training data has covered almost all publicly accessible knowledge on the Internet, and hence, LLMs possess a remarkably extensive knowledge base as experts. However, in practical applications, even for much general knowledge that LLMs have already learned, they cannot always express it in an accurate manner [108, 234, 101].

For more visualization, we first propose the knowledge spectrum to characterize the degree of knowledge mastery of LLMs, as illustrated in Figure 3.1. A substantial portion of the knowledge that LLMs acquire but fail to accurately express is associated with the light green area that represents *weakly known* knowledge pieces, also referred to as the **ambiguous knowledge boundaries** [226]. Therefore, improving the factuality of LLMs can thus be regarded as a knowledge enhancement, in which *weakly known* (light green) knowledge is transformed into *known* (dark green) knowledge, as demonstrated by the transition from (a) to (b) of Figure 3.1.

To address this issue, inspired by LLM alignment methods [142, 9], we propose a factuality alignment strategy, which first improves LLM perception by leveraging uncertainty estimations to represent precise knowledge boundaries, and then explicitly incorporates these representations as input features into prompts for LLMs to align with factual knowledge, succinctly summarized as “**From Perception to Expression**”. Therefore, the proposed method focuses on two pivotal issues: how to capture the knowledge boundary representations and how to align with factuality.

First, we enable LLMs to perceive the knowledge boundaries by uncertainty estimations. We prepare a dataset on knowledge question-answering (QA), by calculating two uncertainty estimations, including confidence score and semantic entropy, to precisely capture the intrinsic perception of knowledge boundary representations. Knowledge boundaries always indicate the known level of fac-

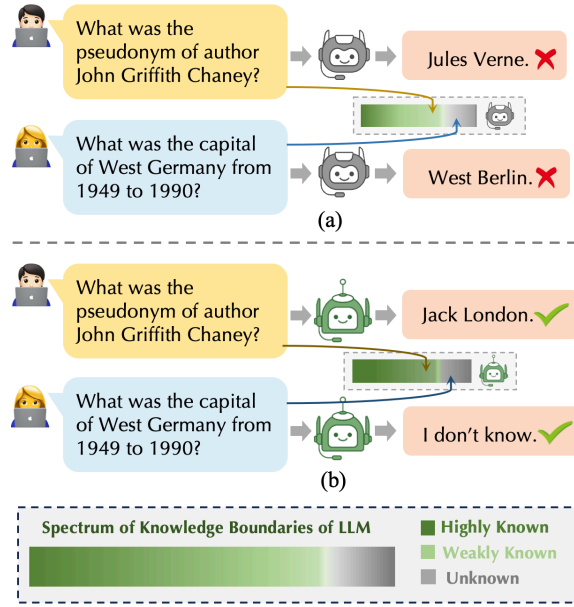


Figure 3.1: Examples of LLMs with (a) ambiguous and (b) explicit knowledge boundaries when answering questions, where the light green, dark green, light gray, and dark gray areas of the spectrum represent the weakly known, known, weakly unknown, and unknown knowledge pieces, respectively.

tual knowledge, generally implemented using uncertainty estimation methods on LLMs [155]. The two measures, as complementary, can reflect the conviction and dispersion of generated responses to a question based on LLMs’ internal knowledge. Questions with at least one correct sampled answer are regarded as “known”, and those with all incorrect sampled responses are considered “unknown”. The ground-truth answers of unknown questions are revised to refusal responses to delineate known and unknown facts [233].

Subsequently, following [142], using the prepared dataset, we explicitly leverage the uncertainty estimations to align with factuality on the prepared dataset using both supervised fine-tuning (SFT) and reinforcement learning (RL). We employ SFT to train two uncertainty estimation models to predict confidence and entropy, and then train a reward model to evaluate the correctness of the generated answer conditioned on the input comprising the question, the generated response, and two uncertainty estimations regarding the knowledge boundary. With the reward

model, we further adopt the Proximal Policy Optimization (PPO) [157] algorithm for LLM alignment by feeding both questions and two measures as prompts to elicit the policy LLM’s factual expressions to improve the factuality.

Experiments are conducted to evaluate in-domain and out-of-domain performance on a range of knowledge QA datasets. The results demonstrate our proposed method significantly enhances the reliability and generalization for LLMs over various baseline methods to accurately express known factual knowledge and refuse unknown questions, suggesting that leveraging the two employed uncertainty estimations in alignment can notably improve LLMs’ factuality.

3.2 Problem Statement

Despite their remarkable proficiency across a diverse range of tasks, LLMs still frequently struggle to accurately express factual knowledge acquired during the pre-training stage, particularly when that knowledge is held but uncertain. In such cases, **the knowledge boundaries are somewhat ambiguous by LLMs, remaining a gap between “known” and “expression”** [108, 234, 101], giving rise to the hallucination issue and undermining the reliability and applicability to users. In other words, although LLMs acquire factual knowledge, they still often struggle to accurately leverage it in downstream tasks, resulting in deficiencies in factuality. This constitutes the key problem addressed in this chapter: **how to enable LLMs to always output their learned knowledge in a factually accurate manner.**

LLMs generate responses (also referred to as “expressions” in this chapter) based on the knowledge distributions internalized during pre-training (“known”). However, a substantial portion of this acquired knowledge exhibits vague boundaries, encompassing numerous learned yet uncertain knowledge pieces (*weakly known*, denoted by the light green area of the spectrum in Figure 3.1 (a)) [45]. Consequently, even when LLMs possess relevant knowledge, they may fail to convey accurate information with appropriate confidence in downstream tasks [234].

Furthermore, LLMs may exhibit overconfidence regarding knowledge they are unfamiliar with (*weakly unknown* or *unknown*, represented by the light and dark gray areas of the spectrum in Figure 3.1 (a)), resulting in hallucinated outputs [233, 112].

This phenomenon stems primarily from the failure of LLMs to reconcile knowledge boundaries with factual accuracy during the alignment stage [107, 212, 177, 108, 234, 222]. Therefore, the key to improving the factuality of LLM expressions is to elicit *weakly known* facts from LLMs by explicitly perceiving the knowledge boundary during alignment. More specifically, LLMs need to precisely leverage the knowledge boundary information to instruct them to confidently express their known yet uncertain information and firmly refuse questions that exceed their knowledge scope, as illustrated in Figure 3.1 (b). By grounding model expression in a more precise perception of its internal knowledge boundary representations, the resulting outputs of LLMs are rendered more factual, thereby narrowing the gap between “known” and “expression” [108, 234, 101].

3.3 Related Work and Preliminary

3.3.1 Knowledge Boundary

Previous works investigate the knowledge boundary to identify the known level of a knowledge piece of LLMs by quantifying the confidence or uncertainty estimations like output consistency [18], prompting methods [155], or knowledge probing [71]. Researchers are examining the limits of parametric knowledge in LLMs with the objective of delineating the extent of the LLMs’ knowledge or capability boundaries. Present studies on the knowledge boundary primarily focus on measuring the knowledge boundaries using confidence or uncertainty estimations on specialized tasks [99, 226]. The ambiguity of knowledge boundaries can be attributed to the imbalanced knowledge distribution learned from the pre-training stage or the influence of external knowledge leading to knowledge conflict [213]

and inconsistency [220].

3.3.2 LLM Alignment

Alignment is a standard procedure to improve LLMs' capability to be consistent with human preference [9]. The main goal of LLM alignment is to guide human preference through Supervised Fine-Tuning (SFT), Reinforcement Learning from Human Feedback (RLHF) [142, 9] or AI feedback [10], which may also guide LLMs to output detailed and lengthy responses [165] but inevitably encourage hallucination. Therefore, many works explore to apply RL to improve LLMs' factuality through Proximal Policy Optimization (PPO) [157] with the trained reward model [107, 212] or Direct Preference Optimization (DPO) [150] with the constructed preference dataset [234, 108] to align with factuality preferences annotated by human beings. [212] encourages LLM to reject unknown questions using the constructed preference data by leveraging knowledge boundary feedback. Some works also explore alignment methods for LLM safety [228, 230, 229]. Distinct from recent studies that apply RL to mitigate LLM hallucinations [234, 108, 107, 212], this work improves LLMs' reliability by explicitly leveraging the uncertainty estimations for LLM alignment.

3.3.3 Specific Notations

The definitions of specific notations of this chapter are summarized in Table 3.1.

Notation	Description
$\mathcal{Q}$	Dataset containing n Question-Answering pairs. ($ \mathcal{Q} = n$)
$\mathcal{P}$	Set of few-shot exemplars.
x_i	The i -th question sample in $\mathcal{Q}$.
$\hat{y}_i$	The i -th ground-truth answer in $\mathcal{Q}$.
$y_i^{(k)}$	The k -th sampled response to the i -th question in $\mathcal{Q}$.
p_k	k -th few-shot exemplar to sample $y_i^{(k)}$.
Y_i	Answering set containing K sampled response $\{y_i^{(k)}\}$ for the i -th question x_i .
$z_i^{(k)}$	The label of $y_i^{(k)}$ ($z_i^{(k)} \in \{0, 1\}$, 1 for <i>True</i> and 0 for <i>False</i>).
Z_i	Label set corresponding to Y_i .
c_i	The confidence score for the i -th question x_i .
e_i	The semantic entropy for the i -th question x_i .
$\mathcal{D}$	Constructed training set containing N tuple samples $(x_i, Y_i, Z_i, \hat{y}_i, c_i, e_i)$.
τ	Uncertainty estimation model to calculate confidence score by feeding x .
μ	Uncertainty estimation model to calculate semantic entropy by feeding x .
θ	Binary classifier by feeding (x, c, e, y) as the reward model.
$\mathcal{L}_{\mathcal{M}}$	Training loss functions for three models respectively where $\mathcal{M} \in \{\tau, \mu, \theta\}$.
r	Final reward signal consisted of reward score r_1 and KL-penalty r_2 .
β	Coefficient for the KL-penalty r_2 .
π_{θ}	Policy model to be optimized using r by PPO.
π_o	Reference model initialized by the original policy.
T	Sampling temperatue.
K	Number of sampled responses.
N	Number of QA pairs.

Table 3.1: Specific notations of Chapter 3.

3.4 Factuality Alignment: From Perception to Expression

We propose a factuality alignment method following the principle of “From Perception to Expression”, referred to as **UAlign**, which leverages Uncertainty estimations to represent knowledge boundary perceptions, and then explicitly incorporates these representations as input features for LLMs to **Align** with factual expressions. First, we enable LLMs’ perception of precise knowledge boundaries in Section 3.4.1. More specifically, we construct a UAlign dataset on QA tasks, including strategies to collect multiple responses, as well as uncertainty measures to capture intrinsic representations of knowledge boundaries by calculating confi-

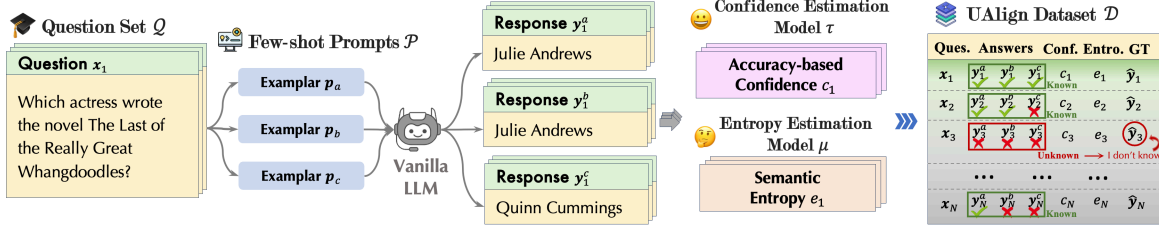


Figure 3.2: Illustration of UAlign dataset preparation process.

dence score and semantic entropy. Then in Section 3.4.2, we utilize the obtained UAlign dataset to train the uncertainty estimation models, which can leverage perceptions of knowledge boundaries for factual expressions. Finally, we explicitly incorporate the estimations as input features to elicit LLMs to generate factual responses using SFT- and PPO-based alignment methods as in Section 3.4.3.

3.4.1 Perception of Knowledge Boundary

Responses Sampling Strategy

As in Figure 3.2 and Algorithm 1, to precisely represent the perception of the knowledge boundaries of the LLM $\mathcal{M}$ given a question, we sample multiple responses by repeating the generation procedure several times. In this phase, the preparation process can be represented in a tuple $(\mathcal{Q}, \mathcal{P}, \mathcal{A})$. $\mathcal{Q}$ contains a batch of N QA pairs $\{(x_i, \hat{y}_i)\}_{i=1}^N$ where x_i and $\hat{y}_i$ denote the i -th question and ground-truth answer respectively. To mitigate context sensitivity, we utilize different few-shot prompts in $\mathcal{P}$ with temperature $T = 0.2$ to make a trade-off between the accuracy and diversity to represent knowledge boundaries [45]. The few-shot prompt set $\mathcal{P}$ consists of K different 1-shot exemplars in this study, which is enough for LLMs to generate answers in the correct format.

In the k -th sampling process for the i -th question x_i , we employ each few-shot exemplar $p_k \in \mathcal{P}$ with the question x_i to the LLM to generate the k -th response $y_i^{(k)}$. By taking K times of the sampling process, we can obtain an answer set $\mathcal{Y}_i =$

$\{y_i^{(k)}\}_{k=1}^K$ to x_i . We set the labels $Z_i = \{z_i^{(k)}\}_{k=1}^K$ by comparing each generated answer $y_i^{(k)}$ with the ground-truth $\hat{y}_i$ to indicate the correctness ($z_i^{(k)} \in \{0, 1\}$, 1 for *True* and 0 for *False*). We collect and format the data in $(x_i, Y_i, Z_i, \hat{y}_i)$ in an extended dataset and calculate the uncertainty measures subsequently. Note that since fine-tuning LLMs on unknown knowledge will encourage hallucinations [233, 45], we revise the ground-truth answer to the question with $z_i^{(k)} = 0, \forall z_i^{(k)} \in Z_i$ to “Sorry, I don’t know.” to teach LLMs to refuse to answer the questions beyond their knowledge scope [233].

Algorithm 1 UAlign Dataset Construction Pipeline

```

1: Input:  $N$  QA pairs  $\mathcal{Q} = \{(x_i, \hat{y}_i)\}_{i=1}^N$ ,  $K$  different 1-shot exemplar prompts  $\mathcal{P} = \{p_k\}_{k=1}^K$ ,
   vanilla LLM  $\mathcal{M}$  with temperature  $T = 0.2$  for sampling.
2: Output: UAlign dataset  $\mathcal{D}$ .
3: Initialize empty dataset  $\mathcal{D} \leftarrow \emptyset$ 
4: for each QA pair  $(x_i, \hat{y}_i) \in \mathcal{Q}$  do
5:   Step 1: Response Sampling
6:   Initialize answer set  $Y_i \leftarrow \emptyset$ 
7:   for  $k = 1$  to  $K$  do
8:      $y_i^{(k)} \leftarrow \mathcal{M}.\text{generate}(x_i, \text{prompt} = p_k, T = 0.2)$ 
9:      $Y_i \leftarrow Y_i \cup \{y_i^{(k)}\}$ 
10:  end for
11:  Initialize label set  $Z_i \leftarrow \emptyset$ 
12:  for  $k = 1$  to  $K$  do
13:     $z_i^{(k)} \leftarrow \mathbb{I}(y_i^{(k)} == \hat{y}_i)$   $\triangleright z_i^{(k)} \in \{0, 1\}$ , 1=correct, 0=incorrect
14:     $Z_i \leftarrow Z_i \cup \{z_i^{(k)}\}$ 
15:  end for
16:  if  $\forall z_i^{(k)} \in Z_i, z_i^{(k)} = 0$  then
17:     $\tilde{y}_i \leftarrow \text{“Sorry, I don’t know.”}$   $\triangleright$  All responses incorrect: mark as unknown
18:  else
19:     $\tilde{y}_i \leftarrow \hat{y}_i$   $\triangleright$  Keep original ground-truth
20:  end if
21:  Step 2: Uncertainty Measures
22:  Calculate accuracy-based confidence  $c_i$  as Equation 3.1.
23:  Cluster responses in  $Y_i$  into semantic equivalence classes  $\{s\}$ .
24:  For each semantic class  $s$ , compute semantic entropy  $e_i$  as Equation 3.2.
25:  Append  $(x_i, Y_i, Z_i, \tilde{y}_i, c_i, e_i)$  to  $\mathcal{D}$ .
26: end for

```

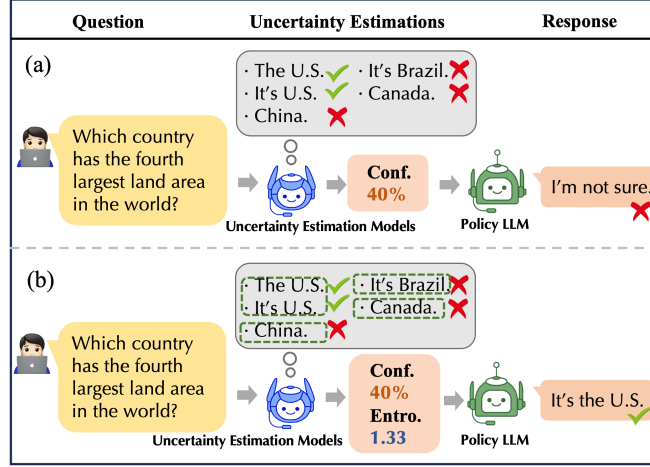


Figure 3.3: Illustration of the effects of different uses of uncertainty estimations under varying knowledge boundaries perceived by LLMs.

Uncertainty Measures

In order to quantify the knowledge boundaries, we can employ some uncertainty estimation methods. The knowledge boundary of LLMs in this work is defined in two aspects. The first involves the prior judgment to a question x_i regardless of the answers [155], which indicates the certainty level of x_i . The second entails the dispersion measure to the distribution of the generated responses in Y_i to x_i . Accordingly, we adopt accuracy-based confidence [211] and semantic entropy [88] to jointly determine and represent precise perceptions of the actual knowledge boundary information.

Accuracy-based Confidence A natural idea of aggregating varied responses is to measure the accuracy among the candidate outputs to denote confidence scores [123, 211]. Given a question x_i , the accuracy is calculated by comparing each candidate response in Y_i with the ground-truth answer $\hat{y}_i$, which can serve as the confidence score c_i , computed as follows.

$$c_i = \text{Conf}(x_i) = \frac{1}{K} \sum_{k=1}^K \mathbb{1}(\tilde{y}_i = y_i^{(k)}) \quad (3.1)$$

Semantic Entropy Due to the variable length and semantically equivalent generated sequences in sentence-level output spaces, [88] proposes semantic entropy to capture uncertainty on the semantic level to quantify the degree of dispersion of sentence meanings. The semantic entropy e_i given $\mathbf{x}_i$ and $\mathbf{Y}_i$ is calculated as

$$p(s|\mathbf{x}_i) = \frac{1}{K} \sum_{k=1}^K \mathbb{1}[\mathbf{y}_i^{(k)} \in s] \quad (3.2)$$

$$e_i = \text{SE}(\mathbf{x}_i) = - \sum_s p(s|\mathbf{x}_i) \log p(s|\mathbf{x}_i) \quad (3.3)$$

where s denotes a set of sentences in semantic equivalent space. As illustrated in Figure 3.3, semantic entropy is calculated by clustering semantically equivalent responses, as a measure to quantify the dispersion of generations to confirm the correct answer despite the low confidence, which will be further analyzed with the experimental results in Section 3.5.3. We calculate the confidence score and semantic entropy for both known and unknown questions. Then we update a UAlign dataset $\mathcal{D}$ by formatting the i -th sample in $(\mathbf{x}_i, \mathbf{Y}_i, \mathbf{Z}_i, \hat{\mathbf{y}}_i, c_i, e_i)$.

Algorithm 2 UAlign Training Algorithm

- 1: **Input:** UAlign dataset $\mathcal{D}$, uncertainty models τ and μ , reward model θ , and initial policy π_o .
 - 2: **Output:** Optimized policy π_θ .
 - 3: **Step 1: Learning Perception of Knowledge Boundaries**
 - 4: Train uncertainty model τ and μ on $\mathcal{D}$ to predict c_i and e_i by feeding $\mathbf{x}_i$ using Equation 3.4 and 3.5, respectively.
 - 5: **Step 2: Alignment via SFT and PPO**
 - 6: Train reward model θ on $\mathcal{D}$ to predict z_i by feeding $\mathbf{x}_i, c_i, e_i, \mathbf{y}_i^{(k)}$ using Equation 3.6.
 - 7: Collect reward r including the reward signal r_1 by θ and KL-penalty r_2 between policy π_θ and initial policy π_o as Equation 3.7.
 - 8: Update policy π_θ using the collected reward r .
-

3.4.2 Leveraging Perception for Factual Expression

The obtained UAlign dataset $\mathcal{D}$ contains precise perceptions of knowledge boundary information given a question. Then we need to enable LLMs to learn to perceive the knowledge boundaries. Therefore, we train two uncertainty estimation models to explicitly learn the two estimations given specific questions, as presented in Figure 3.4 (a) and Algorithm 2. Uncertainty estimation models of τ and μ are utilized to predict the confidence score and semantic entropy, respectively, which are continuously used to train a reward model. When training τ and μ , we only feed a question x_i to the models to generate two uncertainty estimations. The training objectives are to minimize the cross-entropy losses $\mathcal{L}_\tau$ and $\mathcal{L}_\mu$ as

$$\arg \min_{\tau} \mathcal{L}_\tau, \arg \min_{\mu} \mathcal{L}_\mu,$$

$$\mathcal{L}_\tau = -\mathbb{E}_{(x_i, c_i) \sim \mathcal{D}} [\log p_\tau(c_i | x_i)] \quad (3.4)$$

$$\mathcal{L}_\mu = -\mathbb{E}_{(x_i, e_i) \sim \mathcal{D}} [\log p_\mu(e_i | x_i)] \quad (3.5)$$

where the models can explicitly learn and express the uncertainty estimations, which represent more accurate knowledge boundary information.

3.4.3 Aligning Factual Expressions via SFT and RL

Reward Model Training via SFT

Subsequently, the reward model is introduced as a binary evaluator to determine if a generated answer $y_i^{(k)} \in Y_i$ is correctly conditioned on the question x_i , confidence c_i , and entropy e_i . Both c_i and e_i are explicitly used as additional auxiliary features to improve the accuracy of the reward model. The binary cross-

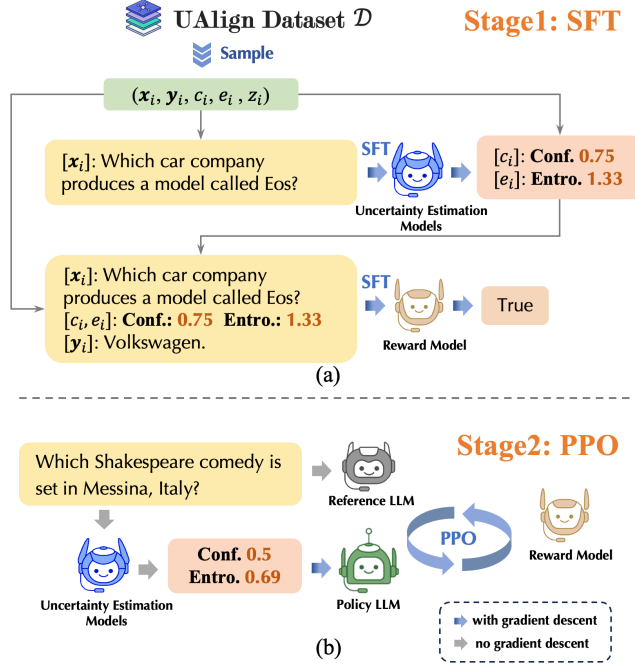


Figure 3.4: Illustration of (a) SFT and (b) PPO alignment processes of **UAlign** framework. Note that for simplicity, we only present one estimation model in the figure but there are actually two.

entropy loss $\mathcal{L}_\theta$ for the reward model θ is minimized as follows.

$$\begin{aligned}
 \arg \min_{\theta} \mathcal{L}_\theta, \mathcal{L}_\theta &= -\mathbb{E}_{(x_i, y_i^{(k)}, z_i^{(k)}, c_i, e_i) \sim \mathcal{D}} [\mathcal{L}_\theta^{(i)}] \\
 \mathcal{L}_\theta^{(i)} &= -z_i^{(k)} \log p_\theta(z_i^{(k)} | x_i, c_i, e_i, y_i^{(k)}) \\
 &\quad - (1 - z_i^{(k)}) \log(1 - p_\theta(z_i^{(k)} | x_i, c_i, e_i, y_i^{(k)}))
 \end{aligned} \tag{3.6}$$

Policy Model Training via PPO

The **UAlign** PPO is to elicit the LLM’s factual expressions to a question with the uncertainty measures using the obtained models. Inspired by the progress of reinforcement learning from human feedback (RLHF) technique [142, 243], we employ proximal policy optimization (PPO) [157] for LLM optimization with the reward model θ . As illustrated in Figure 3.4 (b), the LLM to be optimized is used as the policy π_θ . During this phase, we iteratively feed the question x , and the predicted confidence c and entropy e to both the policy π_θ and the reference π_o ,

and the reward function r will facilitate reliable expressions of $\mathbf{y}$ of the policy model π_θ . The training objective is to maximize the following reward function r as

$$\begin{aligned} & \arg \max_{\pi_\theta} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, c \sim \tau(\mathbf{x}), e \sim \mu(\mathbf{x}), \mathbf{y} \sim \pi_\theta(\mathbf{x}, c, e)} [r] \\ r = & \underbrace{\theta(\mathbf{x}, \mathbf{y}, c, e)}_{r_1} - \beta \underbrace{\text{KL}[\pi_\theta(\mathbf{x}, c, e) || \pi_o(\mathbf{x})]}_{r_2} \end{aligned} \quad (3.7)$$

where the reward function r contains a reward signal r_1 from θ and a KL-penalty r_2 to make sure the generated answers $\mathbf{y}$ by policy π_θ don't diverge too much from the original policy π_o . The hyper-parameter β is the coefficient of KL-penalty.

3.5 Experiments and Analysis

3.5.1 Experimental Setup

Datasets

The **UAlign** training set is comprised of three widely used knowledge QA datasets:

TriviaQA (TVQA) [80] is a comprehensive reading dataset of QA resources distinguished by its complexity and serves as an effective benchmark for evaluating machine comprehension of factual knowledge. We pre-process and extract 76,523 QA samples from the TriviaQA training set and 9,960 from the development set to contribute to the **UAlign** training and in-domain test set respectively.

SciQ [79] contains 13,679 crowd-sourced science exam questions about physics, chemistry and biology, among others. The original dataset was divided, with 11,679 samples allocated as the training set and an additional 1,000 samples designated as the validation set. These were subsequently incorporated into our **UAlign** training set and in-domain test set.

NQ-Open [90] which is constructed by Google Search queries along with annotated short answers or documents. We employ the training and development

set of NQ-Open, which contains 87,925 and 3,610 samples, respectively, to further enhance the data sets.

For **UAlign** testing, we evaluate the in-domain (ID) performance on the corresponding validation/test sets and generalization on an out-of-domain (OOD) test set **LSQA** [219] which contains multilingual language-specific QA pairs covering specific social, geographical, and cultural language contexts for the UK & US, France, China, Japan, and Thailand. In this study, we only employ the QA pairs in English from each LSQA subset, which includes 1,025 samples as the OOD test set.

Evaluation Metrics

To evaluate LLMs’ factuality, we employ two metrics: *Precision* (*Prec.*) and *Truthfulness* (*Truth.*). *Precision* is defined as the proportion of correctly answered questions among all the known questions, representing LLMs’ ability to accurately express their known knowledge. *Truthfulness* represents the proportion of the sum of correctly answered known and refused unknown questions among all questions.

To ascertain the correctness of the LLM-generated answer y with the ground truth $\hat{y}$, we employ a string-matching approach. Exact matching (EM) of $y \equiv \hat{y}$ always misjudges some correct answers with slight distinctions on such QA tasks. Therefore, we replace EM with a variant of $y \in \hat{y} \vee \hat{y} \in y$ to evaluate the accuracy.

Explanations of Precision and Truthfulness We list different answer types to a question regarding whether LLMs know about it and can answer it correctly below.

- KC: Known and answered correctly.
- KI: Known but answered incorrectly.
- KR: Known but refused to answer.
- UC: Unknown but answered correctly.
- UI: Unknown but answered correctly.

- UR: Unknown and refused to answer.

“*Truthfulness*” is the proportion of questions the LLM either the known answered correctly or the unknown refused to answer. Some unknown but correctly guessed answers will not be included. The calculation of *Truthfulness* is as follows.

$$\text{Truthfulness} = \frac{\text{UR} + \text{KC}}{\text{KC} + \text{KI} + \text{KR} + \text{UC} + \text{UI} + \text{UR}} \quad (3.8)$$

Precision is defined as the proportion of correctly answered questions among all the known questions, representing LLMs’ ability to accurately express their known factual knowledge. The equation of *Precision* is as follows.

$$\text{Precision} = \frac{\text{KC}}{\text{KI} + \text{KC} + \text{KR}} \quad (3.9)$$

Clarifications of Use of Truthfulness To avoid the over-conservative problem incurred by using precision only, we employ “*Truthfulness*” as complementary to measure the proportion of questions the model either known answered correctly or unknown refused to answer, which reflects the honesty of the model. The employed two metrics of precision and truthfulness can comprehensively measure the factuality of different methods, thereby comprehensively demonstrating the superiority of our method over other baselines.

For closed-book QA evaluation, we observe that simply applying EM may misjudge the correct answers. We compare several variants of EM as in Table 3.2 and report their successful judgments on responses of 20 selected samples that are misjudged using EM, where PEM, RRM, and PREM indicate Positive-EM, Recall-EM, and Positive-Recall-EM and the mathematical explanations are presented in Table 3.2. Upon human discrimination, EMPR exhibits the lowest failure rate and is therefore selected as the evaluation metric.

Area Under the Receiver Operator Characteristic Curve (AUROC) AUROC assesses the effectiveness of confidence estimation [36] by quantifying how likely a

Variant	Explanation	# Fail
EM	$y \equiv \hat{y}$	20
PEM	$y \in \hat{y}$	16
REM	$\hat{y} \in y.$	6
PREM	$y \in \hat{y} \vee \hat{y} \in y.$	2

Table 3.2: Number of failed judgments by human check for different EM variants.

randomly chosen correct answer possesses a higher confidence score than an incorrect one, yielding a score within the range of $[0, 1]$, implemented by `sklearn` toolkit. A higher AUROC score, implying higher reliability is preferred.

Baselines

We present several baselines in four categories below. To clearly delineate the differences between our proposed method and other baselines, we illustrate all methods in Figure 3.5.

Prompt-based Baselines We present several prompt-based baselines as follows.

- **ICL:** Few-shot prompts containing m examples are utilized for answer generation with temperature $T = 0.2$ where m is set to 2.
- **ICL-IDK:** Two examples are included in the few-shot prompt while one is selected from the ICL-used example, and another is an unknown question whose answer is revised to “*Sorry, I don’t know.*”.
- **ICL-CoT:** We also employ the Chain-of-Thought (CoT) [200] in few-shot examples by recalling the relevant knowledge piece of LLMs and incorporating it into thinking steps before answering the question.

SFT-based Baselines We employ two SFT-based methods as follows.

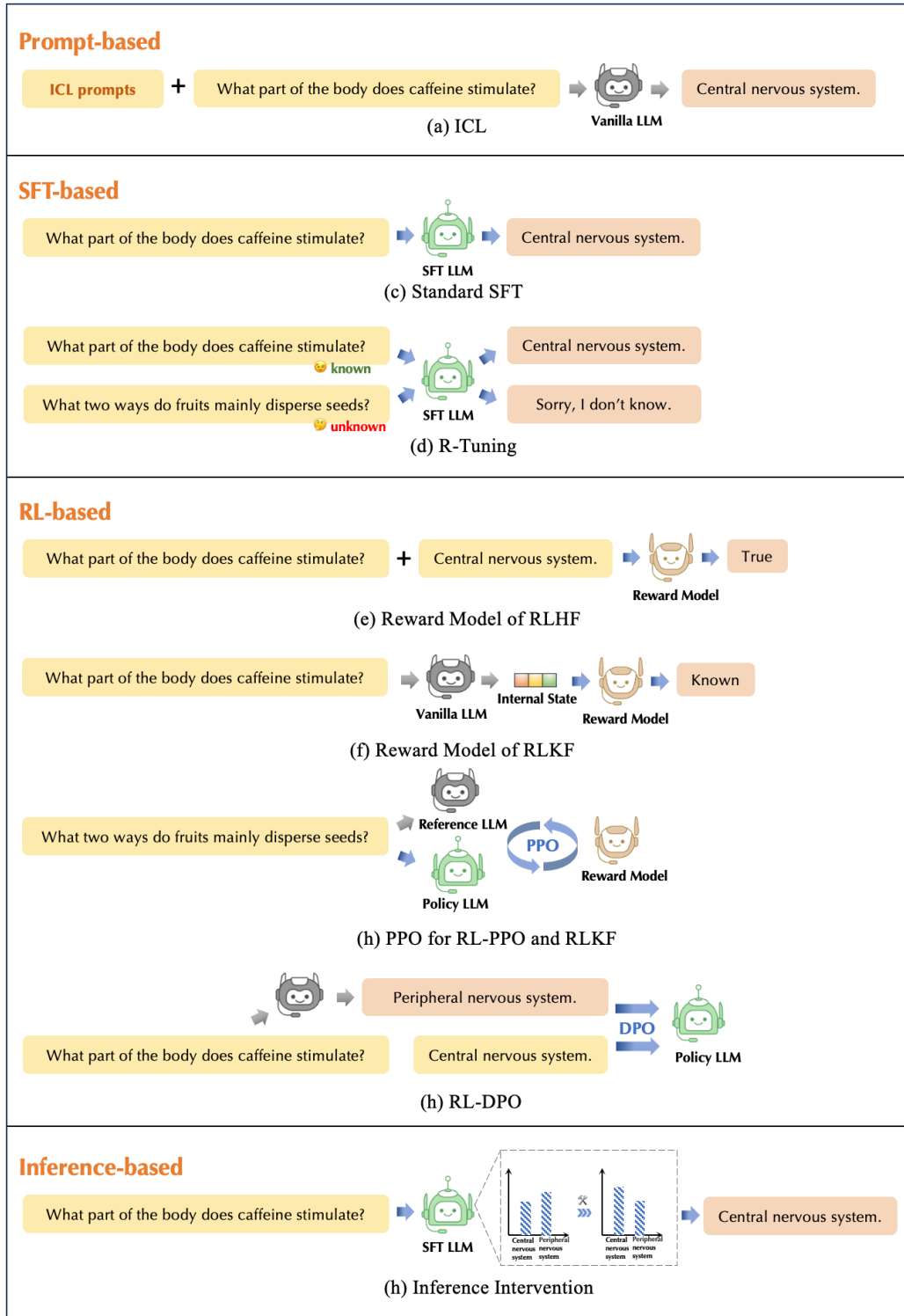


Figure 3.5: Illustration of baselines including prompt-, SFT-, RL-, and Inference-based methods.

- **SFT:** The standard supervised fine-tuning (SFT) is implemented by minimizing the negative log-likelihood of the ground-truth $\hat{y}$ conditioned on input x on model $\mathcal{M}$.

$$\arg \min_{\mathcal{M}} \mathcal{L}_{\text{SFT}} = -\mathbb{E}_{(x_i, \hat{y}_i) \sim \mathcal{D}} [\log p_{\mathcal{M}}(\hat{y}|x)] \quad (3.10)$$

- **R-Tuning:** We also introduce R-Tuning [233] which teaches LLM to refuse their unknown questions. R-Tuning is implemented in the same way as SFT, which only revises the ground-truth label of unknown questions to refusal answers. The unknown questions are those whose sampled responses in the UAlign dataset are incorrect.

RL-based Baselines We employ three RL-based baselines below.

- **RL-PPO:** Following [142], we develop the RL-PPO by training a reward model using the LLM-generated incorrect responses as negative samples. Then we conduct the PPO [157] algorithm with the obtained reward model. In other words, the RL-PPO baseline is a variant of **UAlign** which discards the uncertainty estimations.
- **RLKF:** Following [107], we employ the RLKF baseline by training the reward model on the LLMs’ internal states with the knowledge probes and further conduct PPO using the reward model. The knowledge probing implementations are referred to as [107].
- **RL-DPO:** All [177, 108, 234] focus on long-context generation like biography. We still utilize the LLMs’ generated incorrect responses as negative samples to construct the preference data to conduct the direct policy optimization (DPO) [150] algorithm.

Inference-based Baseline Another branch focuses on shifting the output distribution to improve factuality during inference. We replicate [98] by training a head

probe in the attention layer to intervene in the activations to the “truthfulness” direction as inference-time intervention (ITI). To be consistent with [98], we also train the head on TruthfulQA [110] with our prepared UAlign dataset to decode in the “truthfulness” direction. Then we further train the LLM using LoRA by SFT to adapt QA tasks. Therefore, the replicated ITI can be regarded as conducting SFT on LLMs with an additional “truthfulness” head.

Implementation Details

Experiments are conducted on two LLMs: **Llama-3-8B** (Llama-3) [3] and **Mistral-7B** (Mistral) [75]. When preparing the UAlign dataset, we sample 10 responses for each question on $K = 10$ different 1-shot prompts. The sampling temperature T is set to 0.2 to achieve a trade-off between the diversity and factuality of the answers.

During training, all the LLMs are trained using LoRA [63] with rank $r = 16$. Both the uncertainty estimation models and the reward model utilize the vanilla LLM as their bases and are trained using LoRA with rank $r = 4$. ADAM parameter update is used in a mini-batch mode. Uncertainty estimation models and the reward model are trained using SFT on the UAlign dataset. To conserve memory overhead and accelerate computation, all the models are quantified using `float16` (`fp16`) to load and save parameters during both the training and inference phases. The batch sizes for the LLM, uncertainty estimation models, and reward models are set at 4, 16, and 16, respectively. The initial learning rate of $1e-4$ is utilized with the 0.05 warm-up ratio and 0.01 weight decay of the ADAM optimizer. We set the training epoch to 2 and ensure that all the models can be trained to convergence by increasing additional training steps if necessary. The dropout rate is set at 0.05 during all model updates to reduce overfitting. The PPO training of **UAlign** and all the RL-based baselines are implemented by `trl`.

When decoding, the temperature is also set to 0.2 to be consistent with the sampling setting. All the hyper-parameters related to PPO are default values by

Method	TVQA (ID)		SciQ (ID)		NQ-Open (ID)		Avg. (ID)		LSQA (OOD)	
	Prec. $\uparrow$	Truth. $\uparrow$	Prec. $\uparrow$	Truth. $\uparrow$	Prec. $\uparrow$	Truth. $\uparrow$	Prec. $\uparrow$	Truth. $\uparrow$	Prec. $\uparrow$	Truth. $\uparrow$
Llama-3-8B										
ICL	76.15	56.55	70.43	44.30	50.28	20.11	65.62	40.32	77.35	52.98
ICL-IDK	69.17	54.10	68.36	43.00	45.43	20.72	60.98	39.27	66.67	50.24
ICL-CoT	66.68	53.37	72.34	45.90	57.34	23.60	65.45	40.95	73.96	49.37
SFT	70.80	52.57	72.18	45.40	41.41	16.57	61.46	38.18	68.09	46.63
R-Tuning	72.93	55.44	71.38	44.90	47.81	18.12	64.04	39.48	71.54	52.15
RL-PPO	76.32	55.19	75.70	45.80	54.07	24.19	68.03	41.72	72.18	48.43
RL-DPO	72.08	53.96	71.23	44.20	49.65	19.18	64.32	39.11	71.09	48.88
RLKF	77.12	56.07	72.36	44.90	54.86	22.15	68.11	41.04	74.95	52.46
ITI	71.09	53.97	72.35	43.80	43.20	17.13	62.21	38.30	68.52	46.99
UAlign	79.14	57.04	76.44	48.00	56.60	26.09	70.72	43.71	79.56	55.88
(w/o c)	74.13	54.45	74.05	45.00	54.19	23.60	67.45	41.01	74.25	52.06
(w/o e)	78.43	57.69	75.39	47.50	56.68	27.56	70.16	44.25	76.14	54.43
Mistral-7B										
ICL	77.92	55.14	68.62	42.20	52.09	17.95	66.21	38.43	74.09	47.71
ICL-IDK	72.59	51.37	63.74	39.20	51.13	17.67	62.48	36.20	72.27	47.32
ICL-CoT	76.73	54.78	71.87	44.20	54.47	18.22	67.69	39.06	79.24	52.59
SFT	74.57	54.77	65.85	42.50	50.82	14.42	63.74	37.08	68.33	44.00
R-Tuning	67.70	52.25	64.44	40.10	46.33	15.52	59.49	36.29	64.67	44.05
RL-PPO	79.23	55.08	71.35	44.10	53.76	19.19	68.11	39.45	74.49	49.67
RL-DPO	72.20	52.98	66.44	41.80	50.95	16.42	63.19	37.06	67.82	43.77
RLKF	80.43	56.92	70.66	43.90	52.09	18.24	67.72	39.68	74.19	49.23
ITI	74.65	55.16	66.90	44.90	51.12	16.68	64.22	38.91	67.73	46.20
UAlign	82.10	59.05	73.21	46.70	54.17	19.64	70.82	41.79	76.29	52.89
(w/o c)	76.44	55.13	69.84	43.50	50.30	17.88	65.52	38.83	73.15	47.06
(w/o e)	80.18	57.64	72.90	45.60	52.21	18.44	68.43	40.56	75.34	50.15

Table 3.3: Experiments of Precision (Prec.) and Truthfulness (Truth.) on four datasets on Llama-3 and Mistral. *c* and *e* represent confidence and semantic entropy, respectively.

the `trl PPOConfig` recipe except the epoch, learning rate, and batch size which are set at 2, $1e-5$, and 2, respectively. All the experiments are conducted on $4 \times$ NVIDIA A100-40GB GPUs.

3.5.2 Results of LLM Factuality

We present the results of **UAlign** and several baselines on three ID and one OOD test sets as shown in Table 3.3. Several findings are listed below.

Factuality Significant improvements are consistently achieved on diverse datasets using the proposed **UAlign** framework over other baseline methods on both Llama-3 and Mistral. We highlight the supreme Precision and Truthfulness performance

Conf.	Entro.	ID			OOD LSQA
		TVQA	SciQ	NQ-Open	
Llama-3-8B					
✗	✗	82.31	79.00	67.45	70.12
✓	✗	85.41	84.30	70.37	75.09
✗	✓	82.05	77.90	67.85	70.40
✓	✓	86.73	86.40	72.00	74.59
Mistral-7B					
✗	✗	84.53	77.30	65.24	68.31
✓	✗	86.80	79.50	72.10	72.95
✗	✓	85.24	74.60	66.64	71.22
✓	✓	88.06	79.80	75.14	73.61

Table 3.4: Accuracy of reward model varying different uses of uncertainty measures Conf. and Entro. in **UAlign** dataset on Llama-3 and Mistral.

using grey highlights among the all baselines of each column in Table 3.3. The core idea of our **UAlign** method is the utilization of uncertainty estimation models. Compared with the most relevant baselines of RL-PPO and RLKF, both the reward model and policy model in **UAlign** generate predictions and responses conditioned on uncertainty estimations regarding the knowledge boundaries to questions, thereby yielding better reliability performance. It can be attributed that by explicitly appending uncertainty measures following the question, LLMs can be assisted to precisely perceive their knowledge boundaries and thus elicit more accurate responses based on intrinsic knowledge perceptions.

Generalization We also introduced an OOD test set to assess the generalization capability of the **UAlign** method. The results in Table 3.3 indicate that most training-based baselines (SFT, RL, Inference) are unstable and result in performance decreasing compared with prompt-based baselines when generalizing on the OOD test set. However, with the assistance of accurate perceptions of knowledge boundaries, comparable factuality performances are obtained on two LLMs using the proposed **UAlign** in comparison with prompt-based methods, demonstrating strong generalization capability.

3.5.3 Analysis of Knowledge Boundary Perception

Effects of Knowledge Boundary Perception to Reward Model

Setting To investigate the effects of introducing uncertainty estimations as input features to reward models, we report the accuracy of reward models that vary in different uses of two measures on ID and OOD tasks. The reward models are trained on the UAlign dataset on both Llama-3 and Mistral.

Results As in Table 3.4, we present the results of the accuracy of reward models. Significant accuracy improvements of reward models are obtained that predominantly benefit from the use of confidence scores across both ID and OOD test sets on two LLMs, validating the effectiveness of our proposed **UAlign** method. The isolated use of semantic entropy does not guarantee a stable improvement but may even lead to a performance decrease on some test sets. However, when semantic entropy is employed in combination with confidence measures, it can facilitate further enhancements, achieving optimal results across most test sets as highlighted gray cells for two LLMs.

Analysis In the **UAlign** method, both confidence score and semantic entropy are introduced to quantify the intrinsic knowledge boundary of LLMs to questions. The explicit introduction of the knowledge boundary representations in prompts can be regarded as the added thinking step like CoT. The combined use of confidence and semantic entropy can achieve supreme prediction performance in Table 3.4. It can be assumably attributed that when leveraging both certainty and uncertainty estimations, they can complement each other for more accurate knowledge boundary representations to improve the accuracy of the reward models. We illustrate the mechanism as follows.

As in Figure 3.3 (a), by sampling multiple responses to a question, we can approximate LLM’s intrinsic knowledge boundary, where the certainty level of the

answer “*The U.S.*” is 40%. In previous work [233] which only considers the confidence level, the correct answer that the LLM knows but is not sure will be discarded and the LLM will refuse to answer. However, as in Figure 3.3 (b), the LLM can perceive that even though its certainty level to the correct answer is low, other answers are more uncertain and the dispersion level of answers is relatively high which is quantified by semantic entropy. After PPO of **UAlign**, the ability to generate correct answers conditioned on questions and estimations is well enhanced. As a result, the correct but unsure knowledge will be elicited in the responses.

Effects of Precision of Knowledge Boundary Representations

Setting Evaluating the precision of knowledge boundary representations is essential to the **UAlign** method. We employ two uncertainty estimation methods, confidence score and semantic entropy, to represent the knowledge boundary. Therefore, AUROC is utilized for precision evaluation of uncertainty estimations. We present the AUROC results of two estimations in comparison with three confidence/uncertainty estimation methods (one probability-based method (Prob.), two prompt-based methods including p(True) and verbalized (Verb.) on TriviaQA.

Results In Figure 3.6, both the confidence and entropy prediction consistently outperform other baseline uncertainty estimation methods. Optimal AUROC performances are obtained using confidence on both Llama-3 (80.45) and Mistral (82.19).

Analysis After **UAlign** SFT stage, the uncertainty estimation models are converged on the UAlign dataset to predict both confidence and entropy, indicating the models possess the ability to predict the two measures. Practically, our utilized confidence and semantic entropy incorporate the advantages of both sampling- and training-based uncertainty estimations. Multiple sampling can better approximate the actual knowledge boundaries of LLMs, while the training-based approach enables the LLMs to learn to perceive their intrinsic knowledge boundaries. Com-

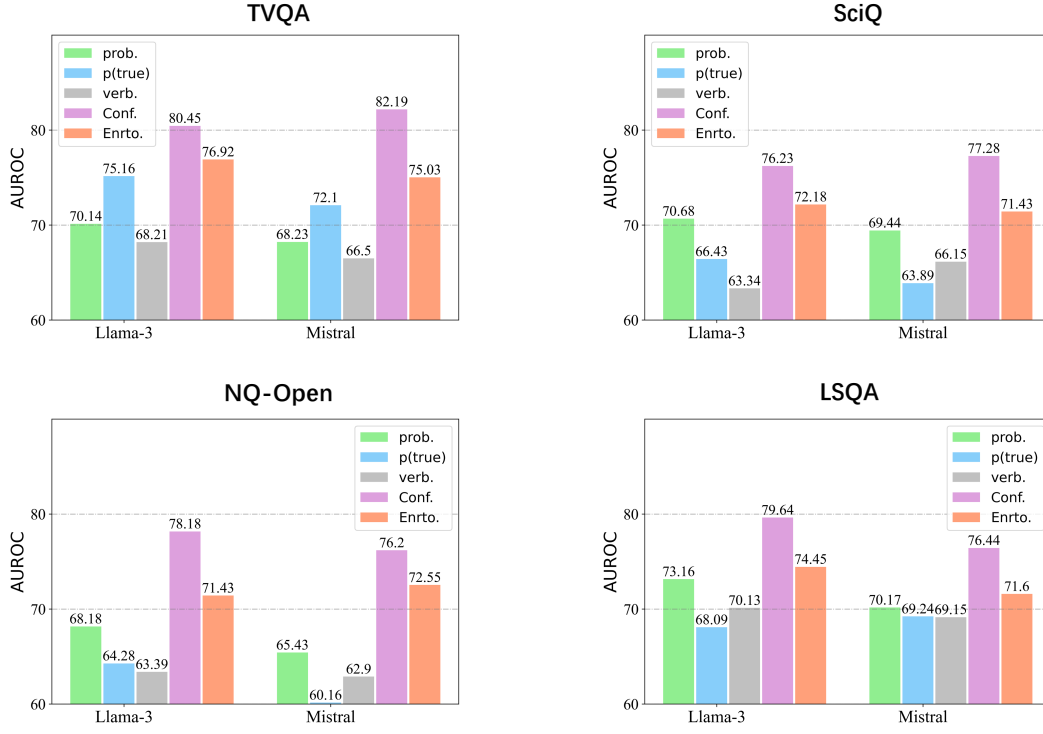


Figure 3.6: Results of AUROC $\uparrow$ of several uncertainty estimation methods on four test sets using Llama-3 and Mistral.

pared to other baselines that suffer from overconfidence issues with low AUROC scores, our methods yield more reliable estimates, thereby ensuring improved performance for both the reward model and the policy model in the following stages.

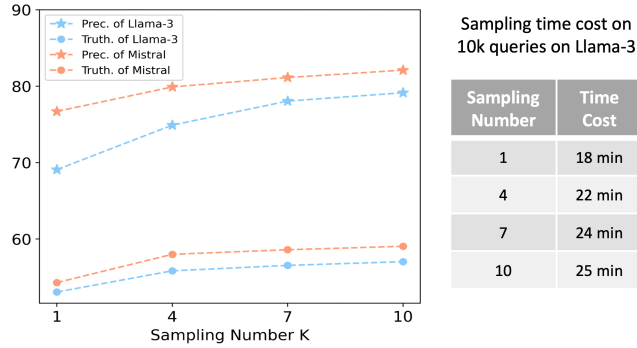


Figure 3.7: Experiments of Prec., Truth. (left), and time costs (right) of various sampling number K of 1, 4, 7, and 10 on TVQA on both Llama-3 and Mistral.

3.5.4 Analysis of Computation Cost

Computation Cost Varying Different Sampling Number

Setting The sampling number K is a crucial hyper-parameter in the **UAlign** method. Different values of K can significantly affect the precision of the knowledge boundary measurements. To evaluate the effects, we compare performances using various K of 1, 4, 7, and 10. Experiments on TVQA are presented in Figure 3.7.

Findings Results in Figure 3.7 indicate that when using small sampling numbers, increasing K leads to significant improvements in both *Prec.* and *Truth.*. However, as K increases, the factuality improvement tends to plateau, exhibiting convergence. Hence, we opt $K = 10$ as the optimal setting and don’t experiment with larger K .

We also report the sampling time costs to construct the training set in Figure 3.7. We showcase that with various acceleration and quantization methods, time costs of **UAlign** can be significantly reduced when scaling to larger models or datasets, exhibiting both great efficiency and efficacy for deployment.

Analysis The results in Figure 3.7 demonstrate that while the sampling number K increases linearly, the performance improvements are non-linear. This may be attributed to utilizing non-linear metrics, or it could suggest that $K = 10$ can approximate the actual knowledge boundaries, resulting in a gradual slowdown in performance gains. Consequently, setting K to 10 in this work makes a trade-off between performance gains and computation expense.

Computation Cost of Constructing UAlign Dataset

Setting We test the time costs to construct the UAlign dataset $\mathcal{D}$ in different sampling numbers. We present different sampling time costs on 10000 QA samples on Llama-3-8B and Llama-3-70B [3] on 4CE40G A100 GPUs loaded in fp16. We have tried to address the computation cost problem by introducing many effective accel-

eration or quantization packages like vllm, bitsandbytes, that are widely used to drive test time scaling law [166].

Findings As presented in Table 3.5, the results demonstrate the efficiency of our proposed **UAlign** method even though scaling on larger models. The dataset construction for Llama-3-70B can be well-controlled and is applicable to larger LLMs.

Model	Sampling Number	Time Cost
Llama-3-8B	1	18 min
	4	22 min
	7	24min
	10	25min
Llama-3-70B	1	1h 18min
	4	1h 33min
	7	1h 40min
	10	2h 12min

Table 3.5: Time cost in different sampling numbers of **UAlign** on Llama-3-8B and Llama-3-70B.

Analysis The relatively low computation costs when sampling can be attributed to the fact that the answer spans of knowledge QA are entity-level, and each answer only contains a few tokens. Since the output form is relatively simple, sampling ten times is sufficient and cost-saving to accurately fit the knowledge boundaries.

Furthermore, Test Time Scaling Law [166] has attracted much attention recently, which proposed to consider allocating more computation resources in inference to generate high-quality responses. These LLMs’ self-generated data can be further used for LLM training to self-improve LLMs [50]. Many works validate that incorporating multiply-sampled data on LLMs in inference can benefit LLMs for further improvements, which is a new trend for LLM training. In this way, our proposed **UAlign** provides a novel insight into the test time scaling law to represent knowledge boundaries by calculating uncertainty estimations on the sampled responses, and further explicitly leverages the uncertainty estimations for factuality alignment, heralding a promising view of the test time scaling law. Although a few

additional computation costs are required, our proposed **UAlign** is still efficient to be utilized practically with significant factuality improvements.

Computation Cost of Inference of UAlign

We subsequently analyze the computation cost during inference of **UAlign**.

Setting Our proposed **UAlign** barely increases additional inference memory and time budget. Uncertainty models also share the base LLMs with their respective plug-in LoRA modules with rank $r=4$. Additional parameters introduced only account for less than 1% of the base model parameters. Second, uncertainty models only predict two tokens of uncertainty estimations in inference.

Findings We report the inference time cost on four test sets of vanilla Llama-3, which only generates answers, and **UAlign** trained Llama-3 which predicts uncertainty estimations and then generates answers on a single A100 GPU. With the slight increase in additional memory and time cost in inference, **UAlign** significantly outperforms other baseline methods, demonstrating the factuality and efficiency on such tasks.

Dataset	Time Cost	
	Vanilla ICL	UAlign
TVQA	58 min	1h 6min
NQ-Open	28 min	32m in
SciQ	6 min	7 min
LSQA	5 min	6 min

Table 3.6: Inference time cost on four test sets of Llama-3 using vanilla ICL prompt-based and **UAlign** methods. Note the inference time costs on all the baseline methods in Figure 3.5 are comparable to the vanilla ICL prompt-based baseline method.

3.5.5 Analysis of Knowledge Distribution Update during PPO

During the PPO process, only the policy model π_θ is optimized while the uncertainty models μ, τ do not need to be updated because the reward model θ updates are offline. As discussed and demonstrated in Section 3.5.3 and Table 3.4, uncertainty models are directly associated with and benefit the reward model. In our UAlign PPO algorithm, by incorporating the two uncertainty estimations, the reward model θ can provide more precise reward scores, thereby guiding LLMs π to generate more factual responses. Since the reward model is offline during PPO, the uncertainty models also do not require online updates.

In addition, due to the KL-divergence constraint, the knowledge distribution of policy LLMs may not diverge too much from the initial policy. Both uncertainty models and reward models are trained on data generated by sampling from the vanilla LLMs, and their combined effect is to elicit the LLMs’ capacity for factual expression, evolving towards improved factuality. During the PPO process, with the KL-divergence constraint in Equation 3.7, the knowledge distribution of policy LLMs may not shift too much from the initial policy. We demonstrate the accuracy-based confidence distribution of Llama-3 before and after **UAlign** training on TriviaQA validation sets as follows.

Conf. Range	Before UAlign	After UAlign
[0, 0.25)	2404	2116
[0.25, 0.5)	1786	1628
[0.5, 0.75)	1509	1747
[0.75, 1.0]	4261	4469

Table 3.7: Accuracy-based confidence distribution of Llama-3 before and after **UAlign** training on TriviaQA validation sets.

Since the knowledge distribution does not shift too much from the initial policy, we can still achieve good performance without updating the uncertainty model for simplicity. Compared to traditional RLHF that solely utilizes reward

models, our proposed **UAlign** introduces uncertainty models that leverage knowledge boundary representations to benefit reward models and finally enhance LLMs, leading to significant improvements in reliability and generalization.

3.6 Summary

In this chapter, we analyze the hallucination issue of non-factual expressions due to the ambiguous knowledge boundary of LLMs. We follow the principle of “From Perception to Expression” and propose a **UAlign** factuality alignment method to explicitly leverage precise knowledge boundary perceptions to elicit LLMs to accurately express factual knowledge that LLMs cannot constantly answer correctly. We introduce the perception strategy of dataset construction and **UAlign** training strategies of factuality alignment by incorporating confidence score and semantic entropy as input features. Experiments on several knowledge QA tasks affirm the efficacy of our proposed **UAlign** to enhance the LLMs’ factuality, which demonstrates significant improvements over various baselines. In the next chapter, we will discuss the honesty evaluation of LLMs.

Chapter 4

Building Reliable LLMs regarding Honesty

Acknowledging what you know and what you don't is true wisdom.

知之為知之，不知為不知，是知也

— Confucius 孔子 《論語·為政》

Chapter 4 is organized as follows. Section 4.1 presents an overview of LLM honesty and our proposed two benchmarks. Section 4.2 conducts a detailed analysis of the factors affecting the honesty of LLMs, and further discusses the challenges associated with two benchmarks. Section 4.3 elaborates on the preliminary knowledge of this chapter, including uncertainty estimation, multilingual LLMs, and mathematical reasoning on LLMs, etc. Section 4.4 presents the first benchmark for multilingual confidence estimation for LLMs. Section 4.5 presents the second benchmark on the awareness of solvability on mathematical reasoning problems for LLMs. Finally, Section 4.6 summarizes the entire chapter.

4.1 Overview

Honesty constitutes a fundamental principle in aligning LLMs with human values, requiring models to accurately distinguish between what they know and do not know, and to express well-calibrated confidence in their outputs [101, 222]. It becomes increasingly crucial for users to directly ascertain how much they can trust a model’s response, and assessing the honesty of outputs can immediately indicate the level of reliability to users [46, 82]. To comprehensively investigate LLM honesty, we define the honesty of LLMs along two key dimensions: **self-awareness** and **confidence estimation**. These constitute the two central aspects of honesty examined in this chapter.

Self-awareness, a concept originating from psychology, is also applicable to LLMs with increasingly advanced intelligence. It is commonly defined as metacognition (“cognition about cognition,”), referring to an LLM’s capacity to understand its own knowledge state of distinguishing between what it knows and does not know, and to assess whether it can successfully complete a given task [37, 38, 44]. This form of self-monitoring serves as a critical indicator of an LLM’s intelligence. Confidence estimation is a quantitative measure, requiring that an LLM’s confidence estimations in completing a task align with the actual probability of correctness. Deviations may result in overconfidence or underconfidence [52, 46].

Unlike factuality, which relies on world knowledge as ground-truth references, the assessment of honesty is inherently more complex due to the diversity of awareness and confidence estimation methods, which may vary across tasks. Consequently, **constructing appropriate benchmarks is fundamental to the study of the honesty of LLMs**. To this end, we propose two benchmarks targeting two key scenarios, respectively addressing metacognitive ability and confidence calibration.

Benchmark on Multilingual Confidence Estimation First, we introduce our proposed benchmark on multilingual confidence estimation. Confidence estimations

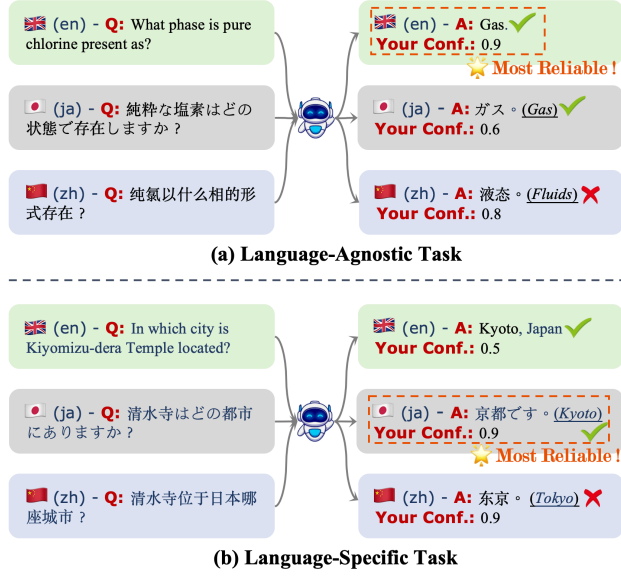


Figure 4.1: Examples of generations and confidence scores of Llama-3.1 given the same inputs in three languages in LA and LS scenarios derived from SciQ and LSQA datasets respectively.

indicating the extent of honesty of the generations become essential. However, existing research on confidence estimation for LLMs has been predominantly limited to English [82, 109, 46, 178]. The confidence estimation primarily developed for English remains a crucial subject for exploration when applied to other languages.

Hence, we propose a benchmark called **MlingConf** to investigate **Multilingual Confidence** estimation performance of several LLM confidence estimation methods on five languages, including *English*, *Japanese*, *Chinese*, *French*, and *Thai*. We focus on both language-agnostic (LA) and language-specific (LS) tasks tailored to specific social, cultural, and geographical contexts of a language, as illustrated in Figure 4.1, to explore the performance and language dominance effects of multilingual confidence estimations. First, we meticulously constructed the **MlingConf** dataset for the benchmark, including five datasets of different tasks in LA and LS scenarios, respectively. The LA involves four different tasks that are widely used in confidence estimation in *English* and are translated into four other languages. We also create a language-specific QA (LSQA) dataset for the LS scenario, including five subsets for

the five investigated languages, comprising QA pairs about social culture, history, geography, and celebrity pertaining to the specific language. To ensure the data quality, we conduct rigorous translation consistency checks and finally employ linguistic experts for human evaluations.

Our experiments reveal that on LA tasks *English* exhibits notable linguistic dominance in confidence estimations than other languages, while on LS tasks, using question-related language to prompt LLMs demonstrates better linguistic dominance in multilingual confidence estimations. The phenomena inspire a simple yet effective native-tone prompting strategy by employing language-specific prompts for LS tasks, effectively improving LLMs’ reliability and accuracy in LS scenarios.

Benchmark on Mathematical Reasoning Solvability Then, we introduce a benchmark on self-awareness by investigating the solvability of mathematical reasoning. LLMs have exhibited impressive capabilities in reasoning tasks [105, 17]. However, they tend to be overconfident in producing deterministic responses to any questions [177]. Hence, as illustrated in Figure 4.2 (a), when confronted with problems that are intrinsically unsolvable or beyond their capabilities [227, 6, 222, 114, 101], LLMs may still attempt to fabricate reasoning steps to output plausible but illogical, nonsensical or misleading answers, severely undermining LLM honesty.

Prior work has mainly examined awareness of answerability on knowledge tasks, leaving math reasoning unexplored due to the dearth of unsolvable math problems. To this end, we first formulate the reliability evaluation on self-awareness for both solvable and unsolvable math problems as follows: a reliable LLM should correctly answer solvable problems or explicitly identify the unsolvability of unsolvable problems after step-by-step reasoning as in Figure 4.2 (b), which is the most preferable and considered as *successful* in Figure 4.2 (c). If the problem falls beyond the LLMs’ capacity, a suboptimal choice would be *refusing* to answer [173, 212] to prevent misleading users. All other responses are considered as *failures*.

We then develop a **ReliableMath** dataset, including both solvable and unsolv-

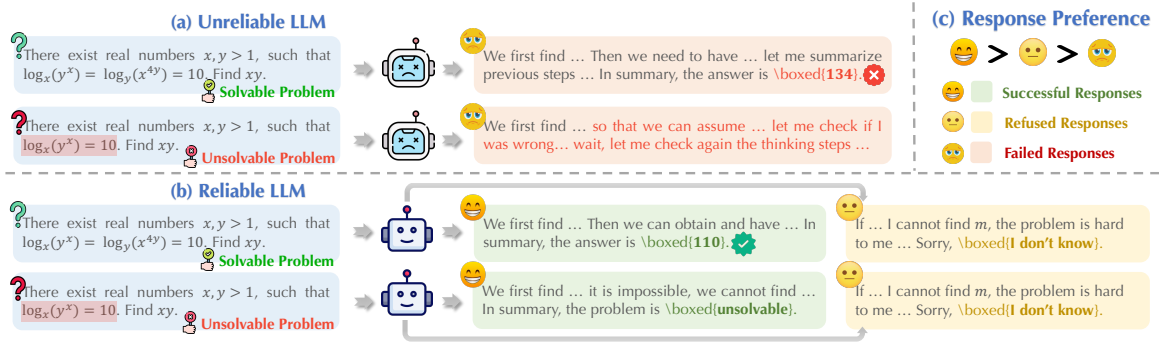


Figure 4.2: Illustrations of LLM reliability evaluation of metacognition on math problems.

able data. Solvable cases are collected from open-source math datasets at different levels. Unsolvable problems are synthesized using our proposed data construction workflow with expert check. The data synthesis workflow is scalable to generate unsolvable math problems at any difficulty level for evaluations.

Experiments on **ReliableMath** dataset are conducted on a series of reasoning LLMs and instruction LLMs. Our analysis yields several key findings: 1) LLMs can occasionally recognize the illogicality of unsolvable cases, but will fabricate reasoning instead of directly identifying the unsolvability. 2) When instructing LLMs to critically identify solvability with reliable prompts, the reliability performance of larger-sized LLMs remains on the solvable, and notably improves on unsolvable problems yet still lags behind on solvable cases. 3) Small LLMs rarely show any progress despite employing reliable prompts. Therefore, we further employ an honesty alignment strategy to enable small LLMs to think critically and identify the solvability of problems before generating responses, which can significantly improve honesty on both in-domain math and out-of-domain knowledge tasks.

4.2 Problem Statement

An honest LLM should acknowledge its limitations when it encounters queries beyond its capabilities, rather than providing misleading information [101]. Nev-

ertheless, since the majority of pre-training data consists of deterministic textual content, LLMs naturally tend to generate deterministic responses [177]. This tendency arises not only from the high prevalence of such patterns in the training data, but also from the models’ optimization toward aligning with human preferences, which increases the risk for “dishonest” responses and finally misleads outputs.

Although incorporating data that encourages truthful responses during post-training can alleviate this issue to some extent [233, 222, 218, 108], the fundamental challenge lies in enabling LLMs with an inherent awareness of maintaining honesty spontaneously. Unlike factuality, which can be grounded in world knowledge as a reference for verification, honesty evaluation is model-specific, whose performance varies across different scenarios and evaluation protocols. Therefore, **designing scenario-specific benchmarks for assessing honesty becomes critical in this area**. We focus on two particularly challenging scenarios: multilingual confidence estimation and solvability of mathematical reasoning tasks.

Multilingual Confidence Estimation The dearth of confidence estimations in languages other than English hinders users from assessing the honesty of LLMs in non-English scenarios, restricting the LLMs’ global deployment. Due to the variations in the quantity and domain coverage of pre-training corpora across different languages, the confidence estimation ability may also presumably vary. To conduct a fine-grained investigation of multilingual tasks, we divide the tasks into language-agnostic (LA) [237] and language-specific (LS) scenarios as in Figure 4.1, considering the effects of linguistic dominance on multilingual confidence estimations. Linguistic dominance refers to that one language holds a superior status over others within a specific social or cultural context [12, 179, 59], and can also exist in confidence estimation ability on different languages.

In this study, the LS refers to the tasks that hold linguistic dominance caused by the knowledge domain coverage varying in different language training corpora, such as questions pertaining to social, cultural, or geographical contexts for a spe-

cific language, while the LA involves linguistic dominance mainly caused by the quantities of training corpora, such as general knowledge, common sense, and reasoning [11, 170]. **This study aims not only to investigate the performance of LLMs in confidence estimation across multilingual tasks, but also to examine, separately for LA and LS, the influence of language dominance, and to propose targeted optimization methods accordingly.**

Solvability of Mathematical Reasoning Prior research assessing LLM metacognition has predominantly focused on LLMs’ abilities to identify their known/unknown questions on knowledge-based tasks [227, 6, 239, 101], lacking attention on reasoning tasks like math [173, 115]. Existing mathematical benchmarks exclusively concentrate on solvable problems [173, 115], while unanswerable/unknown datasets are confined to knowledge tasks [227, 6], leaving a scarcity of high-quality open-source unsolvable math problems, which can be utilized to assess LLM honesty [119].

Generally, educational math problems may be either solvable or unsolvable, depending on whether a definitive solution that satisfies all requirements exists. Unlike knowledge-intensive tasks, in which LLMs only need to determine whether they possess the required knowledge or not [227, 6], mathematical reasoning tasks require the LLM to determine not only whether a definitive solution exists for the math problem but also whether the LLM itself is capable of solving it. Additionally, determining the solvability of problems necessitates step-by-step reasoning, intensifying the challenges of honesty evaluation on math reasoning. In general, real-world unsolvable math problems include both illogical, ill-posed, or undecidable problems (e.g., Find a real solution to an equation with no real solutions.) [125, 203] and unsolved open problems (e.g., Hilbert’s problems) [202]. Since unsolved problems are limited and public, unsolvable problems in our study are confined to ill-posed, illogical, and undecidable problems where no definitive solution exists.

With the soaring popularity of DeepSeek-R1 [25], long-form reasoning has emerged as a new paradigm for LLMs. Accordingly, we can broadly categorize

LLMs into fast-thinking instruction LLMs and slow-thinking reasoning LLMs. For mathematical tasks, particular attention is given to the performance of reasoning LLMs. **This study first formalizes the metacognition evaluation of mathematical reasoning tasks, then constructs dedicated datasets to investigate the behavioral patterns of both instruction LLMs and reasoning LLMs. Based on the empirical findings, an improvement strategy is further proposed.**

4.3 Related Work and Preliminary

4.3.1 Confidence Estimation of LLMs

To alleviate over-confidence and enhance the reliability of LLMs, reliable uncertainty estimation is essential to determine whether a question is known or not to the LLM [46]. Both *Uncertainty* and *Confidence* estimations can indicate the honesty level of the responses generated by LLMs, and are generally used interchangeably [207, 14, 46, 116]. Uncertainty detection is essential for hallucination mitigation on knowledge-intensive tasks [211, 183, 160, 195, 184, 194, 123]. In this part, we investigate several commonly used *confidence & uncertainty* estimation methods for generative LLMs. Specifically, we denote $\text{Conf}(x, y)$ as the confidence score associated with the output sequence $y = [y_1, y_2, \dots, y_N]$ given the input context $x = [x_1, x_2, \dots, x_M]$. We also summarize the methods as well as their disadvantages as illustrated in Figure 4.3.

Likelihood-based Methods: Following model calibration on classification tasks [51], [184, 219, 183, 190] intermediately quantify sentence uncertainty over token probabilities. In traditional discriminative models, except likelihood-based methods, confidence estimations also include ensemble-based and Bayesian methods [93, 40, 217, 188, 39, 1], and density-based methods [94]. However, this likelihood-based method requires access to token probabilities and thus being limited to white-box

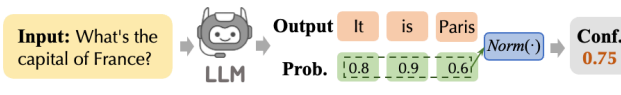
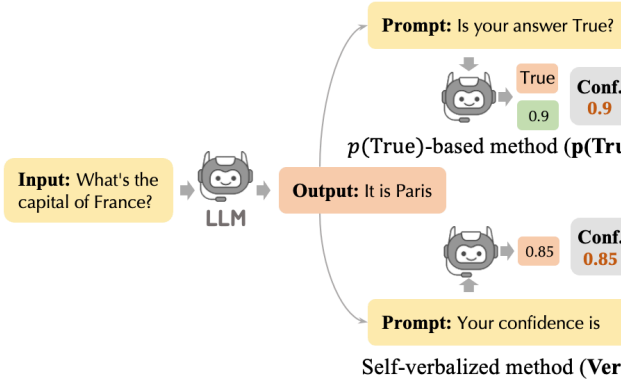
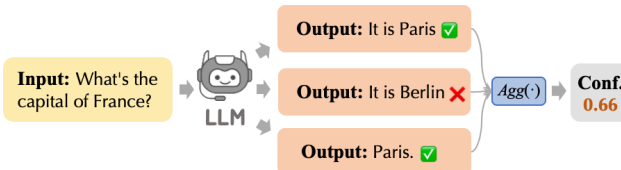
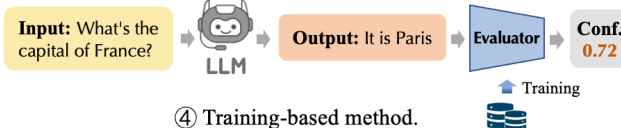
Confidence & Uncertainty Estimation Methods on LLMs	Disadvantages
 ① Likelihood-based method.	a. Requires normalization due to variable sequence length; b. Requires access to token-level probabilities, inapplicable to black-box LLMs; c. Fails to capture semantic meaning over token-level probabilities.
 ② Prompting-based method.	a. Relies on prompting strategies to elicit confidence estimation, varying in different methods (Is <i>True</i> probability, numerical confidence, and word expressions, etc.); b. Cannot improve LLM's intrinsic confidence estimation ability. c. Prone to be over-confident.
 ③ Sampling-based method.	a. Requires additional inference time cost; b. Varying in different aggregation methods; c. Cannot improve LLM's intrinsic confidence estimation ability.
 ④ Training-based method.	a. Requires training an additional evaluator; b. Difficult to learn LLM's intrinsic confidence estimation on unseen domains.

Figure 4.3: An illustration of several confidence estimation methods as well as their drawbacks.

LLMs. The likelihood-based confidence is estimated by calculating the joint token-level probabilities over y conditioned on x . As longer sequences are supposed to have lower joint likelihood probabilities that shrink exponentially with length, the product of conditional token probabilities of the output should be normalized by calculating the geometric mean by the sequence length [130, 121], and the confi-

dence score can be represented as:

$$\text{Conf}(\mathbf{x}, \mathbf{y}) = \left(\prod_i^N p(y_i | \mathbf{y}_{<i}, \mathbf{x}) \right)^{\frac{1}{N}} \quad (4.1)$$

Similarly, the arithmetical average of the token probabilities is adopted in [183]:

$$\text{Conf}(\mathbf{x}, \mathbf{y}) = \frac{1}{N} \sum_i^N p(y_i | \mathbf{y}_{<i}, \mathbf{x}) \quad (4.2)$$

Furthermore, a low probability associated with even one generated token may provide more informative evidence of uncertainty [183]. Hence, the minimum of token probabilities is also employed.

$$\text{Conf}(\mathbf{x}, \mathbf{y}) = \min \{ p(y_1 | \mathbf{x}), \dots, p(y_N | \mathbf{y}_{<N}, \mathbf{x}) \} \quad (4.3)$$

Prompting-based Methods: Recently, LLMs’ remarkable instruction-following ability [13] provides a view of instructing LLMs to self-estimate their confidence level to previous inputs and outputs, including expressing uncertainty in words [109, 241, 178, 211], or instructing the LLM to self-evaluate its correctness on $p(\text{True})$ [82]. The $P(\text{True})$ confidence score is implemented by simply asking the model itself if its first proposed answer $\mathbf{y}$ to the question $\mathbf{x}$ is true [82], and then obtaining the probability $p(\text{True})$ assigned by the model, which can implicitly reflect self-reflected certainty as follows.

$$\text{Conf}(\mathbf{x}, \mathbf{y}) = p(\text{True}) = p(\mathbf{y} \text{ is True?} | \mathbf{x}) \quad (4.4)$$

Another method is to prompt LLMs to linguistically express tokens of confidence scores in verbalized numbers or words [109, 128, 241, 178, 211].

Sampling-based Methods: The sampling-based methods refer to randomly sampling multiple responses given a fixed input $\mathbf{x}$ using beam search or temperature sampling strategies [123, 211, 118]. Various aggregation methods are adopted on

sampled responses to calculate the consistency level as the confidence score. Moreover, some uncertainty quantification methods are used to calculate the entropy, indicating the dispersion level of multiple outputs [88, 111, 134].

Training-based Methods: For training methods, an external evaluator trained on specific datasets is introduced to output a confidence score given an input and an output. The evaluator can be a pre-trained NLI model [128], or a value head connected to the LLM output layer [109, 82], or the LLM itself [57].

Note that sampling- and training-based methods are omitted in this work as they are costly and time-consuming for multilingual confidence estimations. In addition, although *confidence* and *uncertainty* are always used interchangeably, the former *confidence* pertains to the model’s certainty regarding a specific generation, while the latter *uncertainty* denotes the “dispersion” of potential predictions for a given context. In this chapter, semantically equivalent inputs in various languages are thoroughly distinct in token space. Hence, we employ *confidence estimation*, albeit uncertainty estimation methods are still applicable.

4.3.2 Multilingual LLMs

Most recent LLMs primarily pre-trained on English corpora have showcased remarkable capabilities [145, 162, 137]. However, their efficacy in other low-resource languages remains limited. Recent advances in multilingual LLMs have significantly expanded the scope of natural language processing beyond English-centric systems. Early efforts, such as cross-lingual representation learning, enabled knowledge transfer across languages, forming the foundation for modern MLLMs. Recent surveys highlight the emergence of large-scale multilingual pretraining models (e.g., BLOOM [204], LLaMA [3]) that leverage massive multilingual corpora to improve cross-lingual generalization and low-resource language performance [215, 64]. More recent work emphasizes challenges including data imbalance, align-

ment, and cultural bias, as well as the need for better evaluation benchmarks and fairness-aware training strategies. Many research works have extended various tasks in multilingual domains, such as the jailbreak problem [26]. Prior studies have also explored diverse cross-lingual applications [191, 192, 147]. Despite rapid progress, current models still struggle with underrepresented languages and culturally grounded knowledge, motivating ongoing research into more inclusive multilingual datasets. Therefore, our research systematically delves into an under-explored domain of multilingual confidence estimation for LLMs.

4.3.3 Self-awareness of LLMs

Self-awareness ability is a foundational requirement for LLM, requiring generating factually correct, informative, and consistent responses to users [114, 101, 41]. A unified perspective of LLM reliability is that LLMs should be able to identify questions that fall outside their scopes or are unanswerable [227, 6, 196]. Prior work enhances the reliability of knowledge tasks by aligning LLMs with their uncertainty to confidently answer known questions and refuse unknown questions [233] to mitigate hallucinations [218, 222, 212, 108, 239, 219]. However, current research primarily focuses on knowledge-intensive tasks. There is a void of benchmarks on mathematical reasoning tasks.

4.3.4 Mathematical Reasoning on LLMs

Math tasks have long been regarded as effective proxies of the reasoning capabilities of LLMs [86, 127, 173]. These tasks require LLMs to comprehend the semantics and symbols in the question, engage in problem-solving processes step by step, and present the final answers [159, 221]. Open benchmarks range from school-level GSM8K [22] and MATH [61] to more challenging, Olympiad- or college-level AIME [185] and CollegeMath [173]. Recent advanced LLMs like DeepSeek-R1 [25], emulating the slow and deliberate deep Chain-of-Thought (CoT) [200] reasoning

steps (System 2 thinking), have exhibited remarkable math reasoning abilities [17] over previous **foundation and instruction LLMs** like DeepSeek-V3 [24] which execute fast and heuristic-driven generation (System 1 thinking) [105, 68]. LLMs endowed with deep reasoning capabilities for solving complex math problems are commonly referred to as **reasoning LLMs** [17]. However, prior work has not considered the reasoning reliability of LLMs where unreliable LLMs fabricate solutions to unsolvable problems, thus leading to hallucination in reasoning steps.

4.3.5 Specific Notations

The definitions of specific notations of this chapter are summarized in Table 4.1.

4.4 Benchmark of Multilingual Confidence Estimation

4.4.1 Multilingual Dataset Construction

Language Selection

Owing to the lack of multilingual resources to comprehensively exhibit confidence estimation across diverse languages, we construct a high-quality multilingual dataset called MlingConf dataset encompassing five languages: *English* (**en**), *Japanese* (**ja**), *Chinese* (**zh**), *French* (**fr**), and *Thai* (**th**). We consider the effects of both language family and resource level for language selection. The MlingConf dataset includes four tasks for the language-agnostic (LA) scenario and one task for the language-specific (LS) scenario. We specify the data source and construction process of the MlingConf dataset in the following part.

Our study leverages several datasets in *English* (**en**) and translates into four languages, encompassing *Japanese* (**ja**), *Chinese* (**zh**), *French* (**fr**), and *Thai* (**th**). We evaluate the quality of confidence estimations on our proposed MlingConf dataset,

Notation	Description
Section 4.5.1: Metacognition Evaluation on Mathematical Reasoning	
$\mathcal{M}$	LLMs.
$\mathcal{A}$	Solvable mathematical problems.
$\mathcal{U}$	Unsolvable mathematical problems.
$\mathcal{S}$	Successful responses for reliability evaluation.
$\mathcal{R}$	Refused responses for reliability evaluation.
$\mathcal{F}$	Failed responses for reliability evaluation.
Section 4.5.2: Dataset Construction	
$\mathcal{M}_I$	Instruction LLMs.
$\mathcal{M}_R$	LLMs Reasoning LLMs.
$\mathcal{D}_r$	ReliableMath dataset.
$\mathcal{D}_a$	ReliableMath solvable subset.
$\mathcal{D}_u$	ReliableMath unsolvable subset.
x_i	The i -th question sample from $\mathcal{D}_a$.
$\hat{y}_i$	The ground-truth answer to x_i in $\mathcal{D}_a$.
y_i	The LLM generated response to x_i .
c_i^j	The j -th extracted condition of x_i .
$\tilde{x}_i^j$	The j -th rewritten question of x_i .
$\tilde{c}_i^j$	The rewritten condition by model verification of $\tilde{x}_i^j$ where $\tilde{c}_i^j$ entails c_i^j .
$\tilde{a}_i^j$	The unsolvable reason analysis by model verification of $\tilde{x}_i^j$.
d_i^j	The difficulty level by human evaluation of $\tilde{x}_i^j$.
Section 4.5.5: Honesty Alignment	
$\mathcal{M}_T$	Advanced teacher LLM to be distilled.
$\mathcal{M}_S$	Relatively small LLMs to be improved.
$\mathcal{Q}_a^{(t)}$	Solvable problem training set.
$\mathcal{Q}_u^{(t)}$	Unsolvable problem training set.
$\mathcal{D}^{(t)}$	Alignment training set with problems and sampled responses.
p_s	Standard CoT-based mathematical problem-solving prompt.
p_r	Mathematical problem-solving prompt with refusal instruction.
p_u	Mathematical problem-solving prompt with instruction to indicate unsolvability.

Table 4.1: Summarized notations in Chapter 4.

including four datasets for the language-agnostic (LA) scenario and one dataset for the language-specific (LS) scenario.

Data Source

Language-Agnostic (LA) Tasks For LA tasks, we employ the following four datasets that are widely used for confidence estimations in *English* [88, 211]. The datasets

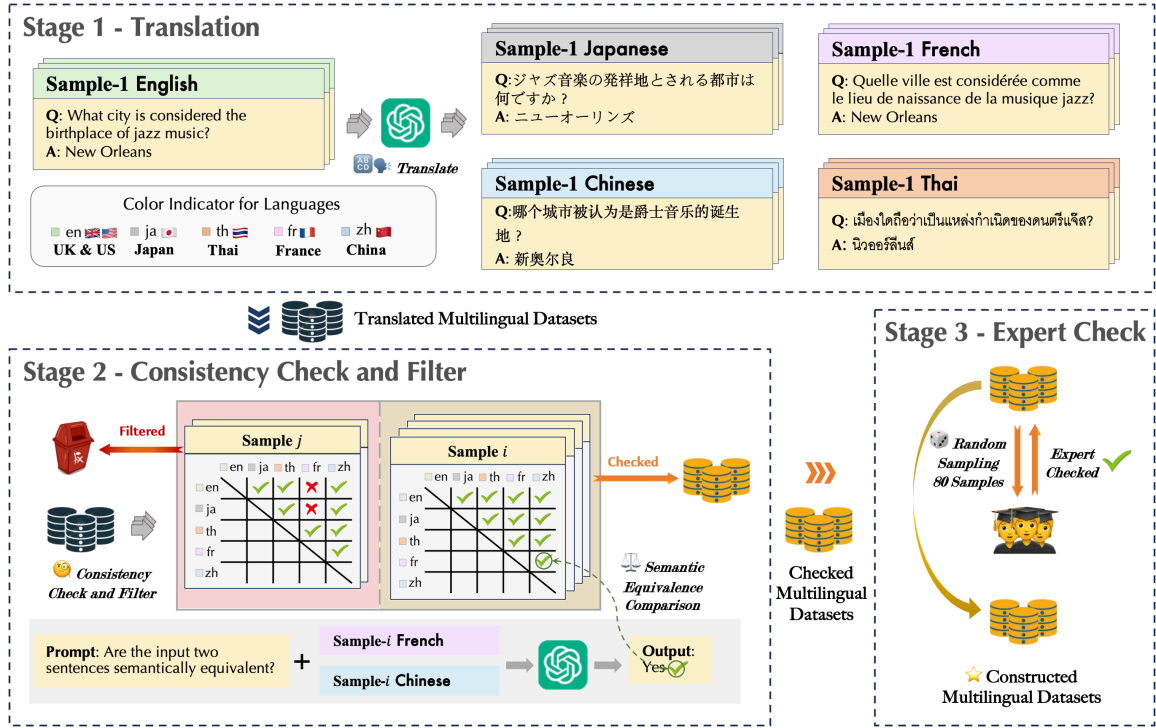


Figure 4.4: Three stages of MlingConf dataset construction.

include 1) **TriviaQA (TVQA)** [80] of closed-book trivia question-answering pairs to gauge models' factual knowledge; 2) **GSM8K** [22] for arithmetic reasoning task of math problems; 3) **CommonsenseQA (CSQA)** [171] of multiple-choice QA pairs requiring different types of commonsense knowledge; 4) **SciQ** [79] requiring scientific professional knowledge. We translate the datasets into four other languages. All the datasets in this chapter are pre-processed in standard QA format.

Language-Specific (LS) Tasks We create **Language-Specific QA (LSQA)** dataset pertaining to language-dominant knowledge covering specific social, geographical, and cultural language contexts for the UK & US, France, China, Japan, and Thailand. We prompt GPT-4 [137] to generate 200 questions pertaining to only one specific language contexts as a language-specific subset. For example, all questions in *Japanese* subset pertain to Japanese social culture, history, geography, etc.

Dataset Construction

The construction of the MlingConf dataset in this study follows an elaborate three-stage procedure as delineated in Figure 4.4.

Stage 1 The QA samples derived from the above datasets are translated into four languages (**ja**, **th**, **zh**, and **fr**) through GPT-4 [137].

Stage 2 We check the consistency of five translated results by comparing the semantic equivalence in pairs in $C_5^2 = 10$ times for each sample. The samples with more than 2 times semantic in-equivalence are treated as noisy data and then filtered. The changes of samples before and after consistency check and filter are in Table 4.3 and more clean multilingual datasets are obtained. Moreover, we present the number of samples for each language-specific LSQA subset as in Table 4.4. For all generations of MlingConf dataset construction, the temperature T is set to 0.

Stage 3 Finally, we employ several experts majoring in linguistics to examine the translation performance across 50 randomly selected samples as shown in Table 4.2. Given the dataset obtained after Stage 2, we randomly select 50 samples from each test set and send them to language experts. Each language expert is required to evaluate the translation results of their respective specialized language in each sample, determining whether the translation is correct (returning 1 for correct and 0 otherwise). We calculate the translation accuracy for each language in each test set and present the results in Table 4.2. Human evaluation results suggest the obtained multilingual datasets are high-quality for further experiments.

Evaluation Metrics

Accuracy (Accu.) We employ a string-matching approach to evaluate the accuracy of generated answers y and compare them with the ground truth $\hat{y}$. Although exact matching (EM) of $y \equiv \hat{y}$ is widely used on GSM8K and CSQA, it always misjudges

Lang.	TVQA	GSM8K	CSQA	SciQ	LSQA
zh	96%	100%	100%	98%	100%
ja	98%	100%	98%	96%	100%
fr	100%	100%	100%	98%	100%
th	96%	100%	98%	94%	100%

Table 4.2: Translation accuracy evaluated by linguistic experts on 50 randomly selected samples.

	TVQA	GSM8K	CSQA	SciQ	LSQA
Before	2000	1319	1221	1000	1000
After	1238	1318	1152	640	857

Table 4.3: Number of samples before and after consistency check and filter.

LSQA	en	zh	ja	fr	th	Total
	185	172	167	179	154	857

Table 4.4: Statistics of samples for each language-specific subset of the LSQA dataset.

some correct answers with slight differences on closed-book QA tasks, to better assess the result accuracy (Accu.), we replace EM with a variant called positive-recall exact matching (PREM) of $\mathbf{y} \in \hat{\mathbf{y}} \vee \hat{\mathbf{y}} \in \mathbf{y}$, same as Factuality in Section 3.5.

Area Under the Receiver Operator Characteristic Curve (AUROC) AUROC assesses the effectiveness of confidence estimation [36] by quantifying how likely a randomly chosen correct answer possesses a higher confidence score than an incorrect one, yielding a score in range of [0, 1], implemented by `sklearn` toolkit.

Expected Calibration Error (ECE) ECE gauges the calibration performance which indicates how well a model’s predicted confidence matches its actual accuracy [52]. For an expected well-calibrated AI system, samples x with confidence of q should also have an average accuracy of q on predictions $\mathbf{y}$ where $P(\mathbf{y} = \hat{\mathbf{y}} | \text{Conf}(x, \mathbf{y}) = q) = q$ with $\text{ECE}=0$. We partition the inference results into M disjoint bins $\{B_m\}_{m=1}^M$ based on the confidence scores $\{q_i\}$, compute the average confidence score in

$\left(\frac{m-1}{M}, \frac{m}{M}\right]$ for the m -th bin B_m , and compare it with the average true accuracy $\text{acc}(B_m)$ of the answers within B_m . The ECE is calculated by:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (4.5)$$

The average accuracy $\text{acc}(B_m)$ and confidence $\text{conf}(B_m)$ of the answers in B_m is obtained by:

$$\text{acc}(B_m) = \frac{1}{|B_m|} \sum_{a_i \in B_m} \mathbb{I}(\hat{b}_i = b_i) \quad (4.6)$$

$$\text{conf}(B_m) = \frac{1}{|B_m|} \sum_{a_i \in B_m} q_i \quad (4.7)$$

where a_i , b_i , $\hat{b}_i$, and q_i indicate the input data, label, prediction result, and confidence score respectively for the i -th sample. ECE is essential for reliable AI systems on prediction tasks like weather forecasting. The smaller the ECE value, the better.

Implementation Details

Experiments are conducted on GPT-3.5-Turbo-0125 (GPT-3.5) and Llama-3.1-8B-Instruct (Llama-3.1) [3]. Few-shot prompts containing N_f examples are utilized for answer generation with greedy decoding, which outperforms temperature decoding on knowledge tasks [168]. N_f is set to 8 for GSM8K and 4 for others.

4.4.2 Experiments on Language-Agnostic (LA) Tasks

To comprehensively investigate LLMs' multilingual confidence estimations on LA tasks, as presented in Table 4.5 and Figure 4.5. Experiments are conducted varying in different confidence estimation methods and languages.

Conf.	en		zh		ja		fr		th		Avg.	
	ARC. $\uparrow$	ECE $\downarrow$	ARC. $\uparrow$	ECE $\downarrow$	ARC. $\uparrow$	ECE $\downarrow$	ARC. $\uparrow$	ECE $\downarrow$	ARC. $\uparrow$	ECE $\downarrow$	ARC. $\uparrow$	ECE $\downarrow$
TVQA on GPT-3.5												
Prob.	76.51	24.36	78.39	32.95	76.90	28.14	72.14	27.39	74.30	40.17	75.65	30.60
$p(\text{True})$	79.64	18.25	82.34	22.94	84.50	29.06	80.59	20.90	81.22	40.87	81.66	26.40
Verb.	80.32	16.52	81.76	24.32	84.61	34.49	83.47	26.53	86.72	38.19	83.38	28.01
TVQA on Llama-3.1												
Prob.	80.74	10.72	80.41	40.39	88.75	26.24	79.05	20.38	89.59	36.77	83.71	26.90
$p(\text{True})$	68.98	18.35	68.10	38.19	52.69	37.85	62.00	22.04	60.55	37.01	62.66	30.69
Verb.	77.18	24.73	63.50	37.64	68.91	34.27	69.90	25.44	73.22	40.19	70.54	32.45
GSM8K on GPT-3.5												
Prob.	54.79	26.48	58.49	27.19	57.09	29.46	57.38	37.29	61.73	41.77	57.90	32.44
$p(\text{True})$	65.25	31.88	62.74	28.65	69.75	19.03	60.14	39.08	61.45	49.88	63.87	33.70
Verb.	62.34	22.17	59.25	28.91	58.34	26.71	66.65	25.14	54.02	45.63	60.12	29.71
GSM8K on Llama-3.1												
Prob.	65.69	22.33	66.37	21.92	69.73	35.69	61.07	29.56	63.22	28.51	65.22	27.60
$p(\text{True})$	61.64	14.49	65.83	17.39	71.40	11.26	57.04	8.02	57.31	12.90	62.64	12.81
Verb.	57.00	50.05	63.04	42.89	58.93	45.33	54.45	35.31	55.30	34.94	57.75	41.70
CSQA on GPT-3.5												
Prob.	59.06	24.45	55.92	38.30	48.01	50.60	55.33	31.12	48.21	41.71	53.31	48.18
$p(\text{True})$	67.13	19.65	58.64	27.08	65.23	19.24	66.33	23.43	59.96	34.47	63.46	24.77
Verb.	69.60	16.84	54.30	21.54	68.34	19.84	61.87	21.81	68.93	21.71	64.73	24.35
CSQA on Llama-3.1												
Prob.	78.06	13.64	64.91	36.33	75.65	30.11	66.12	19.72	77.65	42.18	72.47	28.40
$p(\text{True})$	56.25	34.04	64.24	36.82	66.60	40.32	59.34	27.72	58.07	29.91	60.90	33.76
Verb.	62.42	28.16	54.61	28.84	61.06	37.85	57.97	23.96	71.91	37.15	61.39	31.19
SciQ on GPT-3.5												
Prob.	69.50	32.23	71.29	35.63	78.28	47.06	72.66	34.85	75.13	56.17	73.37	41.19
$p(\text{True})$	72.06	23.15	76.18	30.44	80.16	36.18	71.30	37.85	68.29	41.25	73.60	33.77
Verb.	70.18	20.80	75.50	37.59	77.89	30.33	69.31	32.47	74.85	41.15	73.55	32.47
SciQ on Llama-3.1												
Prob.	74.14	13.40	72.09	32.26	74.48	34.21	77.45	22.76	77.61	36.10	75.15	27.75
$p(\text{True})$	62.38	19.28	64.89	37.01	58.92	36.47	61.01	10.72	51.90	41.06	59.82	28.91
Verb.	62.65	24.10	52.90	32.94	69.10	39.15	59.30	24.92	65.93	40.67	61.98	38.36
Avg. (TVQA, GSM8K, CSQA, SciQ) on GPT-3.5												
Prob.	64.96	26.88	66.02	33.51	65.07	38.71	64.37	32.66	64.88	44.95	65.06	35.34
$p(\text{True})$	70.38	23.23	69.97	27.27	74.91	25.87	69.59	30.31	67.73	41.61	70.65	29.66
Verb.	70.61	19.01	67.70	28.09	72.29	27.84	70.32	26.48	71.13	36.67	70.45	28.64
Overall	68.68	23.04	67.90	29.65	70.75	30.84	68.10	29.82	67.90	41.08	68.66	30.89
Avg. (TVQA, GSM8K, CSQA, SciQ) on Llama-3.1												
Prob.	74.88	15.02	70.94	32.72	77.15	31.56	70.92	23.10	77.01	35.89	74.14	27.66
$p(\text{True})$	62.31	21.54	65.76	32.21	62.40	31.47	59.59	17.12	56.95	30.22	61.51	26.54
Verb.	64.81	31.76	58.51	35.57	64.50	39.15	60.40	27.40	66.59	38.23	62.92	35.93
Overall	67.26	22.77	65.10	33.55	67.97	34.06	63.72	22.58	66.85	34.78	66.19	29.38

Table 4.5: Experimental results of AUROC (ARC.) and ECE of three confidence estimation methods on four LA datasets on GPT-3.5 and Llama-3.1.

Results regarding Different Confidence Estimations on LA Tasks

Findings: Applying prompt-based confidence estimations is preferable in multilingual tasks for LLM with stronger instruction-following ability. The probability method performs better confidence estimations on relatively weak LLM. The findings provide a direct takeaway about selecting optimal confidence estima-

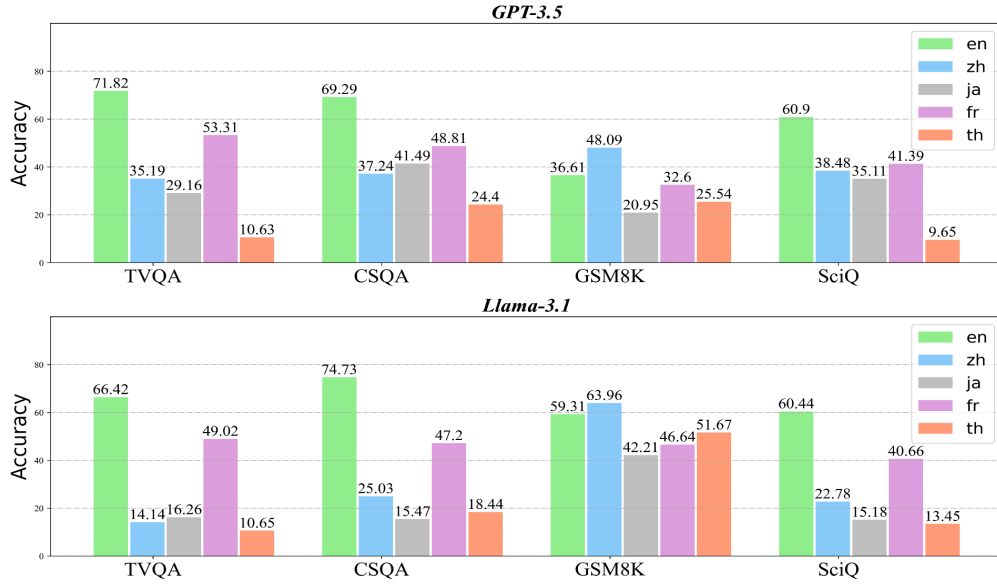


Figure 4.5: Experimental results of Accuracy on four LA datasets on GPT-3.5 and Llama-3.1.

tions of LLMs in multilingual scenarios. In Table 4.5, we highlight the supreme performance in bold among the three methods for each column of each dataset. On GPT-3.5, both p(True) and Verb. yield superior performances than Prob. across all languages averaged on four datasets (ARC.: 70.65, 70.45 vs. 66.02; ECE: 29.66, 28.64 vs. 35.34). p(True) and Verb. have comparable performance in ARC. scores, while Verb. is better calibrated. In contrast, Prob. shows superior performance than p(True) and Verb. and performs more stable on Llama-3.1 (ARC.: 74.14 vs. 61.51 and 62.92; ECE: 27.66 vs. 26.54 and 35.93). p(True) demonstrates better calibration results on languages other than *English*.

Analysis: In Table 4.5, the performance differences between two LLMs can be attributed to GPT-3.5’s strong instruction-following abilities, which benefit the prompt-based multilingual confidence estimation methods Verb. and p(True), but leading to over-confidence in output token probabilities. In contrast, Llama-3.1 cannot stably generate verbalized confidence scores and perform relatively weak instruction-following abilities, but maintains well-calibrated likelihood probabilities during the

pre-training stage for all languages.

Results regarding Different Languages on LA Tasks

Findings: Linguistic dominance is manifested in *English* with superior confidence estimation performances on LA tasks for multilingual LLMs. Prior works only validate the efficacy of prompt-based confidence estimations in *English*. Our findings indicate that the methods are also preferable in other languages and performances fluctuate in different languages. In Table 4.5, ARC. scores are less fluctuating across different languages while ECE in *English* (23.04 and 22.77) performs better than in other languages on both GPT-3.5 and Llama-3.1. We also report the accuracy on LA datasets in Figure 4.5. *English* consistently performs better across all datasets exclusively GSM8K. Generally, prompting in *English* outperforms others, hence responding in *English* on LA tasks can be adequately credible and accurate where linguistic dominance is leading in *English*.

Analysis: Despite the powerful multilingual capacity of LLMs, discrepancies exist in the quantity of distinct linguistic training corpora available for each language. Results in Table 4.5 suggest that ARC. is a metric not significantly related to language usage in LLMs, while the strong performance of ECE in *English* can be attributed to the extensive training corpus or calibrations conducted during training in *English*. As the only middle-resource language, *Thai* exhibits a notably lower level of reliability compared to the other high-resource languages.

Considering consistency check in Table 4.3, the lowest filter rate in GSM8K translation indicates that mathematical reasoning tasks are minimally affected by language bias. As a result, accuracy fluctuations across different languages on GSM8K are relatively small. For that *Chinese* exhibits slightly superior mathematical capabilities compared to *English* on GSM8K on both GPT-3.5 and Llama-3.1 (Accu. 48.09 and 63.96), it is hypothesized that pre-training corpora contain a substantial amount of Chinese mathematical problems.

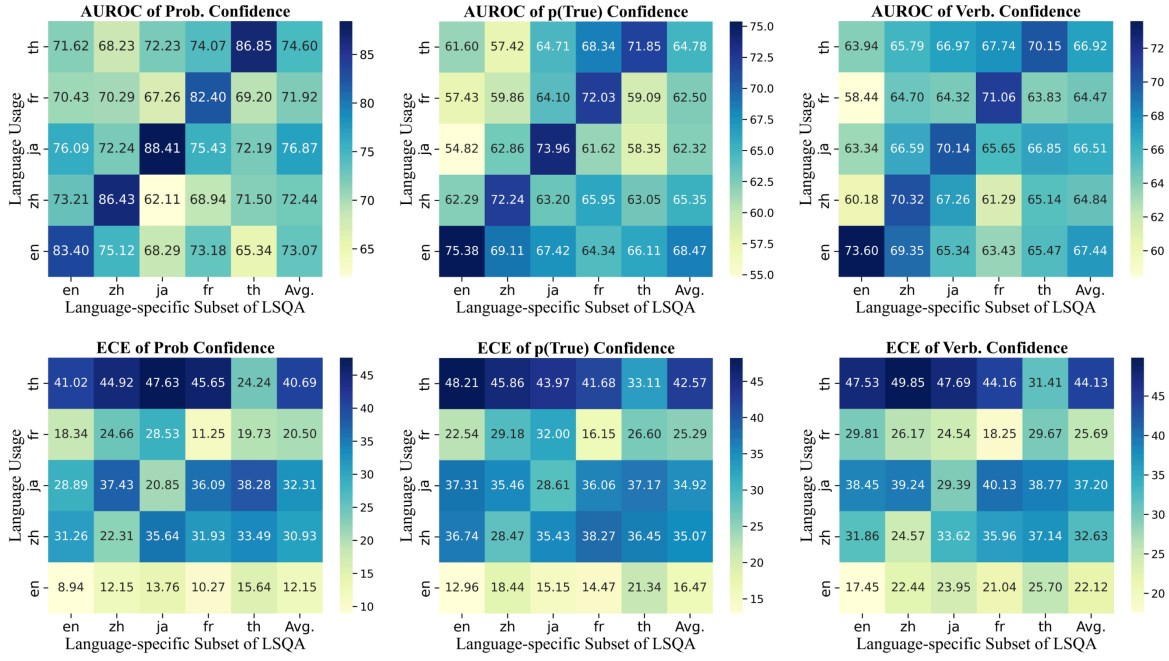


Figure 4.6: Experimental results of AUROC and ECE of three confidence estimation methods on five language-specific subset of LSQA using Llama-3.1.

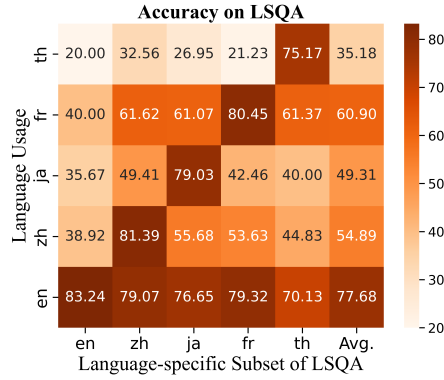


Figure 4.7: Experiments of Accuracy on five LSQA subsets using Llama-3.1.

4.4.3 Experiments on Language-Specific (LS) Tasks

Findings: Applying prompts in query-related language demonstrates linguistic dominance with better multilingual confidence estimations on LS task. In Figure 4.6, the diagonal values of the ECE and ARC. heatmaps of Prob. are more pronounced, indicating that when using Prob. confidence, linguistic dominance

is more apparent compared to $p(\text{True})$ and Verb. . Consequently, we have opted Prob. for subsequent experiments. Additionally, in Figure 4.7, although prompting in *English* performs well and is stable across different subsets, there is a noticeable improvement in accuracy when prompts are related to each subset’s language. While on LA tasks, linguistic dominance is mainly manifested in *English*, on LSQA, linguistic dominance is determined by specific language-related subset.

Analysis: The linguistic dominance of LSQA can be conjectured to stem from the fact that such data pertaining to the language-specific cultural, geographical, or social contexts are already included in the pre-training corpora of their respective languages with higher certainty or confidence, thereby achieving optimal performance when prompting in their respective specific languages.

4.4.4 Native-Tone Prompting for Confidence Calibration

Confidence estimation performance differences caused by linguistic dominance phenomena on the LS task motivate us to explore the improvement method. Inspired by results on each language-specific LSQA subset, we propose a simple yet effective Native-Tone Prompting (NTP) strategy to achieve better confidence estimation performance on the LS task. NTP first prompts LLMs to identify the language context of the question, and then uses that query-related language to answer the question, effectively exhibiting a “native tone” that is more familiar in that language-related context. We present the results of prompting by any single language versus NTP on LSQA in Table 4.6.

Prompt	en	zh	ja	fr	th	NTP
Accu. $\uparrow$	77.68	60.16	44.64	60.90	35.18	79.46
ARC. $\uparrow$	73.07	72.44	76.87	71.92	74.60	77.25
ECE $\downarrow$	12.15	30.93	32.31	20.50	40.69	10.28

Table 4.6: Experimental results of overall Accu., ARC., and ECE on the LSQA dataset by prompting using different languages and our proposed NTP method.

Findings: Prompting LLMs using the query-related language can enhance the reliability of confidence estimations and accuracy on LS tasks, which provides an insight to improve LLMs’ reliability regarding the prompt language usage. In Table 4.6, the experiments demonstrate that NTP better leverages the linguistic dominance, thereby yielding more reliable results than any single language prompt, validating the effectiveness of NTP for calibration on the LS task.

Analysis: Results in Table 4.6 indicate that the multilingual capabilities and reliability of LLMs are still constrained by the imbalanced training corpus among diverse languages. The reliability and accuracy on *English*, serving as the primary training corpora, have not been adequately generalized to other languages. Even for semantically equivalent queries in different languages, the reliability of responses cannot be consistently maintained.

4.5 Benchmark of Mathematical Reasoning Solvability

4.5.1 Honesty on Mathematical Reasoning

Reliable LLMs should identify whether they are able to answer or not [101, 41, 114]. Hence, on knowledge tasks, given a question x , ground truth $\hat{y}$, and an LLM $\mathcal{M}$, identifying known/unknown facts in the generated response $y = \mathcal{M}(x)$ is widely employed for reliability evaluation [227, 6]. However, such evaluations are model-specific on knowledge tasks, while complex reasoning problems may be intrinsically unsolvable. We define LLM reliability on reasoning tasks below.

	Success $\mathcal{S}$	Refusal $\mathcal{R}$	Failure $\mathcal{F}$
Solvable $\mathcal{A}$	$\mathcal{AS}$	$\mathcal{AR}$	$\mathcal{AF}$
Unsolvable $\mathcal{U}$	$\mathcal{US}$	$\mathcal{UR}$	$\mathcal{UF}$

Table 4.7: LLM reliability formulation on reasoning tasks with respect to questions and responses.

As depicted in Figure 4.2 (b) and Table 4.7, we formulate the LLM awareness

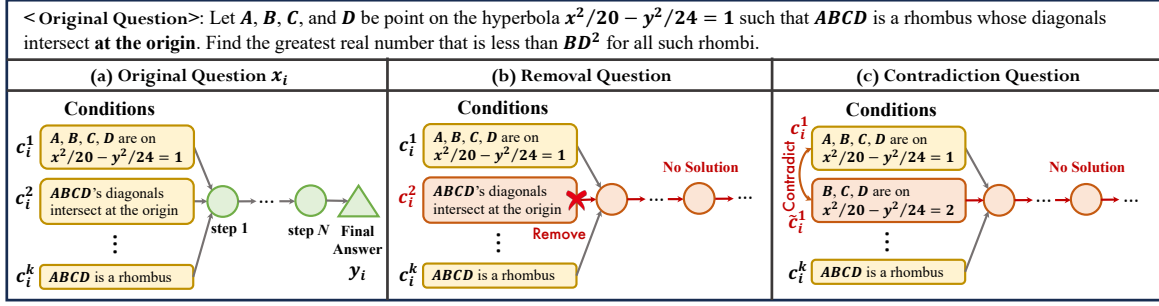


Figure 4.8: Illustrations of (a) a solvable question from AIME and two rewritten schemes by removing one condition (b) or adding one contradictory condition (c).

on reasoning tasks as follows. The questions are categorized along two dimensions - **Solvable** ($\mathcal{A}$) and **Unsolvable** ($\mathcal{U}$) - and LLM responses along three dimensions - **Success** ($\mathcal{S}$), **Refusal** ($\mathcal{R}$), and **Failure** ($\mathcal{F}$). A successful y exactly matches the ground truth $\hat{y}$, which provides the correct answer for $x \in \mathcal{A}$ or stating the problem is unsolvable for $x \in \mathcal{U}$ after step-by-step reasoning. Refused responses express “I don’t know” in y for both $\mathcal{A}$ and $\mathcal{U}$. All other cases are considered as failed. For both $\mathcal{A}$ and $\mathcal{U}$, the preference of y is: $\mathcal{S} > \mathcal{R} > \mathcal{F}$. This formulation offers an insight into further designing evaluation metrics and alignment strategies to enhance LLMs’ reliability in the following parts.

4.5.2 Unsolvable Math Dataset Construction

This section will present the construction process of the **ReliableMath** dataset $\mathcal{D}_r$, including the solvable subset $\mathcal{D}_a$ and construction of the unsolvable subset $\mathcal{D}_u$.

Solvable Data Collection

Mathematical tasks are widely utilized to assess LLMs’ reasoning capabilities [173, 115]. Accordingly, this study employs four representative open-source datasets, spanning from high-school-level data MATH [61] to competitive college-level problems MinervaMath [124] and Olympic-level challenges AIME, AMC [4, 5].

Full sets of 30 AIME and 83 AMC mathematical problems are incorporated, alongside 100 problems randomly sampled from Minerva and 100 from MATH, constituting the solvable subset $\mathcal{D}_a = \{(x_i, \hat{y}_i)\}_{i=1}^N$.

Unsolvable Data Rewriting Types

We develop an unsolvable mathematical dataset for the gap that existing unanswerable problems have exclusively focused on knowledge tasks. To ensure unsolvable problems under the same distribution of solvable data sources, we construct the unsolvable dataset $\mathcal{D}_u$ by rewriting solvable problems in $\mathcal{D}_a$ into unsolvable forms. For a problem x_i with k necessary math conditions, a unique solution $\hat{y}_i$ is derived by reasoning using all conditions $\{c_i^1, \dots, c_i^k\} \in x_i$, as in Figure 4.8 (a). Any alteration to one condition may result in the loss of the unique solution [34]. As in Figure 4.8 (b) and (c), we employ two schemes to synthesize unsolvable cases by 1) *removing* one premise c_i^2 to prevent reasoning steps or 2) introducing a *contradicted* condition $\tilde{c}_i^1$ that leads to logical inconsistency.

Simply modifying a condition cannot ensure that the problems are genuinely unsolvable or that they have solutions that differ from the original. Therefore, we propose a rigorous workflow for unsolvable dataset construction as follows.

Unsolvable Data Construction Workflow

To obtain high-quality unsolvable data, further professional verification is required. Prior work on creating unanswerable knowledge problems has relied solely on human collection [6, 227]. However, math problems present greater challenges, and fully manual rewriting and validation are prohibitively expensive. Hence, we design a data construction workflow with three steps to ensure both efficiency and quality, as illustrated in Figure 4.9.

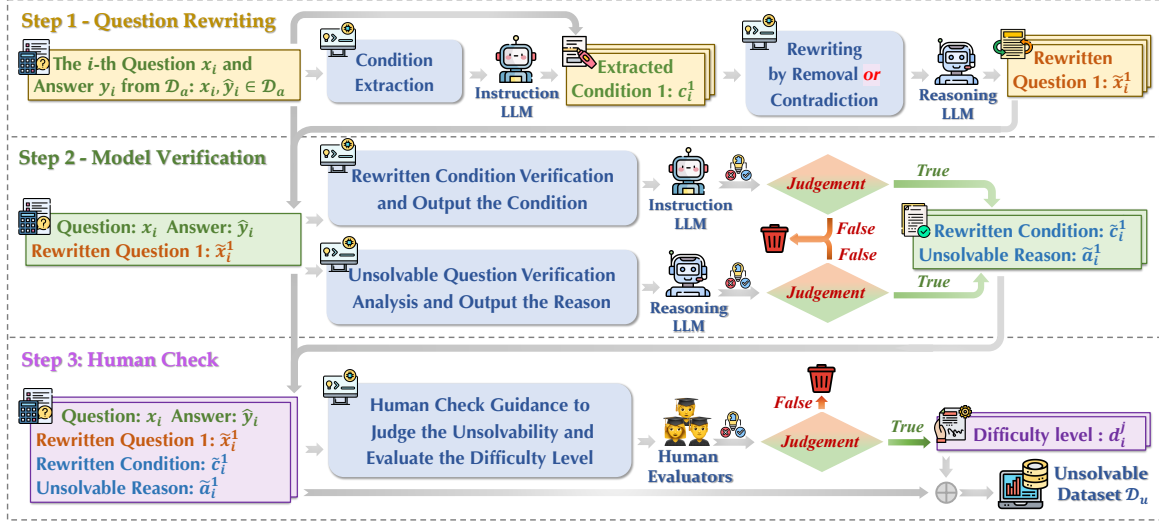


Figure 4.9: Unsolvability data construction workflow.

Step 1: Question Rewriting First, given a solvable question x_i along with the ground truth $\hat{y}_i$ from $\mathcal{D}_a$, we employ an instruction model $\mathcal{M}_I$ (GPT-4o [138]) to extract 1–3 mathematical necessary conditions $\{c_i^j\}_{j=1}$. For each c_i^j , we employ an advanced reasoning model $\mathcal{M}_R$ (Deepseek-R1 [25]) to rewrite x_i by removing the condition c_i^j from x_i *or* adding a condition contradictory to c_i^j into x_i , adhering to (1) keeping other conditions unchanged, and (2) rendering the rewritten problem is unsolvable. $\mathcal{M}_I$ is employed to perform relatively simple tasks such as condition extraction, while $\mathcal{M}_R$ is utilized for complex analytical tasks with chain-of-thought (CoT) [200], ensuring both quality and efficiency of data construction. For each c_i^j , several rewritten questions $\{\tilde{x}_i^j\}_{j=1} \sim \mathcal{M}_R(x_i, \hat{y}_i)$ to x_i are obtained.

Step 2: Model Verification Then, given x_i and $\hat{y}_i$, for each $\tilde{x}_i^j$, $\mathcal{M}_I$ is employed to compare $\tilde{x}_i^j$ and x_i to judge whether $\tilde{x}_i^j$ rewrites other conditions, while $\mathcal{M}_R$ is utilized to analyze whether $\tilde{x}_i^j$ is indeed unsolvable. If both criteria are satisfied, $\tilde{x}_i^j$ is retained; otherwise, filtered out. For retained $\tilde{x}_i^j$, both $\mathcal{M}_I$ and $\mathcal{M}_R$ will output the rewritten condition $\tilde{c}_i^j \sim \mathcal{M}_I(\tilde{x}_i^j, x_i)$ and the unsolvable reason analysis $\tilde{a}_i^j \sim \mathcal{M}_R(\tilde{x}_i^j, x_i, \hat{y}_i)$ to assist human evaluation.

Step 3: Human Check Finally, we format all $\{x_i, y_i, \tilde{x}_i^j, \tilde{c}_i^j, \tilde{a}_i^j\}$ and deliver them to employed experts holding a master’s or higher degree in STEM field for assessment. Experts first determine whether $\tilde{x}_i^j$ meets the two criteria of model verification in Step 2. Then they rate the difficulty level d_i^j of identifying the unsolvability to $\tilde{x}_i^j$: problems with obvious missing or contradictory conditions to determine unsolvability are regarded as simple and set $d_i^j = 0$, while those requiring step-by-step reasoning are regarded as hard with $d_i^j = 1$. The qualified question $\tilde{x}_i^j$ is then formatted and incorporated into the unsolvable subset $\mathcal{D}_u$.

Both $\mathcal{D}_a$ and $\mathcal{D}_u$ are combined to constitute the **ReliableMath** dataset $\mathcal{D}_r$. Data statistics are as follows. We specify detailed principles for difficulty level annotation to minimize human variations arising from subjectivity.

Dataset	$\mathcal{D}_a$	Step 1 & 2			Step 3 ($\mathcal{D}_u$)								
		Rm.	Cont.	Total	Rm.			Cont.			Total		
					$d=0$	$d=1$	Sum	$d=0$	$d=1$	Sum	$d=0$	$d=1$	Sum
AIME	30	70	71	141	24	43	67	22	43	65	46	86	132
AMC	83	185	192	377	47	84	131	60	104	164	107	188	295
MATH	100	233	216	449	77	77	154	75	89	164	152	166	318
Minerva	100	207	201	408	87	98	185	76	96	172	163	192	357
Total	313	695	680	1375	235	302	537	233	332	565	468	632	1102

Table 4.8: Data statistics of $\mathcal{D}_a$, and the removal (Rm.) and contradiction (Cont.) questions after Step 1&2 and 3. In Step 3, we also present the number of data in two difficulty levels ($d=0$ or $d=1$).

4.5.3 Experimental Setup

Models Experiments are conducted on a series of reasoning and instruction LLMs on **ReliableMath** dataset. Reasoning LLMs include **DeepSeek-R1**, **R1-Distill-Qwen** (Distill) 32B, 14B, 7B, 1.5B [25], and **o3-mini** [140]. Instruction LLMs contain **DeepSeek-V3** [24], **Qwen2.5-Math-Instruct** (Qwen2.5) 7B, 1.5B [174], and **GPT-4o** [138]. We employ **standard** math problem-solving prompts and **reliable** prompts that allow identifying unsolvability or refusal. All generations are produced by greedy decoding with Temperature $T = 0$.

Evaluation Metrics We employ two metrics to evaluate LLM reliability on mathematical reasoning tasks: **Success Rate** (*Succ.*) and **Refusal Rate** (*Refu.*). *Succ.* measures the proportion of successful responses $\mathcal{S}$ calculated on solvable ($\mathcal{A}$) and unsolvable ($\mathcal{U}$) questions separately, and then averaging as:

$$Succ.(\mathcal{A}) = \frac{\#\mathcal{AS}}{\#\mathcal{A}}, \quad Succ.(\mathcal{U}) = \frac{\#\mathcal{US}}{\#\mathcal{U}} \quad (4.8)$$

$$Succ. = 0.5 * [Succ.(\mathcal{A}) + Succ.(\mathcal{U})]. \quad (4.9)$$

Refu. represents the proportion of refused responses $\mathcal{R}$ on $\mathcal{A}$ and $\mathcal{U}$ as follows.

$$Refu.(\mathcal{A}) = \frac{\#\mathcal{AR}}{\#\mathcal{A}}, \quad Refu.(\mathcal{U}) = \frac{\#\mathcal{UR}}{\#\mathcal{U}} \quad (4.10)$$

$$Refu. = 0.5 * [Refu.(\mathcal{A}) + Refu.(\mathcal{U})] \quad (4.11)$$

When assessing LLMs' self-awareness on math reasoning, *Succ.* is prioritized, and the higher the *Succ.*, the better the reliability. When achieving comparable *Succ.* for two LLMs, *Refu.* is then considered, with a higher *Refu.* being preferable.

Specifically, assessing the reliability of responses is conducted on both the intermediate reasoning steps $\mathbf{r}$ and the final answer $\mathbf{y}$ for both $\mathcal{AS}$ and $\mathcal{US}$ (referring to Table 4.7). According to [117], we adopt a hybrid strategy of $s = \alpha \cdot s_p + (1 - \alpha) \cdot s_o$ to calculate *Succ.*($\mathcal{A}$) and *Succ.*($\mathcal{U}$), where s_p and s_o denote the model-based process and rule-based outcome assessments respectively. We employ the state-of-the-art GPT-5 [139] as judging model $\mathcal{M}_e$ [48]. For $\mathcal{AS}$, we instruct $\mathcal{M}_e$ to judge the correctness of $\mathbf{r}$ conditioned on $\mathbf{x}$ and $\hat{\mathbf{y}}$ as $s_p = \mathcal{M}_e(\mathbf{r}|\mathbf{x}, \hat{\mathbf{y}})$ where $s_p \in \{0, 1\}$ (1 for True and 0 for False). For $\mathcal{US}$, we guide $\mathcal{M}_e$ to determine whether $\mathbf{r}$ identifies the unreasonableness or illogicality of the unsolvable question given rewritten condition as $s_p \sim \mathcal{M}_e(\mathbf{r}|\tilde{\mathbf{x}}, \mathbf{c})$. The outcome score evaluates the correctness of final answer as $s_o \sim \mathbb{I}(\mathbf{y} \equiv \hat{\mathbf{y}})$ for both $\mathcal{AS}$ and $\mathcal{US}$. To make a balance between s_p and s_o , we set $\alpha = 0.5$ in all experiments. Assessing *Refu.*($\mathcal{A}$) and *Refu.*($\mathcal{U}$) on $\mathcal{AR}$ and $\mathcal{UR}$ is solely confined to detect refusal in $\mathbf{y}$, which can be viewed as giving up the question. We also examine that the GPT-5 judge results are largely consistent with

LLMs	Prompt	Solvable ($\mathcal{A}$)				Unsolvable ($\mathcal{U}$)				Succ.	Refu.
		Succ. ($\mathcal{A}$)		Refu. ($\mathcal{A}$)	Len.	Succ. ($\mathcal{U}$)		Refu. ($\mathcal{U}$)	Len.		
		s_p	s_o			s_p	s_o				
Reasoning LLMs											
DeepSeek-R1	standard	73.81	74.76	0.00	4.09k	59.80	0.00	0.00	6.51k	52.08	0.00
	reliable	73.17	73.49	0.00	3.82k	70.79	54.89	1.76	4.38k	68.08	0.88
o3-mini	standard	64.85	66.44	0.00	1.49k	14.61	0.08	0.00	5.03k	36.50	0.00
	reliable	69.95	71.58	0.64	1.56k	25.78	29.31	0.87	4.18k	49.15	0.75
Distill-32B	standard	69.32	71.24	0.00	4.99k	31.68	0.00	0.00	14.52k	43.06	0.00
	reliable	67.40	68.36	0.00	5.05k	51.53	41.87	0.24	9.36k	57.29	0.12
Distill-14B	standard	66.12	67.08	0.00	6.51k	36.94	0.00	0.00	16.97k	42.52	0.00
	reliable	62.30	62.93	0.00	6.24k	59.61	46.45	0.12	10.94k	57.83	0.06
Distill-7B	standard	60.37	61.02	0.00	6.16k	1.74	0.00	0.00	6.50k	30.77	0.00
	reliable	57.20	57.52	0.00	6.22k	1.99	0.26	0.00	6.59k	29.24	0.00
Distill-1.5B	standard	40.26	41.86	0.00	9.02k	2.74	0.00	0.00	9.55k	21.20	0.00
	reliable	38.98	39.62	0.00	9.35k	2.18	0.00	0.00	9.69k	20.21	0.00
Instruction LLMs											
DeepSeek-V3	standard	64.22	65.49	0.00	1.43k	27.68	0.09	0.00	1.90k	39.37	0.00
	reliable	65.50	66.45	0.00	1.35k	43.29	37.75	0.86	1.54k	53.25	0.43
GPT-4o	standard	42.82	46.01	0.00	0.72k	10.24	0.10	0.00	0.78k	24.79	0.00
	reliable	42.16	46.01	0.64	0.57k	27.34	33.50	6.95	0.65k	37.23	3.79
Qwen2.5-7B	standard	44.73	49.84	0.00	0.86k	1.43	0.00	0.00	0.87k	24.00	0.00
	reliable	44.73	50.47	0.00	0.83k	3.38	2.73	0.00	0.88k	25.32	0.00
Qwen2.5-1.5B	standard	41.22	45.05	0.00	0.71k	1.89	0.00	0.00	0.72k	22.04	0.00
	reliable	39.63	42.18	0.00	0.72k	2.17	1.44	0.00	0.75k	21.36	0.00

Table 4.9: Performance of Success Rate (Succ.) / Refusal Rate (Refu.) and Response Length (Len.) on Solvable ($\mathcal{A}$) and Unsolvable ($\mathcal{U}$) subsets on a series of reasoning and instruction LLMs.

the human assessments, validating the effectiveness of the LLM judge.

4.5.4 Experimental Findings and Analysis

Experimental Findings on ReliableMath Dataset

Experiments are presented in Table 4.9 with several findings listed below.

a. LLMs can occasionally recognize the illogicality of unsolvable problems, but fail to directly identify the unsolvability or refuse to answer, and instead fabricate reasoning steps with substantial tokens, diminishing the reliability and aggravating the overthinking. In Table 4.9, all LLMs using standard prompts rarely exhibit the capability of explicitly identifying unsolvability or refusing, with both s_o of $Succ.(\mathcal{U})$ and $Refu.(\mathcal{U})$ close to 0. The generation lengths of unsolvable problems using standard prompts are especially lengthy. In Table 4.10, we examine

All		Remove		Contradict	
wait	59.59	lack	4.00	contradict-	4.44
alternative	13.60	los-	3.85	opposite	4.25
correct	8.12	mistake	3.05	mistake	3.73
again	4.86	assum-	2.87	incorrect	1.92
change	4.73	miss-	1.83	inconsisten-	0.83

Table 4.10: The keywords statistics and their frequencies in the responses on all experiments using standard prompts, where “All” represents the keywords related to deep thinking behavior, while “Remove” and “Contradict” denote the respective keyword patterns to identify issues in two types of unsolvable problems. *pre-* denotes the words sharing the prefix *pre**. For example, *assum-* includes *assume*, *assumption*, and *assuming*.

the outputs and list the keywords related to deep thinking, reflection behaviors, and typical patterns to recognize unsolvability of “removal” and “contradiction” problems, suggesting that LLMs may recognize potential issues of unsolvable data during reasoning but still fabricate reasoning and hallucinated responses. This behavior stems from the fact that during training, LLMs are always encouraged to provide a certain answer given any questions. Hence they cannot critically identify the unsolvability, making their reliability are highly susceptible to attack.

b. After employing reliable prompts that enable LLMs to critically identify the solvability or refuse to answer, the reliability on solvable problems remains stable, but improves significantly on unsolvable problems, albeit still falling short of solvable problems. As in Table 4.9, after using reliable prompts, $Succ.(A)$ on solvable problems remains or slightly fluctuates due to prompt sensitivity, but significantly increases on unsolvable problems in approximately 20-50 in s_p and s_o of $Succ.(U)$ on typically used LLMs like DeepSeek-R1 and GPT-4o, while $Refu.(A)$ and $Refu.(U)$ also marginally improve. The observation implies that reliable prompts can alleviate the unreliability issue on unsolvable cases without sacrificing the performance on solvable data. Moreover, the sequence lengths after using reliable prompts also decrease noticeably, suggesting that reliable prompts can also mitigate the overthinking issue.

c. For larger-size LLMs, the reliability of reasoning LLMs generally outper-

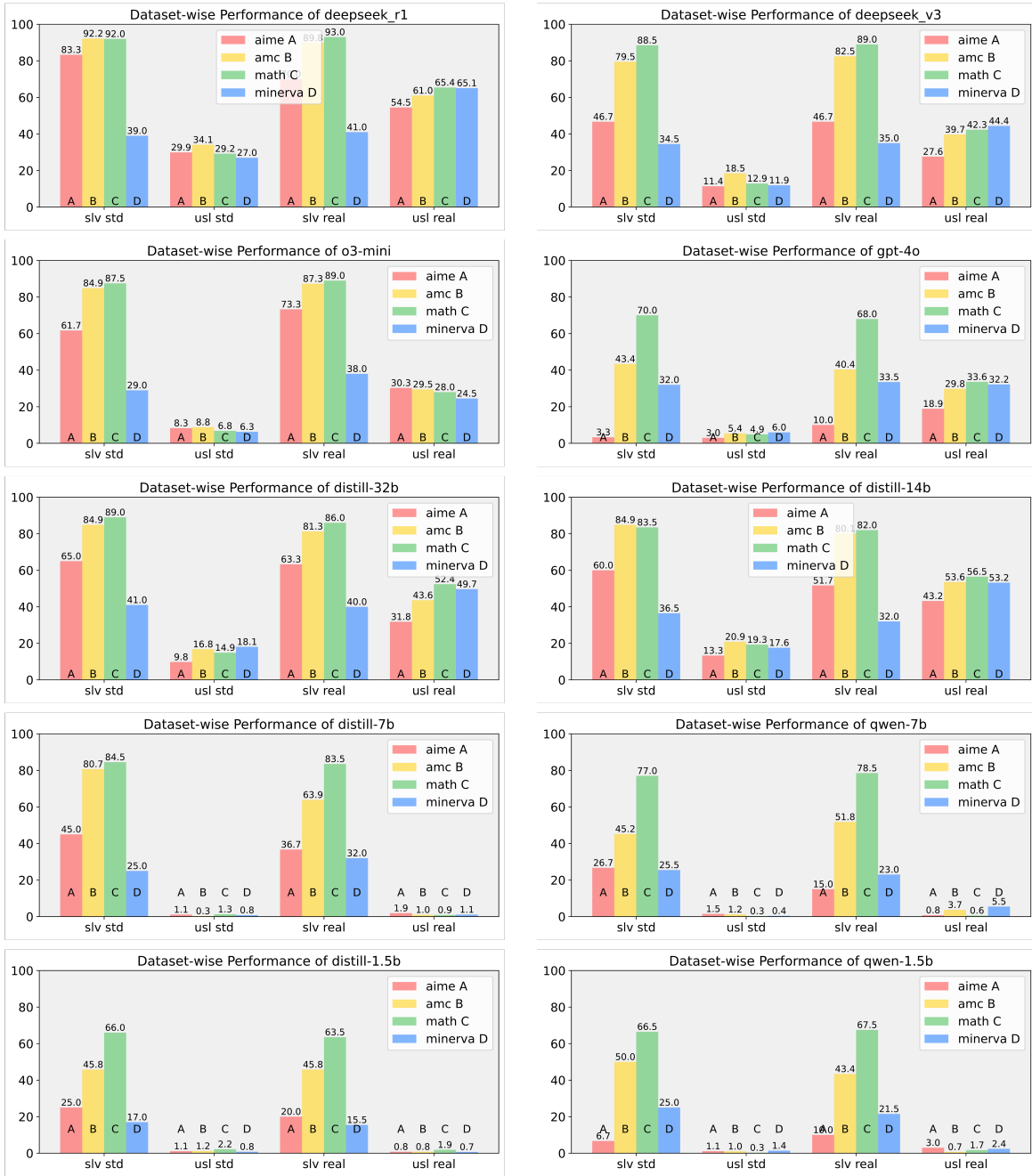


Figure 4.10: Results of Success Rate (Succ.) on o3-mini on four test sets on both solvable (slv) and unsolvable (usl) data using standard (std) and reliable (real) prompts.

forms instruction LLMs when using reliable prompts. Conversely, smaller reasoning LLMs demonstrate inferior reliability to instruction LLMs. As in Table 4.9, DeepSeek-R1 and o3-mini present greater reliability compared to DeepSeek-V3 and

GPT-4o in $Succ(\mathcal{U})$. However, Distill-7B and -1.5B demonstrate slightly weaker reliability than their counterparts, Qwen2.5-7B and -1.5B, particularly in $Refu(\mathcal{U})$. Such relatively small reasoning LLMs are prone to excessive invalid overthinking ($Len.>10k$), severely undermining LLM reliability.

d. Despite performing well on solvable problems, relatively small LLMs can hardly detect unsolvability and show no improvement even when using reliable prompts. Whether small-sized reasoning LLMs (Distill-7B, -1.5B) or the instruction LLMs (Qwen2.5-7B, -1.5B) achieve $Succ(\mathcal{A})$ of 40-60 on solvable problems, but $Succ(\mathcal{U})$ are always almost 0 on unsolvable data even using reliable prompts, suggesting that small LLMs’ general ability and reliability are extremely weak.

e. For math problems at different levels, LLMs exhibit high sensitivity to solvable problems but remain invariance on unsolvable data. In Figure 4.10, for solvable problems (slv), LLMs achieve higher $Succ(\mathcal{A})$ on relatively simple MATH compared with the more challenging AIME, AMC, and Minerva, employing either standard or reliable prompts. In contrast, $Succ(\mathcal{U})$ remain comparable across all levels, suggesting that LLMs’ ability to identify unsolvable data is independent of the levels of the corresponding solvable data.

Dataset Analysis of ReliableMath Dataset

It is observed that LLMs are prone to fail on removal questions or those with difficulty level $d=1$. We thus make some analysis of our constructed unsolvable data regarding different rewriting schemes and difficulty levels, as in Figure 4.11.

Rewriting Schemes (Removal & Contradiction) We present two unsolvable rewriting schemes of removal and contradiction. As in Figure 4.11 (a), we showcase the results of $Succ(\mathcal{U})$ by on two rewriting schemes on several LLMs using reliable prompts. Generally, $Succ(\mathcal{U})$ on the contradiction problems is larger than the removal problems, indicating that LLMs are more adept at identifying the unsolvability of contradiction problems. As observed in generated outputs, LLMs tend to

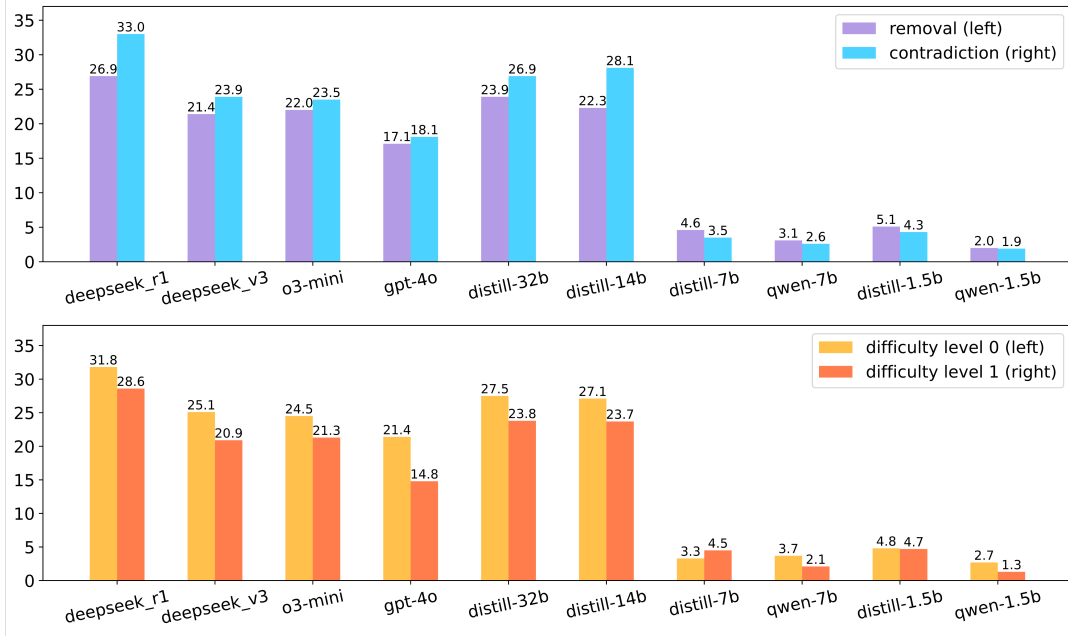


Figure 4.11: Illustrations of $Succ(\mathcal{U})$ of unsolvable problems regarding (a) two rewriting schemes (removal & contradiction), and (b) two difficulty levels labeled by experts (0: simple & 1: hard) on several typically used LLMs using reliable prompts.

fabricate missing conditions for removal problems with lengthy reasoning, resulting in verbose yet hallucinated responses, thus yielding fewer successful responses.

Difficulty Level (0 & 1) As in Step 3 of Section 4.5.2, human experts label problems that are intuitively judged as unsolvable with a difficulty level of 0, whereas problems with a difficulty level of 1 require step-by-step reasoning to determine the unsolvability. In Figure 4.11 (b), we demonstrate $Succ(\mathcal{U})$ with two difficulty levels (0: simple & 1: hard). $Succ(\mathcal{U})$ with difficulty level $d = 1$ is lower than problems of $d = 0$ across all LLMs, suggesting that LLMs are consistently correlated with human experts in identifying unsolvability of problems.

4.5.5 Honesty Alignment for Math Reasoning

Since such small LLMs exhibit worse reliability, we further propose an alignment strategy and conduct experiments in the following part.

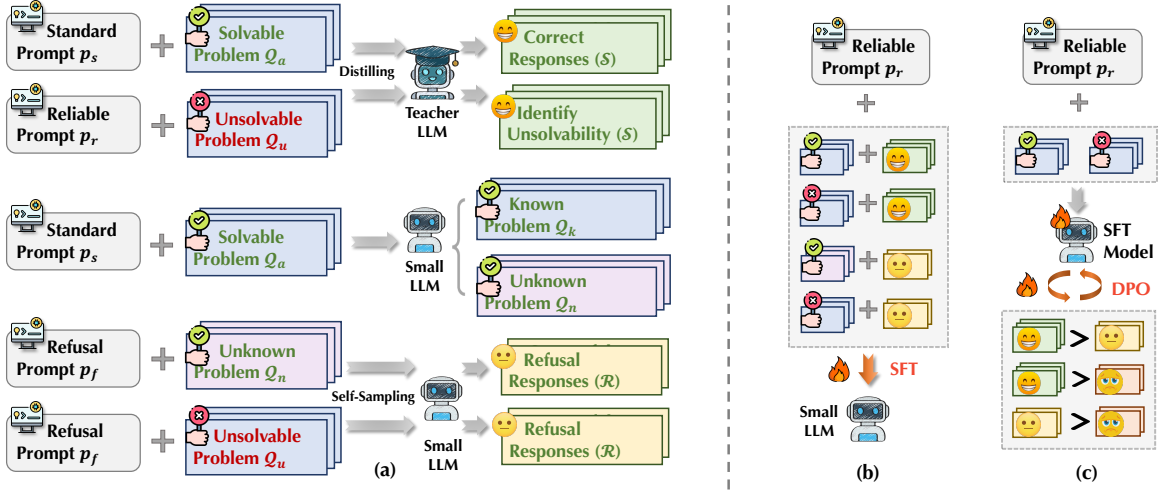


Figure 4.12: Illustrations of (a) reliability alignment training data generation for both solvable and unsolvable problems, (b) SFT-based alignment using the obtained training data, and (c) DPO-based alignment on the obtained SFT model.

Honesty Alignment Strategy for Mathematical Reasoning

To enhance the reliability of small LLMs $\mathcal{M}_S$, we first obtain alignment training data. Given the training set Q_a with open-source solvable math problems, we synthesize unsolvable problems in Q_u using the unsolvable data construction process. Due to the high cost of human checks on substantial training data, Step 3 is omitted. The pass rate of Step 3 is approximately 80.14%; therefore, the quality of synthesized unsolvable problems after Step 2 is regarded as applicable enough. We then generate successful and refused responses for problems in Q_a and Q_u .

Due to the limited ability of $\mathcal{M}_S$ to generate successful responses including high-quality reasoning paths, we employ an advanced teacher LLM $\mathcal{M}_T$ (DeepSeek-R1) for distillation [214]. As in Figure 4.12 (a), we employ the standard prompt p_s for Q_a and the reliable prompt p_r for Q_u to distill successful responses $\mathcal{S}$ from $\mathcal{M}_T$, with a sampling number $K = 4$ to ensure that most problems yield at least one $\mathcal{S}$ response. To prevent LLM from refusing to answer questions it can solve, for Q_a , we first infer on small LLMs to select questions that are incorrectly answered and add to the unknown set Q_n . Then we employ rejection sampling [232] on $\mathcal{M}_S$ to

Methods	In-Domain (ReliableMath)						Out-of-Domain (KUQ)				
	Succ.($\mathcal{A}$)		Refu.($\mathcal{A}$)	Succ.($\mathcal{U}$)		Refu.($\mathcal{U}$)	Succ.($\mathcal{A}$)	Refu.($\mathcal{A}$)	Succ.($\mathcal{U}$)		Refu.($\mathcal{U}$)
	s_p	s_o		s_p	s_o				s_p	s_o	
DeepSeek-R1											
reliable	73.17	73.49	0.00	70.79	54.89	1.76	48.45	8.16	71.66	8.79	57.92
Qwen2.5-1.5B											
standard	41.22	45.05	0.00	1.89	0.00	0.00	4.74	0.07	17.08	17.76	0.12
reliable	39.63	42.18	0.00	2.17	1.44	0.00	3.69	2.50	23.33	4.64	5.45
align-sft	48.24	48.89	9.58 [†]	17.15	22.69 [†]	9.56 [†]	3.16	10.87 [†]	31.36 [†]	25.23 [†]	12.48 [†]
align-dpo	46.00	46.64	5.11 [†]	21.87 [†]	27.22 [†]	8.31 [†]	1.57	8.37 [†]	25.44 [†]	30.26 [†]	7.88 [†]
Distill-1.5B											
standard	40.26	41.86	0.00	2.74	0.00	0.00	0.13	0.00	11.70	0.06	0.00
reliable	38.98	39.62	0.00	2.18	0.00	0.00	0.26	0.07	11.88	0.12	0.00
align-sft	55.27 [†]	56.23 [†]	7.99 [†]	26.13	25.05 [†]	9.25 [†]	2.51	9.86 [†]	16.74 [†]	21.65 [†]	10.14 [†]
align-dpo	53.05	53.99	4.79	27.95 [†]	29.31 [†]	8.60	0.43	6.88 [†]	22.14 [†]	24.45 [†]	7.12 [†]

Table 4.11: Results of Success Rate (Succ.) and Refusal Rate (Refu.) on both in-domain math task and out-of-domain knowledge QA task including both solvable and unsolvable problems. Two relatively small LLMs of Distill-1.5B and Qwen2.5-1.5B are employed using standard and reliable prompts, as well as our proposed alignment strategy including both SFT and DPO. DeepSeek-R1 is the teacher model. † denotes that significant improvements are obtained using the alignment method over two baseline prompting methods.

sample model-specific refusal responses $\mathcal{R}$ for $\mathcal{Q}_n$ and $\mathcal{Q}_u$ with $K = 1$ and a 2-shot refusal prompt p_f to accelerate sampling. Since $\mathcal{S}$ is more preferable than $\mathcal{F}$, we only need a few refusal responses to enable $\mathcal{M}_s$ to appropriately learn the refusal capability. The generated data are incorporated into the training set $\mathcal{D}^{(t)}$ to align with the reliability for $\mathcal{M}_s$ as follows.

In Figure 4.12 (b), the constructed reliable supervision data $\mathcal{D}^{(t)}$ is used for Supervised Fine-Tuning (SFT) to train $\mathcal{M}_s$ to critically identify the unsolvability or refuse to answer while keeping their ability on solvable data. Following the initial SFT phase, as in Figure 4.12 (c), we sample and synthesize high-quality preference data based on LLMs’ outputs, which is subsequently utilized to further refine LLM using Direct Preference Optimization (DPO) [149], progressively aligning its behavior with the desired reliability preferences.

Honesty Alignment Setup for Mathematical Reasoning

We conduct experiments on Qwen2.5-1.5B and Distill-1.5B using the proposed alignment method. We collect solvable problems from open-source AIME, AMC,

and MATH training data into $\mathcal{Q}_a$. To avoid potential data leakage due to the fundamental relation between train and test data which could be a confounder, questions in the $\mathcal{Q}_a$ with high similarity to the solvable test set $\mathcal{D}_a$ were filtered out. For each $x_t \in \mathcal{Q}_a$, we calculate the cosine similarity scores $\{s_i\}_{i=1}^N$ of x_t with all $\{x_i | x_i \in \mathcal{D}_a\}_{i=1}^N$, and if $\max\{s_i\}_{i=1}^N > 0.7$, x_t is removed. The threshold 0.7 is set empirically to filter out the top 10% problems with higher similar scores. When constructing unsolvable problems of $\mathcal{Q}_u$, only one condition is extracted to maintain the size of $\mathcal{Q}_u$ comparable to $\mathcal{Q}_a$. Distilled responses with reasoning steps are used to train Distill-1.5B, and we remove the reasoning steps to train Qwen2.5-1.5B. The problems are formatted to constitute the alignment set $\mathcal{D}^{(t)}$ ($|\mathcal{D}^{(t)}| = 10k$).

LLMs after alignment are tested on both in-domain (ID) math tasks of **ReliableMath** and out-of-domain (OOD) knowledge QA task derived from the **KUQ** dataset [6]. KUQ contains open-source answerable knowledge QAs like TriviaQA [80], as well as real-world unknown questions collected from crowd-source workers. Specifically, we omit categories of controversial and ambiguous questions as they are not actually unanswerable but have various candidate answers. When inference, we also employ standard and reliable knowledge QA prompts for inference. Known questions can be directly evaluated using the F1 score on the outputs, while unknown questions remain essential to assess whether LLMs’ reasoning process can identify the unsolvability aspects as math problems.

Experiments on Mathematical Reasoning and Knowledge QA

Reliability improvements suggest that the proposed alignment method enables LLMs not only to identify the unsolvability, but also to critically think and identify their knowledge or capability boundary before answering any questions. In Table 4.11, both SFT and DPO can effectively improve the reliability of two LLMs on either ID math or OOD knowledge tasks, with significant improvements of 20 $\sim$ 30 in $Succ.(\mathcal{U})$ over two baseline prompting methods. Advancements

in $Refu.(\mathcal{A})$ and $Refu.(\mathcal{U})$ indicate that aligned LLMs acquire the refusal ability on all problems, which is rare when using prompting methods. In addition, although DPO alignment exhibits larger gains on $Succ.(\mathcal{U})$, it presents inferior generalization on other reliability metrics compared with SFT.

4.6 Summary

In this chapter, we analyze the issue of dishonesty of LLMs. We formulate the honesty evaluation in two aspects of self-awareness and confidence estimation, and emphasize the importance of building specific benchmarks for honesty evaluation for LLMs. We propose two high-quality and challenging benchmarks regarding multilingual confidence estimation (**MlingConf**) and mathematical reasoning solvability (**ReliableMath**) using two data construction workflows with expert check. The proposed two benchmarks serve as a valuable and noteworthy benchmark with extensive key findings by conducting experiments on various LLMs. Finally, we propose two respective optimization methods of native-tone prompts and an honesty alignment strategy to improve the honesty on two benchmarks. The introduction of two benchmarks and insights paves the way for future research on LLM honesty. In the next chapter, we will introduce the consistency of LLMs.

Chapter 5

Building Reliable LLMs regarding Consistency

One principle runs through all.

一以貫之

— Confucius 孔子《論語·里仁》

Chapter 5 is organized as follows. Section 5.1 presents an overview of the LLM inconsistency issue and our investigations on LLM-based search agents. Section 5.2 examines the contextual inconsistency in detail. Section 5.3 presents the preliminary knowledge of this chapter, including LLM-based search agents, previous methods to mitigate contextual inconsistency, Markov Decision Process for Multi-turn LLM Generation, etc. Section 5.4 demonstrates our strategy to track the cause of the contextual inconsistency problem and existing mitigation methods of LLM search agents. Section 5.5 proposes to incorporate context refinement into RL training pipelines of search agents. Finally, Section 5.6 summarizes the chapter.

5.1 Overview

Generally, LLMs suffer from two types of inconsistency issues in one generation. The first arises from knowledge conflicts inherent in the queries or prompts, which interfere with the LLMs’ generation process [213, 220]. The second refers to the unfaithfulness issue, where LLMs fail to follow user instructions and deviate from the original intent of prompts [2, 106]. With the advancement of LLM-based agents, LLMs need to engage in multiple interactions with external environments within a single task [53, 49]. Such multi-turn generation further exacerbates the inconsistency problem: the external information received at each step may itself be conflicting, and the LLM’s outputs across steps may also become inconsistent with its prior responses, also referred to as **contextual inconsistency** [69, 213]. Such contextual inconsistencies constitute a primary factor limiting the upper bound of capabilities to build reliable LLM-based agents. Therefore, in this chapter, we investigate contextual inconsistency in one typical agent scenario as follows.

Recently, LLMs are empowered to invoke search engines to retrieve external knowledge, essentially training them as multi-turn search agents that iterate on retrieval and generation until tasks are solved [189, 74, 67]. However, the contexts of multi-turn search agents are lengthy and complex, encompassing the question, multi-round search queries, retrieved documents, and reasoning steps [78], which may include inconsistent information from the external environment. For example, the retrieved set of documents in each turn would inevitably introduce irrelevant or conflicting information that distracts LLMs from the correct documents, resulting in contextual inconsistency, as illustrated in Figure 5.1.

Therefore, we conduct a systematic study on contextual inconsistency in multi-turn search agents. First, we investigate *which parts of the context of search agents will contribute to the contextual inconsistency*. Then *we introduce several methods to explore how to refine the contexts to mitigate the inconsistency*. Finally, *we leverage context refinement in RL training pipelines of search agents for further performance improvements*.



Figure 5.1: Examples of the search agent with (a) contextual inconsistency in the retrieved documents and (b) refined contexts. The question derives from PopQA [122], and we employ Qwen2.5-7b-Instruct [148] as the foundation LLM of the search agent.

In light of the above questions, we investigate the context inconsistency issue on multi-turn LLM-based search agents with respect to **accuracy** and **efficiency** across a series of closed-book QA benchmarks. We first compare the performances of multi-turn search agents with different inputs by masking specific parts of history contexts. Results identify that contextual inconsistency primarily derives from the latest retrieved document of search agents and slightly arises from previous search queries and documents. Then, we explore a series of contextual inconsistency mitigation strategies. Based on the exploration, we propose to distill a context refinement dataset comprising both retrieved documents and refined texts, which contains the most critical information in documents to the search query. Therefore, we train a context refiner using the dataset, which can mitigate contextual inconsis-

tency for multi-turn search agents as exemplified in Figure 5.1 and improve both efficiency and accuracy. Finally, we further incorporate the context refiner into the reinforcement learning (RL) training pipelines of search agents. Experiments demonstrate that training multi-turn search agents with dynamically refined context achieves significant performance improvements regarding both the accuracy and efficiency over other training baselines, effectively mitigating the contextual inconsistency issue of LLM-based search agents.

5.2 Problem Statement

The contexts of multi-turn search agents are lengthy and complex, encompassing the question, multi-round search queries, retrieved documents, and reasoning steps [78], which may include inconsistent information from the external environment. For instance, the retriever always returns a set of documents to ensure coverage of the search query, which also introduces irrelevant or conflicting documents into the context [29], resulting in contextual inconsistency in multi-turn generations [23, 58, 54, 77].” The context inconsistency presented in each round may distract LLMs from the correct documents and persistently degrade subsequent generation quality [104, 92], thereby hindering the efficiency and accuracy of LLM-based search agents during multi-turn generations.

As in Figure 5.2, the gaps between “**recall rate**” and “**recall accuracy**” demonstrate that search agents always retrieve the documents containing the useful information but fail to generate the correct answer, highlighting the context interference effect of accurate knowledge expression of search agents. “recall rate” denotes the questions calling external knowledge containing helpful information, while “recall accuracy” refers to actually solved questions using external knowledge, and “accuracy” represents successfully answered questions through the combined use of LLM internal knowledge and external knowledge. The gaps between “recall rate” and “recall accuracy” / “accuracy” exemplify the negative impact of context

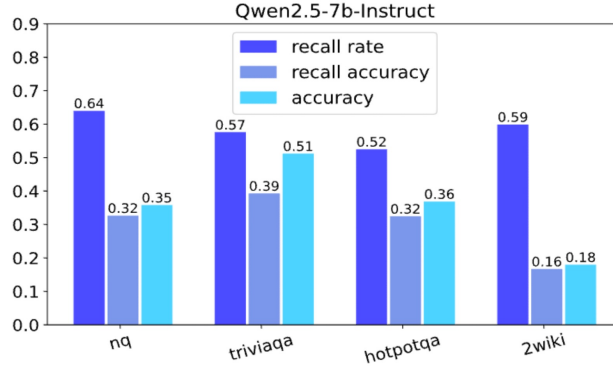


Figure 5.2: Demonstration of how context inconsistency affects the performance of LLM-based search agents. (“recall rate”= N_r/N denotes the proportion of questions for which the retrieved documents contain the correct answer (N_r) among all questions (N). “recall accuracy”= N_{rc}/N refers to the proportion of correctly answered questions (N_{rc}) in N_r , out of all questions (N). “accuracy” represents the proportion of all correctly answered questions N_c out of the questions (N).)

interference. Moreover, distractions from irrelevant information will incur extra retrievals and inference costs, degrading the efficiency of search agents.

Prior studies to mitigate context interference have predominantly centered on dialogue systems [69] and retrieval-augmented generation (RAG) [47, 133, 231], while largely overlooking techniques on multi-turn search agent settings. Therefore, we systematically study the *contextual inconsistency* issue on search agents with three research questions (**RQ**):

- i) Which parts of contexts will contribute to contextual inconsistency for search agents?
- ii) How to refine contexts of search agents to mitigate contextual inconsistency?
- iii) Can leveraging context refinement in RL training pipelines of search agents yield further performance improvements to mitigate contextual inconsistency?

5.3 Related Work and Preliminary

5.3.1 LLM-based Search Agent

Although LLMs exhibit impressive capabilities, they often lack updated or domain-specific knowledge [144, 103], which undermines LLMs’ reliability. Therefore, search engines [236] are widely integrated to provide external evidence. A common paradigm is retrieval-augmented generation (RAG) [43, 97], in which a search engine retrieves documents relevant to the search query and feeds them into LLMs. More recent work treats search engines as interactive tools [156], prompting or fine-tuning LLMs to act as search agents [210, 35, 197, 100]. Approaches such as IRCoT [181] and ReAct [224] employ prompting to interleave reasoning with iterative search calls, whereas Toolformer [156] applies supervised fine-tuning (SFT) to strengthen search capabilities. Search-R1 [78] optimizes LLMs to produce high-quality trajectories through real-time, multi-turn search interactions using RL. Despite these advances, prior studies have largely focused on eliciting LLMs to output accurate knowledge in reasoning paths despite given lengthy contexts with noise, while overlooking the contextual inconsistency in multi-turn search interactions, which may cascade across subsequent actions, degrading the reliability and efficiency of search agents.

5.3.2 Contextual Inconsistency Mitigation

LLMs are highly sensitive to input contexts, leaving them vulnerable to noise or irrelevant content that can degrade output quality [208, 154, 197]. Existing strategies to mitigate this issue fall into three main categories. **(1) Key Information Extraction** identifies and preserves the most critical or relevant information in contents as in dialogue state tracking [69] or RAG reranking [47, 133, 231]; **(2) Compression method** condenses lengthy input sequences into summaries or latent states to weaken noise [76, 225, 104]. **(3) Prompt-based method** explicitly instructs

LLMs to disregard irrelevant content [151]. However, Key Information Extraction relies on a task-specific schema. Compression may distort key information and incur extra compression cost. Prompt-based methods are sensitive to prompt design. More importantly, these issues are exacerbated in multi-turn scenarios, where the large volume of retrieved documents after several turns introduces more noise.

5.3.3 Specific Notations

The definitions of specific notations of this chapter are summarized in Table 5.1.

5.3.4 Markov Decision Process for LLMs

The process of an LLM agent $\mathcal{M}_\theta$ with $\mathcal{K}_I \in \theta$ interacting with an environment E (referred to $\mathcal{K}_E$) to complete a task can be regarded as a Markov Decision Process: $(\mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{O}, \mathcal{R})$. Initially, a specific task x is provided as the initial environmental state. Assuming the interaction proceeds in N rounds and in round n ($n < N$), the LLM agent receives observation $\mathbf{o}_n \in \mathcal{O}$ and takes action $\mathbf{a}_n \in \mathcal{A}$. The state $\mathbf{s}_n$ at round n is the history context of all preceding concatenated sequences $\mathbf{s}_n = (x, \mathbf{a}_0, \mathbf{o}_1, \mathbf{a}_1, \dots, \mathbf{a}_{n-1}, \mathbf{o}_n) \in \mathcal{S}$. $\mathcal{M}_\theta$ is responsible for deciding $\mathbf{a}_n$ based on $\mathbf{s}_n$: $\mathbf{a}_n \sim \mathcal{M}_\theta(\cdot | \mathbf{s}_n)$, and a retriever $\mathcal{E}$ interacts with E to determine the state transition $\mathcal{T}$. Upon task completion after N rounds, the trajectory is characterized as $\tau = (x, \mathbf{s}_0, \mathbf{a}_0, \mathbf{o}_1, \dots, \mathbf{a}_{N-1}, \mathbf{o}_N, \mathbf{a}_N)$ where the final response $\mathbf{y}$ is parsed from $\mathbf{a}_N$. Finally, a reward score r is provided by the reward model $\mathcal{R}$ by comparing the generated answer $\mathbf{y}$ and the ground-truth reference $\hat{\mathbf{y}}$ as $r = \mathcal{R}(\mathbf{y}, \hat{\mathbf{y}})$.

For multi-turn search agents, the i -th observation $\mathbf{o}_i$ is generally a list of retrieved Top- K documents $\mathbf{d}_i = [\mathbf{d}_{i,1}, \mathbf{d}_{i,2}, \dots, \mathbf{d}_{i,K}]$ returned by the retriever $\mathcal{E}(\mathbf{q}_{i-1} | \mathcal{K}_b)$ where $\mathbf{q}_{i-1}$ is the search query generated in the previous round and $\mathcal{K}_b$ is the external knowledge base as $\mathcal{K}_E$. The action $\mathbf{a}_i$ includes a thinking step $\mathbf{t}_i$ and a search query $\mathbf{q}_i$. Following [78], the system instruction guides the LLM to encaps-

Notation	Description
Markov Decision Process of Agent (Section 5.3.4)	
$\mathcal{S}$	Set of states where i -th state $s_i \in \mathcal{S}$.
$\mathcal{A}$	Set of actions where i -th action $a_i \in \mathcal{A}$.
$\mathcal{O}$	Set of observations where i -th observation $o_i \in \mathcal{O}$.
$\mathcal{T}$	Set of state transition functions where $T(s_{n+1} s_n, a_n) \in \mathcal{T}$.
$\mathcal{R}$	Reward model where $r = \mathcal{R}(\cdot)$.
$\mathcal{E}$	External retriever or search engine.
x	Input query or task.
s_i	i -th state including query, and previous actions and observations.
τ	Completed trajectory to the query by LLM.
y	LLM generated final answer of τ .
$\hat{y}$	Ground-truth reference answer to query x .
K_b	External knowledge base.
d_i	Retrieved Top-K documents $[d_{i,1}, \dots, d_{i,K}]$ at round i .
q_i	Search query of a_i in round i .
p_i	Thinking step of a_i in round i .
Context Refiner for Search Agent (Section 5.5.3)	
$\mathcal{F}$	Context refiner.
$\tilde{d}_i$	Refined documents with critical information extracted from d_i to q_{i-1} .
$\mathcal{D}_d$	Distill dataset.
$\mathcal{D}_c$	Context refinement dataset.
$\mathcal{M}_\pi$	Foundation LLM.
$\mathcal{M}_T$	Advanced teacher model.
$\mathcal{L}_\pi^{\text{SFT}}$	Loss function to train context refiner using SFT.
Context Refinement for Search Agent Training (Section 5.5)	
$\mathcal{L}_\pi^{\text{CRRL}}$	Loss function to train policy model based on GRPO.
$\mathcal{M}_\pi$	Policy model.
$\mathcal{M}_{\text{ref}}$	Reference model.
G	Number of rollouts in one group.
ϵ	Clipping ratio.
β	Coefficient for the KL divergence.
A_j	Advantage estimate based on the group-relative rewards of the trajectory.
$\mathcal{D}_t$	Training dataset for GRPO.

Table 5.1: Summarized notations in Chapter 5.

sulate its search query between specially designated search tokens (`<search> ... </search>`) and enclose retrieved documents within retrieval tokens (`<information> ...</information>`). The process terminates once LLM produces its final re-

sponse enclosed between the specified answer tokens (`<answer> ...</answer>`). Both q_i and d_i are appended to the context in each turn. When generating t_i , all the preceding sequences in s_i are fed into $\mathcal{M}_\theta$ and contribute to a_i . However, not all previous documents and search queries are pertinent to the current thinking step, and the inclusion of irrelevant content can introduce context inconsistency, impairing the LLMs’ efficiency and reliability to accurately express knowledge.

5.4 Analysis of Context Inconsistency on Search Agent

This section first details the evaluation setting in Section 5.4.1, then analyzes context inconsistency effects in different parts of contexts of search agents in Section 5.4.2, and finally proposes the context refiner in Section 5.4.3.

5.4.1 Evaluation Settings

Dataset Experiments are conducted to evaluate the performance of search agents on various closed-book QA datasets, which necessitate extra retrieval to address, encompassing both single- and multi-hop scenarios. **Single-hop QA** includes: 1) **Natural Questions (NQ)** [89], which is constructed by Google Search queries along with annotated short answers; 2) **TriviaQA** [81], which contains closed-book trivia QA pairs to gauge models’ factual knowledge; and 3) **PopQA** [122], which consists of entity-centric QA pairs converted from a knowledge tuple retrieved in Wikidata. **Multi-hop QA** includes: 1) **HotpotQA** [223], the first large-scale dataset requiring reasoning across multiple Wikipedia paragraphs; 2) **2WikiMultiHopQA (2Wiki)** [62], which provides evidence information containing a reasoning path for multi-hop questions; 3) **MuSiQue** [180], which features more difficult 2-4 hop questions; and 4) **Bamboogle** [146], which is made up only of complex questions that Google answers incorrectly.

Search Agent We employ two foundation LLMs $\mathcal{M}$ for search agents: **Qwen-2.5-7b-Instruct** and **Qwen-2.5-3b-Instruct** [148]. For retrieval, we leverage E5 [193] as the retriever $\mathcal{E}$ and 2018 Wikipedia dump [85] as the knowledge base K_b . The number of retrieved passages K is set to 3 across all retrieval-based methods.

Evaluation Metrics We employ several metrics to evaluate both the **reliability** and **efficiency** of search agents. For reliability, we assess the correctness of the generated answer y with the reference $\hat{y}$ using **Exact Match (EM)** [167, 78], a standard string-matching metric that checks whether $y \equiv \hat{y}$, which is a percentage that represents the proportion of correctly answered questions out of all questions. Efficiency is evaluated regarding different aspects. Since the time cost of search agents is primarily determined by the number of retrieval operations, the **average retrieval times (ART)**, which denotes the average number of retrievals required per question, is employed to intuitively evaluate the efficiency of the generation processes. In addition, the **average context length (Len.)** in multi-turn generations and the **average inference time per question (AIT)** (/seconds) are also employed to assist efficiency assessments for search agents.

5.4.2 Context inconsistency in Different Parts of Context

Background To figure out context inconsistency effects of different parts in the contexts of search agents, we compare the performance of multi-turn search agents with different input contexts by masking specific segments (actions and observations in preceding rounds) of the history state. The history state s_i includes input question x and a series of actions $a_{0:i-1}$ and observations $o_{1:i}$. Specifically, x representing the initial state is fixed. p_i in a_i generally involves summarizing and reasoning from o_i . q_i is to interact with K_b to get retrieved documents in o_{i+1} . We present several inference methods for search agents as follows.

Inference Methods We employ direct inference (**Direct**) and Chain-of-Thought (**CoT**) reasoning [200] as two retrieval-free baselines, which represent LLMs’ internal knowledge $\mathcal{K}_I$ to answer questions. For retrieval-based settings, we employ Information Retrieval with CoT (**IRCoT**) [182], which enables LLMs to actively call the retriever for questions beyond their knowledge scope after thinking.

To understand the effect of different parts of context, several variants based on **IRCoT** are developed. Since search agents typically decompose a complex question into a set of sub-questions [78, 167], the generated search queries and retrieved documents in a history state are generally mutually independent, exhibiting rarely sequential dependencies. Therefore, we can specifically mask different parts in the states as follows. Several **IRCoT** variants that attempt to mitigate context inconsistency are also considered. 1) To investigate inconsistency in previous documents, **IRCoT-o (w/o $o_{:-1}$)** only incorporates the latest observation o_i of retrieved documents in context $[x, p_0, q_1, p_1, q_1, \dots, q_{i-1}, o_i]$ when generating a_i . 2) For inconsistency in search queries, **IRCoT-oq (w/o $o_{:-1}, q_{:-1}$)** with context $[x, p_0, p_1, \dots, p_{i-1}, q_{i-1}, o_i]$ is also employed. 3) For previous thinking steps, we utilize **IRCoT-oqp (w/o $o_{:-1}, q_{:-1}, p_{:-1}$)** relies exclusively on the latest thinking, search query, and observation as context $[x, p_{i-1}, q_{i-1}, o_i]$ when generating a_i .

Analysis and Findings As presented in Table 5.2, benefiting from K_b , all retrieval-based methods outperform retrieval-free baselines. Moreover, **IRCoT-o** outperforms **IRCoT** in both reliability and efficiency, suggesting that previous retrieved documents before round i contain context inconsistency for generating a_i . **IRCoT-oq** marginally outperforms **IRCoT-o**, indicating that previous search queries $q_{:-1}$ also carry slight inconsistency, although removing them may incur little extra retrieval. **IRCoT-oqp** exhibits a slight drop in reliability and a notable decrease in efficiency compared with others, implying that previous thinking steps store key information for future steps. LLMs may repeat previous search after masking $p_{:-1}$ and thus add retrieval costs. Consequently, **context inconsistency when generat-**

ing a_i may arise from both previous search queries and documents.

Methods	Single-Hop QA			Multi-Hop QA				Avg.
	NQ	TriviaQA	PopQA	HotpotQA	2wiki	Musique	Bamboogle	
Qwen2.5-7b-Instruct								
Direct	17.7 / 0.0	44.2 / 0.0	15.5 / 0.0	17.9 / 0.0	24.0 / 0.0	3.9 / 0.0	8.0 / 0.0	18.7 / 0.0
CoT	17.7 / 0.0	47.0 / 0.0	13.4 / 0.0	21.0 / 0.0	23.6 / 0.0	4.7 / 0.0	30.4 / 0.0	22.5 / 0.0
IRCoT	30.6 / 2.0	51.2 / 1.8	31.7 / 2.1	24.6 / 3.0	18.0 / 3.4	9.8 / 3.1	28.8 / 2.5	27.5 / 2.6
IRCoT-o	29.8 / 1.8	51.3 / 1.7	32.7 / 1.7	25.3 / 2.7	22.7 / 3.1	11.2 / 2.9	34.4 / 1.9	29.6 / 2.3
IRCoT-oq	32.8 / 2.0	51.7 / 2.0	33.0 / 2.0	26.0 / 3.0	23.6 / 3.1	11.4 / 3.1	34.4 / 1.7	30.4 / 2.4
IRCoT-oqp	32.9 / 2.6	51.5 / 2.7	32.7 / 2.8	25.4 / 2.6	22.8 / 3.1	10.6 / 2.8	36.0 / 2.5	30.3 / 2.7
Qwen2.5-3b-Instruct								
Direct	9.7 / 0.0	25.7 / 0.0	7.7 / 0.0	13.5 / 0.0	17.1 / 0.0	1.7 / 0.0	3.2 / 0.0	11.2 / 0.0
CoT	12.4 / 0.0	35.4 / 0.0	9.4 / 0.0	15.5 / 0.0	16.4 / 0.0	2.6 / 0.0	20.0 / 0.0	16.0 / 0.0
IRCoT	21.6 / 1.2	45.2 / 1.1	29.6 / 1.1	24.1 / 1.6	23.5 / 1.9	6.8 / 1.6	19.2 / 1.4	24.3 / 1.4
IRCoT-o	22.4 / 1.2	47.0 / 1.1	31.8 / 1.1	24.7 / 1.5	21.3 / 1.7	8.0 / 1.5	19.4 / 1.2	24.9 / 1.3
IRCoT-oq	23.5 / 1.3	47.2 / 1.2	29.9 / 1.4	24.9 / 1.6	24.2 / 1.7	7.0 / 1.6	19.4 / 1.2	25.2 / 1.4
IRCoT-oqp	24.6 / 1.5	47.7 / 1.3	30.5 / 1.5	24.5 / 1.8	22.5 / 1.7	5.1 / 1.5	19.0 / 1.2	24.8 / 1.5

Table 5.2: Performance (EM/ART) of different inference methods for search agents on test sets, measured by Exact Match (EM, for reliability) and Average Retrieval Times (ART, for efficiency).

However, as presented in Figure 5.3, although the above **IRCoT** variants gain slight improvements in accuracy, the gap between “recall rate” and “recall accuracy” remains considerable, indicating context inconsistency in previous rounds is not the dominant factor. Since the latest thinking p_{i-1} and search query q_{i-1} mostly encapsulate summaries and reasoning of previous context, it can be inferred that **the latest observation o_i is subject to the primary cause of context inconsistency.** To mitigate the inconsistency, further context refinement methods for the latest observed documents are required for search agents.

5.4.3 Construction of Context Refiner for Search Agent

Background Our preliminary findings and analysis in Section 5.4.2 demonstrate that we can marginally mitigate context inconsistency by removing irrelevant information like previous documents and search queries in multi-turn search agents but not enough, which highlights the necessity of further capturing critical information and filtering noise in the latest retrieved documents. More generally, we desire to develop a context refiner $\mathcal{F}$ that, in each round i , refine the context to preserve

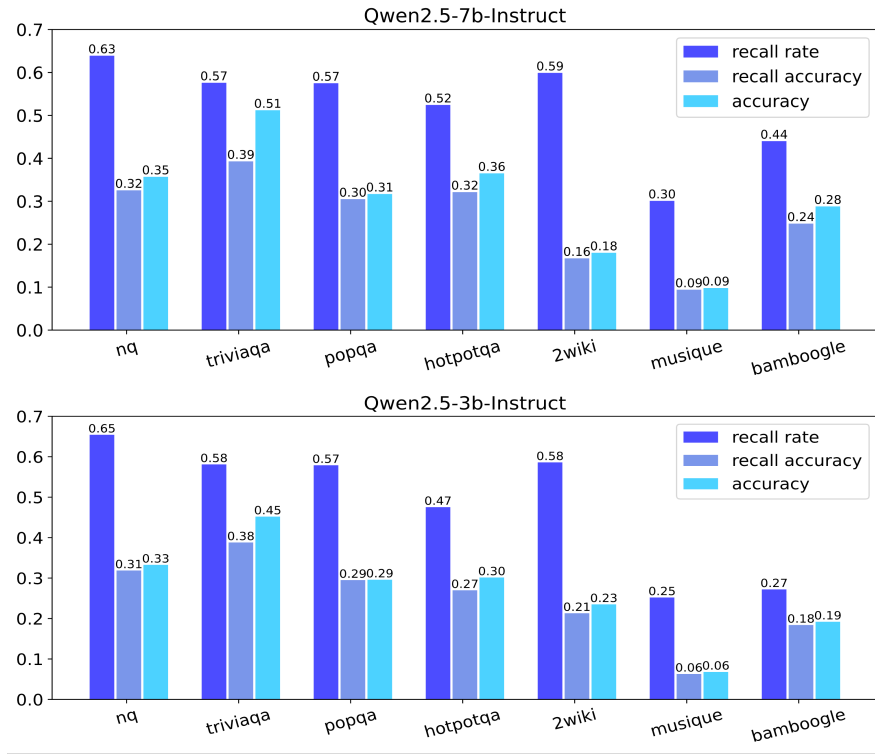


Figure 5.3: Demonstrations of context interference effects on four IRCoT variants of search agents. Results are averaged on all QA test sets; metrics follow the definitions in Figure 5.2. Demonstration of how contextual interference affects the performance of LLM search agents. “recall rate”= N_r/N denotes the proportion of questions for which the retrieved documents contain the correct answer (N_r) among all questions (N). “recall accuracy”= N_{rc}/N refers to the proportion of correctly answered questions (N_{rc}) in N_r , relative to all questions (N). “accuracy” represents the proportion of all correctly answered questions N_c out of the questions (N).

the most relevant information $\tilde{d}_i = \mathcal{F}(q_{i-1}, d_i)$ to search query q_{i-1} from the latest retrieved documents d_i . $\tilde{d}_i$ is served as the i -th observation o_i and then appended to s_i to generate a_i .

However, directly designing precise schemes to capture key information from numerous documents is challenging [47]. A compression model may focus on summarizing global contents, potentially leading to information loss or the introduction of extraneous knowledge [104]. Relatively small LLMs exhibit limited capability for key information extraction (in Table 5.3). Therefore, we propose to distill a dataset for context refinement from advanced LLMs, enabling relatively weak models to

refine context to mitigate inconsistency.

Methods	Single-Hop QA			Multi-Hop QA				Avg.
	NQ	TriviaQA	PopQA	HotpotQA	2wiki	Musique	Bamboogle	
Qwen2.5-7b-Instruct								
IRCoT	30.6 / 2.0	51.2 / 1.8	31.7 / 2.1	24.6 / 3.0	18.0 / 3.4	9.8 / 3.1	28.8 / 2.5	27.5 / 2.6
GPT-Compress	32.5 / 0.9	55.5 / 1.0	34.8 / 1.1	29.8 / 1.6	25.0 / 1.5	9.6 / 1.3	26.4 / 1.1	30.5 / 1.2
GPT-Refine	34.6 / 1.1	57.5 / 0.9	37.0 / 1.0	33.0 / 1.3	27.6 / 1.8	9.3 / 1.4	33.6 / 1.0	33.2 / 1.2
Self-Refine	32.5 / 1.5	53.3 / 0.9	33.0 / 1.0	27.0 / 1.4	21.6 / 1.5	8.4 / 1.2	28.8 / 1.0	29.2 / 1.2
Context Refiner	34.0 / 1.2	56.9 / 0.9	36.4 / 1.0	32.0 / 1.4	27.2 / 1.6	8.6 / 1.2	30.4 / 1.0	32.2 / 1.2
Qwen2.5-3b-Instruct								
IRCoT	21.6 / 1.2	45.2 / 1.1	29.6 / 1.1	24.1 / 1.6	23.5 / 1.9	6.8 / 1.6	19.2 / 1.4	24.3 / 1.4
GPT-Compress	31.3 / 0.9	51.5 / 0.9	29.9 / 1.0	24.1 / 1.1	22.7 / 1.2	7.2 / 1.0	18.4 / 0.9	26.4 / 1.0
GPT-Refine	32.6 / 1.0	54.5 / 0.8	35.6 / 0.9	24.4 / 1.0	22.5 / 1.3	7.5 / 0.9	24.0 / 0.9	28.7 / 1.0
Self-Refine	23.7 / 1.1	47.6 / 0.9	30.8 / 0.9	23.1 / 1.0	21.4 / 1.2	5.0 / 1.1	20.0 / 0.9	24.5 / 1.0
Context Refiner	30.7 / 1.0	50.8 / 0.9	32.0 / 0.9	24.1 / 1.0	22.2 / 1.2	7.2 / 1.0	19.4 / 0.9	26.6 / 1.0

Table 5.3: Performance (EM/ART) of context refinement methods for mitigating interference, measured by Exact Match (EM) and Average Retrieval Times (ART) across QA test sets.

Context Refiner Given a distill dataset $\mathcal{D}_d = \{x_i, \hat{y}_i\}_{i=1}^N$, a foundation LLM $\mathcal{M}_\pi$, an advanced teacher LLM $\mathcal{M}_T$, a retriever $\mathcal{E}$, and a knowledge base K_b , we infer each query x on $\mathcal{M}_T$ using IRCoT [181]. In i -th round of inference for x , the retriever return documents $d_i = \mathcal{E}(q_{i-1}|K_b)$. Then we instruct the teacher model $\mathcal{M}_T$ to specifically extract only critical information related to q_{i-1} from d_i . Extracted information $\tilde{d}_i = \mathcal{M}_T(q_{i-1}, d_i)$ is then appended into context s_i to generate the next-step action a_i . For each correct trajectory with $y \equiv \hat{y}$, we retain all step-wise pairs of extracted data $(\tilde{d}_i, \langle d_i, q_{i-1} \rangle)$ and employ an entailment model to verify that $\tilde{d}_i$ is entirely encompassed within d_i and does not introduce extra knowledge. Finally, all qualified data points are re-formatted and incorporated into the context refinement dataset $\mathcal{D}_c = \{\tilde{d}_j, d_i, q_i\}_{i=1}^M$.

Given $\mathcal{D}_c$, we train the model $\mathcal{M}_\pi$ to enable its ability of context refinement

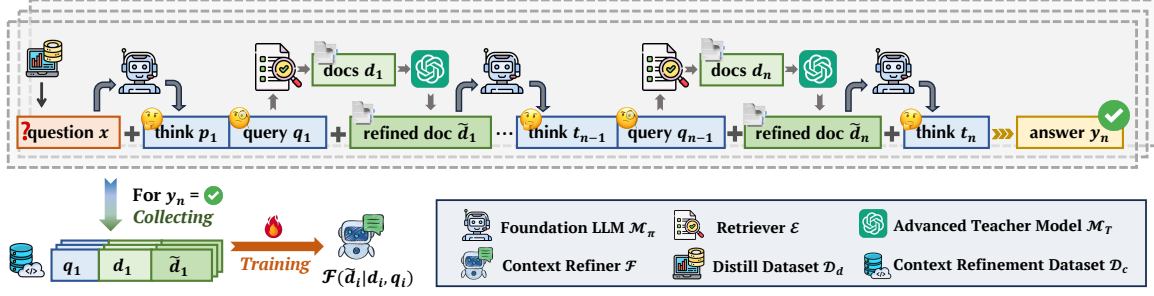


Figure 5.4: Training pipeline for the distill-based context refiner. On the constructed context refinement dataset, we first use an advanced teacher LLM (M_T) to construct a context refinement dataset (D_c). We then fine-tune a foundation LLM (M_π) on D_c via supervised fine-tuning (SFT).

using supervised fine-tuning (SFT) as follows.

$$\pi^* = \arg \min_{\pi} \mathcal{L}_{\pi}^{\text{SFT}} \quad (5.1)$$

$$\mathcal{L}_{\pi}^{\text{SFT}} = -\frac{1}{M} \sum_{i=1}^M \mathbb{E}_{(\tilde{d}_i, d_i, q_i) \sim \mathcal{D}_c} \mathcal{L}_{\pi}^{(i)} \quad (5.2)$$

$$\mathcal{L}_{\pi}^{(i)} = \log \mathcal{M}_{\pi}(\tilde{d}_i \mid d_i, q_i) \quad (5.3)$$

$\mathcal{M}_{\pi}$ is trained to generate refined documents $\tilde{d}_i$, yielding a context refiner $\mathcal{F} = \mathcal{M}_{\pi^*}$ for dynamic context refinement in multi-turn search agents. The teacher model $\mathcal{M}_T$ is GPT-4 and the base model $\mathcal{M}_{\pi}$ of context refiner $\mathcal{F}$ is Qwen2.5-7b-Instruct or Qwen2.5-3b-Instruct, which is the same as their respective base models.

Context Refinement Methods We employ several existing context refinement methods as comparisons: 1) We utilize the compression method by introducing GPT-4 [136] to summarize the previous contexts (**GPT-Compress**); 2) We employ GPT-4 to dynamically refine the latest Top-K retrieved documents d_i in each round based on the search query q_i (**GPT-Refine**). 3) We also employ the foundation LLM itself $\mathcal{M}_{\pi}$ to dynamically refine the latest documents d_i using q_i (**Self-Refine**).

Results and Analysis As in Table 5.3 and Figure 5.5, **IRCoT** baseline is also presented for intuitive comparisons of context inconsistency mitigation. **GPT-Refine**

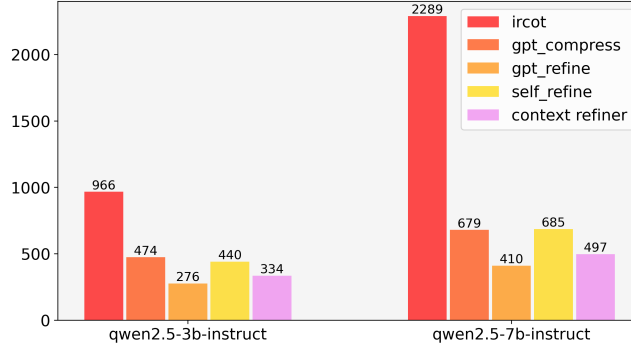


Figure 5.5: Averaged context lengths of search agents on test sets, comparing IRCoT (baseline), three context refinement methods, and the proposed context refiner for efficiency assessments.

consistently outperforms **GPT-Compression** in terms of reliability, while being slightly inferior in search times and context length, indicating that extracting and preserving search query-relevant key information is more effective in mitigating contextual inconsistency than general summarization. These two baselines can effectively reduce the retrieval wastes of search agents on irrelevant information in context over **IRCoT**. **Self-Refine**, by contrast, does not yield performance improvements and even fails to reduce context length, which can be attributed to the lack of information extraction capability of the foundation model $\mathcal{M}_\pi$. In comparison, **Context Refiner** trained on the context refinement dataset achieves close performance to **GPT-Refine**, suggesting that even relatively weak LLMs can also acquire the ability to refine context through fine-tuning, without relying on external models for search agents. In addition, Context Refiner achieves marginally competitive performance to prompt-driven GPT-Refine which represents the performance of the teacher model. This suggests that Context Refiner can refine context to mitigate context inconsistency and outperforms other compression and self-refine baselines.

To further explore the potential of mitigating context inconsistency in search agent training pipelines (as **RQ iii**), we extend our context refiner to the RL training pipeline of search agents, proposing a novel **Context-Refined Reinforcement Learning** (CRRL) framework to dynamically refine context during training rollouts,

reducing context inconsistency in trajectory quality as follows.

5.5 Context Refinement for Search Agent Training

5.5.1 Context-Refined Reinforcement Learning

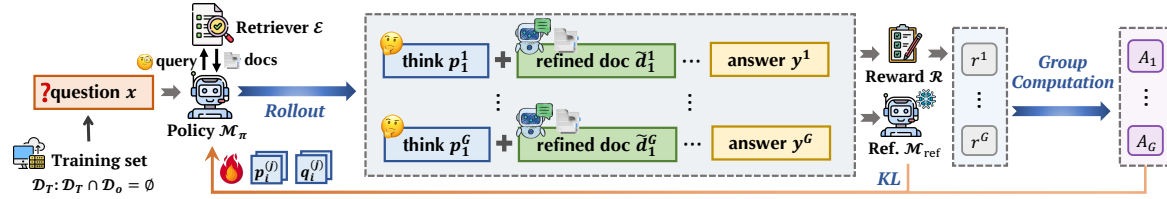


Figure 5.6: Demonstration of our proposed CRRL method.

Reinforcement learning (RL) [83] has emerged as a paradigm for search agent training [167, 78, 16] using PPO [157] and GRPO [158]. During rollouts, these RL algorithms enable LLMs to iteratively interact with search engines and append the retrieved documents to contexts, potentially introducing inconsistency in RL training and resulting in suboptimal performance of RL. Nonetheless, current RL methods overlook the impact of context inconsistency of rollouts. Derived from Section 5.4, we have figured out that removing previous documents and search queries, as well as leveraging context refiner for the latest documents in contexts, can eliminate context inconsistency for multi-turn search agents, which can also improve the quality of rollouts. Therefore, we propose the CRRL algorithm.

Current RL pipelines for search agents are mainly based on Proximal Policy Optimization (PPO) [157] and Group Relative Policy Optimization (GRPO) [158]. GRPO performs multiple rollouts per task and calculates the relative reward within the group as the advantage, which is more lightweight without the value model and demonstrates comparable performance with PPO [78]. Hence, this paper adopts GRPO as the default RL algorithm, and the proposed CRRL is also based on GRPO,

which optimizes the policy as follows:

$$\mathcal{L}^{\text{GRPO}}(\theta) = -\mathbb{E}_{x \sim \mathcal{D}, \{y_i\}_{i=1}^G \sim \pi_{\text{ref}}(\cdot|x)} [\mathcal{R}_\theta] \quad (5.4)$$

$$\mathcal{R}_\theta = \frac{1}{|y|} \sum_{t=1}^{|y|} \min(r_\theta A_t, \text{clip}(r_\theta, 1 - \epsilon, 1 + \epsilon) A_t) \quad (5.5)$$

$$r_\theta = \frac{\pi_\theta(y_t|x, y_{<t})}{\pi_{\text{ref}}(y_t|x, y_{<t})} \quad (5.6)$$

To mitigate context inconsistency of GRPO rollouts and obtain high-quality trajectories, our CRRL dynamically refines the input context during rollout of multi-turn search agents with the context refiner $\mathcal{F}$. As shown in Figure 5.6, during rollout, the trajectories of CRRL only contain thinking steps and refined documents, which indicates that the action $a_i^j = [p_i^j, q_i^j]$ is produced with the context $s_i^j = [x, p_{0:i-1}^j, q_{i-1}^j, \tilde{d}_i^j]$ where $\tilde{d}_i^j = \mathcal{F}(\tilde{d}_i^j)$, $\tilde{d}_i^j = \mathcal{E}(q_{i-1}^j)$. RL pipelines for search agents explicitly incorporate retrieval interleaved reasoning, and the token-level losses are only computed over the rollouts of LLM-generated tokens, including both search queries and thinking steps, where loss masking is introduced for retrieved tokens, ensuring the stabilization of training while preserving the ability to adaptively retrieve. When optimizing the policy $\mathcal{M}_\pi$ on $a_i = [p_i, q_i]$, the CRRL algorithm can be represented as follows:

$$\mathcal{L}^{\text{CRRL}}_{\mathcal{M}_\pi} = -\mathbb{E}_{\{a_i^j\}_{j=1}^G \sim \mathcal{M}_{\text{ref}}(\cdot|s_i^j)} [\mathcal{G}_\pi - \beta \text{KL}] \quad (5.7)$$

$$\mathcal{G}_\pi = \frac{1}{G} \sum_{j=1}^G \frac{1}{\sum_{k=1}^{N-1} |a_{i,k}^j|} [\mathcal{R}_\pi^j] \quad (5.8)$$

$$\mathcal{R}_{\pi,i,k}^j = \sum_k^{N-1} \min(r_{\pi,i,k}^j A_j, \text{clip}(r_{\pi,i,k}^j, 1 - \epsilon, 1 + \epsilon) A_j) \quad (5.9)$$

$$r_{\pi,i,k}^j = \frac{\mathcal{M}_\pi(a_{i,k}^j|x, a_{i,<k}^j, s_i^j)}{\mathcal{M}_{\text{ref}}(a_{i,k}^j|x, a_{i,<k}^j, s_i^j)} \quad (5.10)$$

where $\mathcal{M}_{\text{ref}}$ represents reference model. The term ϵ is a clipping ratio. β is the coefficient for the KL divergence. The advantage estimate is computed on the

Methods	Single-Hop QA			Multi-Hop QA				Avg.
	NQ	TriviaQA	PopQA	HotpotQA	2wiki	Musique	Bamboogle	
Qwen2.5-7b-Instruct								
IRCoT	30.6 / 2.0	51.2 / 1.8	31.7 / 2.1	24.6 / 3.0	18.0 / 3.4	9.8 / 3.1	28.8 / 2.5	27.5 / 2.6
SFT	26.7 / 0.0	40.5 / 0.0	15.3 / 0.0	22.4 / 0.0	20.5 / 0.0	6.3 / 0.0	11.2 / 0.0	20.4 / 0.0
R1	24.7 / 0.0	55.4 / 0.0	20.0 / 0.0	19.4 / 0.0	17.5 / 0.0	7.8 / 0.0	22.4 / 0.0	23.9 / 0.0
RFT	34.4 / 2.2	57.5 / 2.0	36.3 / 2.4	31.8 / 3.0	26.5 / 3.6	11.5 / 3.2	33.6 / 2.7	33.1 / 2.7
Search-R1	35.5 / 1.7	58.4 / 1.4	38.5 / 1.8	35.6 / 2.7	27.0 / 3.0	11.5 / 2.3	36.0 / 1.8	34.6 / 2.1
Search-o1	37.5 / 1.6	60.4 / 1.4	39.2 / 1.8	37.8 / 2.3	29.6 / 2.6	13.4 / 2.0	36.8 / 1.6	36.2 / 1.9
CRRL	38.1 / 1.4	60.1 / 1.2	39.4 / 1.6	38.2 / 2.0	30.5 / 2.5	13.2 / 1.9	36.8 / 1.5	36.6 / 1.7
Qwen2.5-3b-Instruct								
IRCoT	21.6 / 1.2	45.2 / 1.1	29.6 / 1.1	24.1 / 1.6	23.5 / 1.9	6.8 / 1.6	19.2 / 1.4	24.3 / 1.4
SFT	23.8 / 0.0	36.2 / 0.0	11.4 / 0.0	18.6 / 0.0	22.8 / 0.0	4.8 / 0.0	8.0 / 0.0	17.9 / 0.0
R1	19.6 / 0.0	42.9 / 0.0	15.2 / 0.0	18.5 / 0.0	26.5 / 0.0	8.0 / 0.0	19.2 / 0.0	21.4 / 0.0
RFT	29.4 / 1.3	47.5 / 1.2	32.5 / 1.1	26.0 / 1.8	24.4 / 2.2	6.4 / 1.8	21.0 / 1.8	26.7 / 1.6
Search-R1	33.7 / 1.4	52.1 / 1.1	34.2 / 1.1	27.1 / 1.6	26.5 / 1.6	11.8 / 1.7	23.2 / 1.5	29.8 / 1.4
Search-o1	34.7 / 1.1	53.2 / 1.1	33.5 / 1.1	27.0 / 1.3	28.6 / 1.5	12.0 / 1.6	23.2 / 1.3	30.3 / 1.3
CRRL	34.7 / 1.1	54.5 / 1.1	36.8 / 1.0	29.0 / 1.4	30.0 / 1.5	12.4 / 1.6	23.2 / 1.4	31.5 / 1.3

Table 5.4: Performance results of EM/ART across QA test sets on several baselines as well as our proposed CRRL method, measured by Exact Match (EM) and Average Retrieval Times (ART).

group-relative rewards of trajectory τ^j as

$$A_j = \frac{r^j - \mu^j}{\sigma^j} \quad (5.11)$$

where $r^j = \mathcal{R}(y^j)$ is the final reward, and μ^j and σ^j denote the mean and standard deviation of the rewards within the group.

5.5.2 Training Setup

Implementation Details We sample 40k data from the training sets of NQ and HotpotQA and merge them into $\mathcal{D}_t$ where $\mathcal{D}_t \cap \mathcal{D}_d = \emptyset$. Evaluation is conducted on seven QA datasets to assess both in-domain and out-of-domain performance. GRPO implementation is based on verl [161]. The batch size is set at 32, with a mini-batch size of 8 and a micro-batch size of 4. The maximum input sequence length and generation length are set to 2048 and 500 respectively. We enable gradient checkpointing and use Fully Sharded Data Parallel (FSDP) with CPU offloading. For efficient LLM rollouts, we adopt vLLM [91] with a tensor parallel size of 1 and

GPU memory utilization ratio of 0.7. The rollout sampling temperature is set to 0.7 and the top-p value to 1.0. The KL divergence regularization coefficient β and clip ratio ϵ are set to 0.001 and 0.2. The maximum action budget B is set to 8. In cases where training diverges, we evaluate at the most recent stable checkpoint according to the training reward curve; otherwise, the final checkpoint is used for evaluation.

For training-based baselines, due to the computational resource limitation with only 4×40G A100 GPU cards, fine-tuning on the all corpus of 160k samples of the training corpus is expensive. Therefore, we employ a subset with 60k samples randomly sampled from the original training corpus. All other training settings are maintained. Experiments conducted using CRRL are to validate the effectiveness of incorporating context refinement in RL pipelines for search agent training, which is regardless of the training corpus quantity.

Baselines We employ two types of baselines as comparisons. 1) **Retrieval-free Fine-Tuning**: We train LLMs using both supervised fine-tuning (SFT) and (GRPO)-based RL [158] methods, which only contain reasoning and answer steps. 2) **Retrieval-based Fine-Tuning**: To obtain the trajectories with the retriever, we utilize rejection sampling to generate several candidate responses by interacting with the search engine for each question from the training set $\mathcal{D}_t$. Then we collect paths including correct answers for rejection fine-tuning (RFT) [232]. We also employ the RL pipeline for multi-turn search agents in **Search-GRPO** following [78] and **Search-o1** following [102], which optimizes LLM rollouts to autonomously call the retriever when they lack relevant knowledge.

5.5.3 Results and Analysis

In Table 5.4, experimental results demonstrate that our proposed CRRL method outperforms other baselines in EM scores, suggesting that incorporating context refinement into the RL training pipeline of search agents can effectively mitigate

LLMs	Methods	ART	Len.	AIT
Qwen2.5-7b	IRCoT	2.6	2.3k	22.4s
	Search-GRPO	2.1	0.9k	17.7s
	CRRL	1.6	0.7k	16.9s
Qwen2.5-3b	IRCoT	1.4	0.9k	7.5s
	Search-GRPO	1.4	0.7k	4.5s
	CRRL	1.2	0.5k	4.2s

Table 5.5: Efficiency comparisons of baselines and our proposed CRRL during inference on test sets of average retrieval times (ART), average context length (Len.), and average inference time per question (AIT).

context inconsistency and improve the reliability of generations.

In Table 5.5, during the inference phase, our proposed CRRL achieves a significant reduction in both ART and Len. compared to IRCoT and Search-R1, which compensate for context refiner overhead arising from the extra inference costs, and result in lower AIT, suggesting that introducing a context refiner to RL training to relax contextual inconsistency can also improve the efficiency of the search agent.

5.6 Summary

In this chapter, we investigate the context inconsistency issue on LLM-based search agents. We first demonstrate that context inconsistency, largely stemming from the latest retrieved documents, poses a key challenge to multi-turn search agents. Then, we present a distill-based context refiner to dynamically mitigate inconsistency in multi-turn search agents and thus boost both accuracy and efficiency. Furthermore, we propose a context refiner and incorporate it into the RL training pipelines of search agents, which can further yield performance improvements. These findings highlight the importance of context refinement to mitigate context inconsistency to construct reliable and efficient LLM-based search agents, paving the way for a new paradigm of dynamic “refine context and then generate” for AI agents in future work.

Chapter 6

Conclusion

This chapter summarizes the research on the reliability of LLMs conducted in this dissertation in Section 6.1 and reflects on its limitations in Section 6.2. Furthermore, two promising research directions are extended from this thesis in Section 6.3: the first is to construct reliable LLM agents by developing an integrated framework that unifies factuality, honesty, and consistency; the second is to explore approaches for enhancing the reliability of multimodal large models.

6.1 Thesis Summary

This dissertation focuses on addressing the critical hallucination problem of LLMs, which has become a major bottleneck restricting the deployment and trustworthiness of LLMs. Hallucinations originate from the inherent limitations of LLM training mechanisms, which lead to three core unreliability issues of LLMs: lack of honesty in response to unknown queries, factual errors in single responses, and contextual inconsistency in multi-turn interactions. To tackle these issues comprehensively, this thesis proposes a systematic framework for constructing reliable LLMs, centered on the three core dimensions of Factuality, Honesty, and Consistency, and conducts in-depth research and validation in each dimension.

First, regarding the factuality of LLMs, this study identifies that ambiguous knowledge boundaries are the root cause of non-factual responses. To address this, we propose a novel factuality alignment framework characterized by "from perception to expression". This framework enables LLMs to precisely perceive their knowledge boundary representations by combining confidence and semantic entropy, and then aligns their outputs with factual expressions through PPO. Experimental results demonstrate that the proposed framework significantly improves the factuality of LLM outputs compared to baseline methods. Importantly, this work pioneers a new "from perception to expression" paradigm that provides a new direction for future LLM alignment research.

Second, focusing on LLM honesty, this study explores two key aspects: self-awareness and confidence estimation. Recognizing the lack of benchmarks covering complex practical scenarios for evaluating LLM honesty, we introduce two challenging and underexplored benchmarks: the multilingual confidence estimation benchmark **MlingConf** and the mathematical reasoning solvability benchmark **ReliableMath**. The **MlingConf** benchmark includes language-agnostic and language-specific scenarios, revealing the language dominance effect of English in multilingual confidence estimation. Based on this, we propose a native-tone prompting method that effectively improves calibration of LLMs in multilingual scenarios, facilitating the global deployment of reliable LLMs. The **ReliableMath** benchmark, as the first benchmark to evaluate the reliability of step-by-step reasoning on both solvable and unsolvable mathematical problems, provides valuable insights into LLM performance in complex reasoning tasks. We further propose an honesty alignment strategy that enhances LLMs' critical thinking abilities in both mathematical and out-of-domain knowledge tasks, contributing to the improvement of LLM reliability in complex scenarios.

Third, targeting the contextual inconsistency issue of LLM-based search agents in multi-turn interactions with external environments, this study identifies that the latest documents in multi-turn retrieval are the main source of inconsistency. We

propose a distillation-based context refiner to dynamically eliminate contextual inconsistencies, and integrate this refiner into the reinforcement learning (RL) training pipeline of search agents. Experimental results show that this approach not only improves the contextual consistency but also enhances their accuracy and efficiency, inspiring a new paradigm of “refine context and then generate” for the development of LLM agents.

Overall, this dissertation systematically addresses LLM hallucinations by constructing a reliable LLM framework covering factuality, honesty, and consistency. The main contributions of this work are threefold: first, a novel factuality alignment framework and a new research paradigm for LLM alignment; second, two innovative benchmarks that fill the gap in complex scenario evaluation of LLM honesty, along with corresponding optimization strategies; third, a context refinement method for the RL training pipeline of LLM agents, promoting a new research paradigm of LLM-based intelligent agents. These contributions not only effectively alleviate the hallucination problem of LLMs but also enrich the research landscape of LLM reliability and hallucination mitigation, laying a solid theoretical and experimental foundation for the future development of trustworthy AI systems.

6.2 Reflections

Despite the achievements made in this study, there are still several limitations.

- Regarding LLM factuality, the factual alignment method proposed in this work was developed in 2024, where the perception stage relies on measuring representations of the LLM’s knowledge boundaries. However, such measurement becomes significantly more difficult in long-context tasks, and due to resource limitations, we were unable to further scale this approach. Recent factuality alignment methods, while still generally following the paradigm of “from perception to expression,” achieve perception by leveraging RL to elicit

the LLM’s long chain-of-thought (CoT) reasoning capabilities, rather than explicitly constructing datasets to measure knowledge boundaries. As a result, these methods continue to perform well even on long-text tasks. Nevertheless, the proposed factuality alignment approach regarding the knowledge boundary in this thesis offers strong interpretability. It therefore remains highly meaningful for future research on how to leverage knowledge boundaries to improve factuality under long CoT reasoning settings.

- Regarding LLM honesty, the two proposed benchmarks cover multilingual and mathematical reasoning scenarios, but there are still many practical scenarios (e.g., medical, legal) that need to be explored to build more comprehensive evaluation systems for LLM honesty. What is urgently required currently is an honesty assessment specifically tailored to LLM agents. Owing to the additional interactions with the external environment, agents must possess an accurate understanding of the boundaries between their internal knowledge and external knowledge to determine their awareness and confidence.
- Regarding LLM consistency, the context refiner proposed in this thesis is applied to LLM-based search agents; future work can extend this method to other types of LLM agents (e.g., dialogue agents, task planning agents) to enhance their contextual consistency in more complex interaction scenarios. The principal factors of contextual inconsistency in various tasks may differ, necessitating specific mitigation strategies to further improve the reliability and efficiency of AI agents. Current research on intelligent agents predominantly adopts summarization and compression to process long-context input. By contrast, this work argues for maximizing the utilization of historical information, wherein the information bottleneck serves as a well-suited constraint and a promising way for this purpose.

6.3 Future Work

6.3.1 Unified Framework for Reliable LLM Agents

While this thesis has advanced LLM hallucination mitigation by optimizing factuality, honesty, and consistency separately, unifying these three core dimensions into an integrated framework for LLM agents remains unaddressed. Current fragmented optimization and evaluation hinder real-world deployment of reliable agents. Based on this thesis, the following streamlined future directions focus on achieving this unification.

Unified Framework for Multi-objective Optimization Future research will focus on multi-objective optimization alignment to integrate factuality, honesty, and consistency of LLMs into a unified framework, addressing trade-offs of independent optimization. First, a unified reliability scoring system will be established to model the three dimensions as a multi-objective optimization problem. Then, a synergistic alignment mechanism via extended RL pipelines can be designed with constraint-based optimization. Additionally, dynamic adaptation is also supposed to handle diverse scenarios, enhance framework interpretability, extend it to complex LLM systems, and enrich multi-objective reliability benchmarks. This aims to comprehensively mitigate LLM hallucinations and lay a foundation for trustworthy AI.

Holistic Evaluation Benchmark and Metrics A key focus will be on developing a holistic evaluation benchmark and metrics that simultaneously assess factuality, honesty, and consistency. This benchmark will cover diverse tasks and real-world scenarios, integrating human evaluation with automatic metrics to capture both objective accuracy and subjective appropriateness. Unified metrics should be designed to quantify synergies and trade-offs between the three dimensions, enabling rigorous evaluation of multi-objective alignment frameworks and facilitating fair comparison across different methods.

6.3.2 Reliable Multi-modal LLMs

The framework established in this dissertation lays a solid foundation for constructing reliable LLMs along the dimensions of factuality, honesty, and consistency. However, the rapid proliferation of Multimodal Large Language Models (MLLMs)—systems capable of jointly processing and generating text, images, audio, and video—introduces a new and substantially more complex landscape of hallucination challenges. Extending the reliability framework proposed herein to the multi-modal setting constitutes a compelling and urgent direction for future research.

Multimodal Factuality: Cross-Modal Knowledge Boundary Perception. The “perception to expression” paradigm proposed in this thesis for textual factuality alignment opens a promising avenue for multimodal extension. In MLLMs, factual errors frequently arise not only from ambiguous textual knowledge boundaries but from misalignment between visual and linguistic representations. A natural future direction is to extend the knowledge boundary perception mechanism to cross-modal settings, enabling MLLMs to jointly model their uncertainty over both visual and textual knowledge. This would require developing cross-modal boundary representations that capture the degree to which a model’s visual perception is consistent with its linguistic expression, and designing alignment objectives that explicitly penalize visually ungrounded generation. Such a framework would represent a multimodal instantiation of the “from perception to expression” paradigm and address one of the most prevalent failure modes of current MLLMs.

Multimodal Honesty: Calibrated Uncertainty Across Modalities. The benchmarks and optimization strategies for multilingual confidence estimation and mathematical reasoning solvability developed in this thesis reveal that honest uncertainty expression is both task-dependent and language-dependent. In multimodal contexts, the challenge of calibration becomes substantially more complex: an LLM’s uncertainty over a visual question-answering query depends not only on

the LLM’s linguistic knowledge but on the reliability of its visual encoder, the quality of the input image, and the inherent ambiguity of the visual content. Future work could extend the honesty evaluation benchmarks to multimodal scenarios. The honesty alignment method to determine self-awareness in this thesis could be generalized to multimodal reasoning tasks, such as chart understanding or scientific diagram interpretation, where an LLM should ideally recognize the limits of its visual comprehension before committing to a response.

Multimodal Consistency: Long-Horizon Cross-Modal Context Coherence. The contextual inconsistency problem addressed in this thesis within the setting of LLM-based search agents becomes even more pronounced in multimodal agentic systems that interact with heterogeneous external environments —processing sequences of images, documents, and tool outputs across extended multi-turn interactions. In such settings, inconsistency can manifest not only within the textual context but across modalities: a model may contradict earlier visual observations with later textual claims, or fail to maintain coherent entity references across image-text turn boundaries. The context refiner and RL training pipeline developed in this work provide a conceptual blueprint for addressing this challenge; future research could extend this framework to multimodal context refinement, designing mechanisms that selectively compress, align, and verify cross-modal context across long horizons. This is particularly relevant for real-world deployment scenarios such as multimodal medical reports or embodied agents navigating visual environments.

In summary, while this thesis establishes a principled and empirically validated framework for reliable LLMs in the textual domain, the transition to trustworthy multimodal AI systems demands a rethinking of factuality, confidence calibration, and contextual consistency in heterogeneous cross-modal inputs. The theoretical foundations, alignment paradigms, and benchmarking methodologies developed herein offer a strong point of departure for this next frontier of reliable AI research.

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